Probability for Finance

Studiabile — Chapters 1–5, through HW 53

Theory, Expanded Proofs, Worked Homework, Past and AI-Generated Exams, and More

These unrevised notes were generated using artificial intelligence, following the student's preferred style and, where possible, the style of Professor Tiziano De Angelis and former tutor Andrea Bovo. Additional proofs and explanations not presented or proved by the professor have been included at the student's discretion. The material has not undergone a complete independent review and may contain errors.

These notes follow the lecturer's order and explain the hypotheses and steps of each proof. Every homework retains its own statement and solution beside the relevant theory. A later homework is anticipated only when the lecturer's own proof takes that exact fact as given; its proof remains at the homework itself. All mathematical displays are unnumbered.

Contents

Theory, Expanded Proofs, Worked Homework, Past and AI-Generated Exams, and More . . . 1

1. Measure Spaces . . . 4
- Measure theory and continuous functions . . . 4
- Def 1.1. Algebra . . . 5
- HW 1 — Closure under finite intersections . . . 5
- HW 2 — Finite unions and intersections . . . 5
- Def 1.2. σ-algebra . . . 5
- HW 3 — Countable intersections . . . 6
- Measurable spaces, generated sigma-algebras and Borel sets . . . 6
  - Notation: . . . 6
  - Measurable sets: . . . 6
  - Measurable space: . . . 6
- Prop 1.1 . . . 6
- Set functions . . . 7
- Additive and countably additive set functions . . . 7
- HW 4 — Finite additivity by induction . . . 7
- HW 5 — Countable additivity implies additivity . . . 8
- Measure . . . 8
  - Measure space: . . . 8
  - Null set: . . . 8
  - Complete measure space: . . . 8
- Definition 1.3. π-system . . . 8
- HW 6 — Lower rays form a pi-system . . . 8
- Lemma 1.2 . . . 8
- Theorem 1.3 (Carathéodory's Theorem) . . . 9
- Lebesgue measure . . . 9
- HW 7 — The interval algebra . . . 9
- HW 8 — Finite additivity of interval length . . . 10
- HW 9 — Borel singletons . . . 11
- HW 10 — Lebesgue measure of a singleton . . . 11
- Lemma 1.4 . . . 11
- HW 11 — Subadditivity and the union formula . . . 11

HW 12 — Monotonicity of measures . . . 12
HW 13 — Measure of a difference . . . 12
Lemma 1.5. Continuity of measure from below . . . 12
HW 14 — Disjoint pieces . . . 13
HW 15 — Unions of the disjoint pieces . . . 13
Lemma 1.6. Continuity of measure from above . . . 13
HW 16 — Continuity from above . . . 13
Hint: . . . 13
Lemma 1.7 . . . 14
HW 17 — Rational points form a null set . . . 14
2. Events . . . 15
Probability spaces . . . 15
Example 1 (Simple coin tossing models) . . . 15
HW 18 — Sure events and null complements . . . 15
Proposition 2.1 . . . 15
Def lim inf and lim sup . . . 16
HW 19 — Tail infima . . . 16
Definition of convergence through liminf and limsup . . . 16
HW 20 — An oscillating sequence . . . 16
HW 21 — Threshold properties of limits . . . 16
HW 22 — Negation, suprema and infima . . . 17
Def limsup of collection of sets . . . 18
HW 23 — Measurability of the limsup event . . . 18
Lemma 2.2 (Reverse Fatou's lemma for sets) . . . 18
Lemma 2.3 (First Borel–Cantelli's lemma) . . . 18
Def liminf of collection of sets . . . 19
HW 24 — Measurability of the liminf event . . . 19
HW 25 — Complement relation of event limits . . . 19
Lemma 2.4 (Fatou's lemma for sets) . . . 19
HW 26 — Fatou inequality by complements . . . 20
Hint: . . . 20
Definition of indicator function . . . 20
HW 27 — Indicators and limsup . . . 20
Hint: . . . 20
3. Random Variables . . . 22
Random variables and pre-images . . . 22
Def — pre-image of a set . . . 22
Def 3.1 (Measurable functions) . . . 22
Notation: . . . 22
Borel-measurable functions: . . . 22
Properties of pre-image mapping . . . 22
HW 28 — Pre-images of complements and intersections . . . 22
Lemma 3.1 — The pre-image of Borel sets form a σ-algebra . . . 23
Rmk: . . . 23
Problem: . . . 23
Lemma 3.2. Measurability through a generating class . . . 23
Remarks . . . 24
HW 29 — A Borel indicator . . . 24
HW 30 — Equivalent threshold tests . . . 25
Lemma 3.3. Composition of functions . . . 25

Lemma 3.4 . . . 26
Lemma 3.5. Measurability of infimum, supremum, liminf and limsup . . . 26
HW 31 — Superlevel sets of an infimum . . . 27
Corollary 3.6 — Measurability of the Convergence Set . . . 27
Def: Random variable . . . 28
Example 1(b) — Frequency event . . . 28
Def Law of a random variable . . . 28
HW 32 — Pre-image of a countable union . . . 29
Why is it sufficient to specify the law on a π-system . . . 30
Definition CDF . . . 30
Properties of cdf . . . 30
HW 33 — CDF tails with mass at infinity . . . 31
HW 34 — Left limits and atoms . . . 32
Definition — Probability Density Function (pdf) . . . 33
Properties of pdf . . . 33
Theorem – Existence of a r.v. with a prescribed CDF . . . 33
Construction — Inverse Transform Method . . . 34
Lemma 3.7 — The σ-Algebra Generated by a Random Variable . . . 34
HW 35 — Prove That $\pi(Y)$ Is a π-System on Ω . . . 36
Properties — Measurable Functions of Random Variables . . . 36
Financial Interpretation . . . 37

4. Independence . . . 38
Definition 4.1 — Independent Sigma-Algebras . . . 38
Definition 4.2 — Independent Random Variables . . . 38
Definition 4.3 — Independent Events . . . 38
HW 36 — σ-Algebra Generated by an Indicator Function . . . 38
Connection with Independence of Events . . . 39
Lemma 4.1 — Independence of σ-Algebras Generated by π-Systems . . . 39
HW 37 — Prove That Two Set Functions Define Measures . . . 40
Corollary — Characterisation of Independence via Threshold Events . . . 41
Generalisation — Independence of Finitely Many Random Variables . . . 42
HW 38 — Independence of Three σ-Algebras Generated by π-Systems . . . 42
HW 38 — Part II: Why Is the Assumption $\Omega \in \mathcal{I}_k$ Necessary? . . . 44
Counterexample 1 — Pairwise Independence Does Not Imply Joint Independence . . . 44
Counterexample 2 — Necessity of $\Omega \in \mathcal{I}_k$ . . . 45
Lemma 4.2 — Second Borel–Cantelli's Lemma . . . 47
Example 2 (Williams, Example 4.4) — Application of the Borel–Cantelli Lemmas . . . 49
HW 39 — Formula (4.3) . . . 52
Fundamental Question for Modelling . . . 53
Stochastic Processes . . . 53
Existence of Stochastic Processes . . . 53
Example 3 — Markov Chains . . . 54
A Monkey Types Shakespeare . . . 56
HW 40 — Set-Theoretic Verification of the Events in the Monkey Example . . . 58
HW 41 — Independence of Products of Random Variables . . . 61

5. Integration . . . 63
Motivation — Expectation as an Integral . . . 63
Recall — Riemann Integral . . . 63

Notation — Lebesgue Integral with Respect to a Measure . . . 63
Example — Counting Measure and Infinite Series . . . 64
Integrals of Positive Simple Functions . . . 64
Definition — Integral of an Indicator Function . . . 64
Definition — Positive Simple Function . . . 64
HW 42 — Measurability of a Simple Function . . . 64
Different Representations and Canonical Form of Simple Functions . . . 65
Definition — Integral of a Positive Simple Function . . . 65
Observation — The Convention $0 \cdot \infty = 0$ . . . 66
HW 43 — Well-Definedness of the Integral of Simple Functions . . . 66
HW 44 — Properties of Integrals of Simple Functions . . . 68
Recall — μ-almost everywhere. . . . 68
Example — Equality Lebesgue-Almost Everywhere . . . 69
Definition — Integral of a Positive Measurable Function . . . 70
Observation — Agreement with the Simple Integral . . . 70
HW 45 — Monotonicity of the Positive Lebesgue Integral . . . 70
Lemma 5.1 — Zero Integral Implies Zero Almost Everywhere . . . 70
HW 46 — Level-Set Representation of $\{f > 0\}$ . . . 71
Example — Application of Lemma 5.1 . . . 72
Observation — Reduction from Almost-Everywhere to Pointwise Convergence 72
Theorem 5.2 — Monotone Convergence Theorem (MON) . . . 73
HW 47 — Increasing Sets in MON . . . 74
Approximation by Positive Simple Functions . . . 75
Lemma 5.3 — Equality Almost Everywhere . . . 76
Lemma 5.4 — Fatou's Lemma for Functions . . . 77
Lemma 5.5 — Reverse Fatou's Lemma for Functions . . . 77
HW 48 — Reverse Fatou's Lemma for Functions . . . 78
Observation. . . . 78
Lemma 5.6 — Linearity of the Positive Lebesgue Integral . . . 79
Positive and Negative Parts of a Measurable Function . . . 79
HW 49 — Positive and Negative Parts . . . 80
Definition — Lebesgue Integral of a Signed Measurable Function . . . 80
Definition 5.1 — L^1-Space . . . 80
HW 50 — Absolute Value of the Integral . . . 81
HW 51 — Closure of L^1 under Linear Combinations . . . 81
Linearity of the Integral on L^1 . . . 81
Theorem 5.7 — Dominated Convergence (DOM) . . . 82
HW 52 — Why the Domination Hypothesis Is Necessary . . . 83
The Standard Machinery . . . 84
A Measure Defined by a Nonnegative Density . . . 84
HW 53 — Countable Additivity of λ . . . 84
Theorem 5.8 — Radon–Nikodym Theorem . . . 85
Recall. . . . 85

1. Measure Spaces

Measure theory and continuous functions

A function f from a topological set X into another topological set Y (denoted as $f : X \to Y$) is said to be continuous if the pre-images of open sets in Y are open sets in X. In formulae $f^{-1}(A) = \{x \in X : f(x) \in A\}$ is open in X whenever A is open in Y.

In Measure Theory the focus is on measurable functions. A function $f : X \to Y$ is measurable if $f^{-1}(A) = \{x \in X : f(x) \in A\}$ is a measurable set in X whenever A is a measurable set in Y.

Def 1.1. Algebra

Let S be a set. A collection Σ_0 of subsets of S is called an algebra if

(i) $S \in \Sigma_0$,

(ii) $A \in \Sigma_0 \Rightarrow A^c \in \Sigma_0$,

(iii) $A, B \in \Sigma_0 \Rightarrow A \cup B \in \Sigma_0$.

HW 1 — Closure under finite intersections

Show that $A, B \in \Sigma_0 \Rightarrow A \cap B \in \Sigma_0$.

Proof.

Let $A, B \in \Sigma_0$. It is enough to show that $(A \cap B)^c \in \Sigma_0$ since Σ_0 is an algebra.

$$(A \cap B)^c = \{x : x \notin (A \cap B)\} = \{x : x \notin A \text{ or } x \notin B\} = A^c \cup B^c.$$

By the properties **(ii)** and **(iii)** of an algebra we have that $(A \cap B)^c \in \Sigma_0$ and so $(A \cap B) \in \Sigma_0$ by **(ii)**.

HW 2 — Finite unions and intersections

Show that $(A_n)_{n=1}^N \subseteq \Sigma_0$ implies $\bigcup_{n=1}^N A_n \in \Sigma_0$ and $\bigcap_{n=1}^N A_n \in \Sigma_0$.

Proof.

By definition of algebra we know that $A, B \in \Sigma_0 \Rightarrow A \cup B \in \Sigma_0$. We prove the result by induction on N. **Base case:** $N = 1$ is trivial. **Inductive hypothesis:** Assume that for some $N \geq 1$, $\bigcup_{n=1}^N A_n \in \Sigma_0$. **Inductive step:** We want to prove that $\bigcup_{n=1}^{N+1} A_n \in \Sigma_0$. Notice that

$$\bigcup_{n=1}^{N+1} A_n = \left(\bigcup_{n=1}^{N} A_n \right) \cup A_{N+1} \in \Sigma_0.$$

Therefore by inductive hypothesis and by assumption $A_{N+1} \in \Sigma_0$ the closure of an algebra under finite unions holds. For finite intersection by De Morgan's law

$$\bigcap_{n=1}^{N} A_n = \left(\bigcup_{n=1}^{N} A_n^c \right)^c.$$

Hence by the closure under complements $A_n^c \in \Sigma_0$ for $n = 1, \ldots, N$ and by the closure under finite unions also $\bigcup_{n=1}^N A_n^c \in \Sigma_0$ and therefore

$$\left(\bigcup_{n=1}^{N} A_n^c \right)^c = \bigcap_{n=1}^{N} A_n \in \Sigma_0$$

by closure under complements.

Def 1.2. σ-algebra

Let S be a set. A collection Σ of subsets of S is called a σ-algebra if

(i) $S \in \Sigma$,

(ii) $A \in \Sigma \Rightarrow A^c \in \Sigma$,

(iii) $(A_n)_{n=1}^\infty \subseteq \Sigma \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \Sigma$.

HW 3 — Countable intersections

Show that $(A_n)_{n=1}^{\infty} \subseteq \Sigma \Rightarrow \bigcap_{n=1}^{\infty} A_n \in \Sigma$.

Proof.

Let $(A_n)_{n\in\mathbb{N}} \subseteq \Sigma$. Since Σ is a σ-algebra it is closed under complements. Therefore $A_n^c \in \Sigma \; \forall n \in \mathbb{N}$. By closure under countable unions $\bigcup_{n=1}^{\infty} A_n^c \in \Sigma$ and therefore its complement $(\bigcup_{n=1}^{\infty} A_n^c)^c \in \Sigma$, which is $\bigcap_{n=1}^{\infty} A_n$ by De Morgan's law.

Measurable spaces, generated sigma-algebras and Borel sets

In the probabilistic formulation of a random experiment we will use σ-algebras to describe all possible events. However, working with σ-algebras is difficult because it is often difficult to describe ALL sets in a σ-algebra. It is often easier to handle algebras and we will soon learn about π-systems.

Notation: From now on S will always denote a set, Σ_0 an algebra of subsets of S and Σ a σ-algebra of subsets of S.

Measurable sets: any $A \in \Sigma$ is said to be a Σ-measurable set.

Measurable space: a pair (S, Σ) is called a measurable space.

Letting $\mathcal{C}$ be a class of subsets of S we denote by $\sigma(\mathcal{C})$ the smallest σ-algebra that contains all sets in $\mathcal{C}$. We call it the σ-algebra generated by $\mathcal{C}$. Notice that, in particular, if $\mathcal{C} \subseteq \Sigma$ then $\sigma(\mathcal{C}) \subseteq \Sigma$.

Given a topological space X the Borel σ-algebra $\mathcal{B}(X)$ is the smallest σ-algebra that contains all the open sets of X. That is

$$\mathcal{B}(X) = \sigma(\text{open sets in } X).$$

In particular we will often work with $X = \mathbb{R}$ and $\mathcal{B}(\mathbb{R})$ generated by open sets on the real line. In that context a very useful class of subsets is the following (example of π-system)

$$\pi(\mathbb{R}) = \{(-\infty, x], \; x \in \mathbb{R}\}.$$

Indeed it is easy to describe all the set contained in $\pi(\mathbb{R})$ and the next proposition allow us to connect with $\mathcal{B}(\mathbb{R})$.

Prop 1.1

It holds $\sigma(\pi(\mathbb{R})) = \mathcal{B}(\mathbb{R})$.

Proof.

To prove the proposition we will show that $\sigma(\pi(\mathbb{R})) \subseteq \mathcal{B}(\mathbb{R})$ and $\sigma(\pi(\mathbb{R})) \supseteq \mathcal{B}(\mathbb{R})$. A generic set in $\pi(\mathbb{R})$ is of the form $(-\infty, x]$ for some $x \in \mathbb{R}$ and it can be written as

$$(-\infty, x] = \bigcap_{n=1}^{\infty} (-\infty, x + \tfrac{1}{n}) \in \mathcal{B}(\mathbb{R}),$$

A point $y \leq x$ belongs to every open ray on the right. Conversely, if $y > x$, choose n with $1/n < y - x$; then $y \notin (-\infty, x + 1/n)$. This proves the equality. Each ray is open, and the Borel sigma-algebra is closed under countable intersections, proving the final membership. *Every element of $\pi(\mathbb{R})$ is also an element of $\mathcal{B}(\mathbb{R})$.* Therefore $\pi(\mathbb{R}) \subseteq \mathcal{B}(\mathbb{R})$ which implies $\sigma(\pi(\mathbb{R})) \subseteq \mathcal{B}(\mathbb{R})$. *For the reverse inclusion, we show that every open set belongs to $\sigma(\pi(\mathbb{R}))$.* We notice that any open set in $\mathbb{R}$ can be written as countable union of disjoint open intervals. Therefore it suffices to show that any open interval (a, b), allowing infinite endpoints, belongs to $\sigma(\pi(\mathbb{R}))$. If both endpoints are infinite, the interval is $\mathbb{R}$, which belongs to every σ-algebra on $\mathbb{R}$. For the remaining cases the following identities apply, omitting a redundant ray when an endpoint is infinite. Notice that

$$(a, b) = (-\infty, b) \cap (a, \infty),$$
$$(a, +\infty) = (-\infty, a]^c \in \sigma(\pi(\mathbb{R})),$$

because it is the complement of a set in $\sigma(\pi(\mathbb{R}))$.

$$(-\infty, b) = \bigcup_{n=1}^{\infty} (-\infty, b - \tfrac{1}{n}] \in \sigma(\pi(\mathbb{R}))$$

For this equality, every point in a ray on the right is strictly below b. If $y < b$, choose n such that $1/n \leq b - y$; then $y \leq b - 1/n$ and belongs to the union. Each closed ray belongs to the generating class, so countable-union closure proves the membership. Since $(a, +\infty)$ and $(-\infty, b) \in \sigma(\pi(\mathbb{R}))$ then $(a, b) \in \sigma(\pi(\mathbb{R}))$. For an arbitrary open set $A \subseteq \mathbb{R}$ we have a sequence $(a_n, b_n)_{n\in\mathbb{N}}$ such that

$$A = \bigcup_{n=1}^{\infty} (a_n, b_n) \in \sigma(\pi(\mathbb{R})),$$

where the final membership is by countable union of elements of a σ-alg. Therefore any open set in $\mathbb{R}$ belongs to $\sigma(\pi(\mathbb{R}))$ and thus we conclude that $\mathcal{B}(\mathbb{R}) \subseteq \sigma(\pi(\mathbb{R}))$.

Set functions

A set function is a mapping from an algebra to the positive extended real line $\mu_0 : \Sigma_0 \to [0, +\infty]$.

Additive and countably additive set functions

A set function $\mu_0 : \Sigma_0 \to [0, +\infty]$ is additive if $\mu_0(\emptyset) = 0$ and

$$\mu_0(A \cup B) = \mu_0(A) + \mu_0(B)$$

whenever $A, B \in \Sigma_0$ are disjoint. A set function is countably additive if $\mu_0(\emptyset) = 0$ and for any $(A_n)_{n=1}^{\infty} \subseteq \Sigma_0$ such that $A_i \cap A_j = \emptyset \ \forall i \neq j$ and $\bigcup_{n=1}^{\infty} A_n \in \Sigma_0$ it holds

$$\mu_0\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu_0(A_n).$$

HW 4 — Finite additivity by induction

Let μ_0 be an additive set function. Show that for any $(A_n)_{n=1}^{N} \subseteq \Sigma_0$ such that $A_i \cap A_j = \emptyset$ for all $i \neq j$, it holds

$$\mu_0\left(\bigcup_{n=1}^{N} A_n\right) = \sum_{n=1}^{N} \mu_0(A_n).$$

Proof.

We prove the statement by induction on N. **Base case:** $N = 1$ is trivial $\mu_0(A_1) = \mu_0(A_1)$. **Inductive hypothesis:** Assume the statement holds for some $N \geq 1$. **Inductive step:** We want to prove that the statement holds for $N + 1$.

Notice that

$$\bigcup_{n=1}^{N+1} A_n = \bigcup_{n=1}^{N} A_n \cup A_{N+1} \in \Sigma_0$$

since Σ_0 is an algebra (finite unions).

$$\left(\bigcup_{n=1}^{N} A_n\right) \cap A_{N+1} = \emptyset$$

since sets are pairwise disjoint. By additivity of μ_0

$$\begin{aligned}\mu_0\left(\bigcup_{n=1}^{N+1} A_n\right) = \mu_0\left(\bigcup_{n=1}^{N} A_n \cup A_{N+1}\right) &= \mu_0\left(\bigcup_{n=1}^{N} A_n\right) + \mu_0(A_{N+1}) \\ &= \sum_{n=1}^{N} \mu_0(A_n) + \mu_0(A_{N+1}) = \sum_{n=1}^{N+1} \mu_0(A_n).\end{aligned}$$

The first equality uses the decomposition of the union, the second finite additivity for the two disjoint measurable sets, and the third the inductive hypothesis. The final equality is the definition of the finite sum. This completes the induction.

HW 5 — Countable additivity implies additivity

Show that a countably additive set function is also additive.

Proof.

Let $\mu_0 : \Sigma_0 \to [0, +\infty]$ a countably additive set function. By definition of countable additivity, $\mu_0(\emptyset) = 0$. Let $A, B \in \Sigma_0$ be two disjoint sets, we define the sequence $(C_n)_{n\in\mathbb{N}}$ as follows: $C_1 = A$, $C_2 = B$, $C_n = \emptyset\ \forall n \geq 3$. Notice that $(C_n)_{n\in\mathbb{N}}$ are pairwise disjoint and $\bigcup_{n=1}^{\infty} C_n = A \cup B \in \Sigma_0$. Therefore by countable additivity

$$\mu_0(A \cup B) = \mu_0\left(\bigcup_{n=1}^{\infty} C_n\right) = \sum_{n=1}^{\infty} \mu_0(C_n) = \mu_0(A) + \mu_0(B) + \sum_{n=3}^{\infty} \mu_0(\emptyset) = \mu_0(A) + \mu_0(B).$$

The first equality uses the union of the C_n, the second countable additivity, and the remaining equalities use their definitions and $\mu_0(\emptyset) = 0$.

Measure

Given a measurable space (S, Σ) a function $\mu : \Sigma \to [0, +\infty]$ is called measure if it is countably additive.

Measure space: The triple (S, Σ, μ) define a measure space.

A measure is finite if $\mu(S) < \infty$.

A measure is σ-finite if there exists $(A_n)_{n\in\mathbb{N}} \subseteq \Sigma$ such that $\bigcup_{n=1}^{\infty} A_n = S$ and $\mu(A_n) < \infty$ for each $n \in \mathbb{N}$.

Null set: a set $A \in \Sigma$ is a null set if $\mu(A) = 0$. Actually, to be formal the collection of null sets is $\mathcal{N} = \{A \subseteq S : A \subseteq B \in \Sigma,\ \mu(B) = 0\}$. Notice that in the definition of $\mathcal{N}$ the sets A are not necessarily measurable but they are subsets of measurable sets.

Complete measure space: a measure space is complete if $\mathcal{N} \subseteq \Sigma$.

A statement $H(s)$ about points $s \in S$ is said to hold almost everywhere (almost surely in case of probability measures) if its failure set is contained in a measurable set of measure zero. In particular, when the failure set is measurable, this means $\mu(F) = 0$, where $F = \{s \in S : H(s) \text{ is false}\}$.

Definition 1.3. π-system

A collection $\mathcal{I}$ of subsets of S such that $I_1, I_2 \in \mathcal{I} \Rightarrow I_1 \cap I_2 \in \mathcal{I}$ is called π-system.

HW 6 — Lower rays form a pi-system

Prove that $\pi(\mathbb{R})$ is a π-system. Recall the definition of $\pi(\mathbb{R}) = \{(-\infty, x],\ x \in \mathbb{R}\}$.

Proof.

By definition a collection $\mathcal{I}$ is a π-system if $A, B \in \mathcal{I} \Rightarrow A \cap B \in \mathcal{I}$. Let $A, B \in \pi(\mathbb{R})$ then there exists $x, y \in \mathbb{R}$ such that $A = (-\infty, x]$, $B = (-\infty, y]$.

$$A \cap B = (-\infty, \min\{x, y\}] \in \pi(\mathbb{R}).$$

Indeed, a real number belongs to both lower rays exactly when it is at most both endpoints, equivalently at most their minimum. The resulting ray is again an element of $\pi(\mathbb{R})$.

Lemma 1.2

Let $\Sigma = \sigma(\mathcal{I})$ where $\mathcal{I}$ is a π-system on S. Suppose that μ_1 and μ_2 are measures on (S, Σ) such that $\mu_1(S) = \mu_2(S) < \infty$ and $\mu_1(I) = \mu_2(I)$ for every $I \in \mathcal{I}$. Then $\mu_1(A) = \mu_2(A)\ \forall A \in \Sigma$.

Theorem 1.3 (Carathéodory's Theorem)

Let S be a set, Σ_0 an algebra on S and $\Sigma = \sigma(\Sigma_0)$. If a set function $\mu_0 : \Sigma_0 \to [0, +\infty]$ is countably additive, then there exists a measure $\mu : \Sigma \to [0, +\infty]$ such that $\mu(A) = \mu_0(A)$ for any $A \in \Sigma_0$. Moreover if $\mu_0(S) < \infty$ then μ is uniquely defined. We say that μ in the theorem above is an extension of μ_0 to the σ-algebra Σ.

Lebesgue measure

Let $S = (0, 1]$, $\Sigma = \mathcal{B}((0, 1])$ and

$$\Sigma_0 = \left\{ F : F = \bigcup_{i=1}^{n} (a_i, b_i], \text{ for pairwise disjoint sets } (a_i, b_i] \subseteq (0, 1] \text{ and some } n \in \mathbb{N} \right\}.$$

Then $\sigma(\Sigma_0) = \Sigma$. Define μ_0 for $F \in \Sigma_0$ as $\mu_0(F) = \sum_{i=1}^{n} (b_i - a_i)$. μ_0 is countably additive on Σ_0 and $\mu_0(S) = 1 \Rightarrow \exists$ a measure μ on Σ that extends μ_0. We call it Lebesgue measure, denoted Leb. An analogous construction defines Lebesgue measure on $\mathbb{R}$ and on $\mathbb{R}^d$. The countable-additivity step is taken as given here; HW 8 asks only for finite additivity.

HW 7 — The interval algebra

Let $S = (0, 1]$ and let Σ_0 be the collection of finite unions of pairwise disjoint intervals $(a_i, b_i] \subseteq S$, with $0 \le a_i \le b_i \le 1$. Prove that Σ_0 is an algebra on S.

Proof.

First,

$$\emptyset \in \Sigma_0, \qquad S = (0, 1] \in \Sigma_0,$$

since $(x, x] = \emptyset$ for every $x \in (0, 1]$, while S is itself an interval of the required form.

Let $A \in \Sigma_0$ be nonempty. Without loss of generality, we can assume that

$$A = \bigcup_{i=1}^{n} (a_i, b_i],$$

with

$$0 \le a_1 < b_1 \le a_2 < b_2 \le \cdots \le a_n < b_n \le 1,$$

since the nonempty intervals are pairwise disjoint and can be ordered by their left endpoints. We obtain

$$A^c = (0, a_1] \cup \left(\bigcup_{i=1}^{n-1} (b_i, a_{i+1}] \right) \cup (b_n, 1].$$

These intervals are precisely the gaps between the intervals forming A, including possible gaps at the endpoints of S. Since they are finitely many and pairwise disjoint, $A^c \in \Sigma_0$. If $A = \emptyset$, then $A^c = S \in \Sigma_0$.

Finally, let $A, B \in \Sigma_0$, with

$$A = \bigcup_{i=1}^{n} (a_i, b_i], \qquad B = \bigcup_{j=1}^{m} (c_j, d_j].$$

The intervals from the two collections may overlap. To obtain a finite partition, collect all their endpoints a_i, b_i, c_j, d_j, together with 0 and 1, remove repetitions and arrange them in increasing order:

$$0 = t_0 < t_1 < \cdots < t_r = 1.$$

Define

$$D_\ell = (t_{\ell-1}, t_\ell], \qquad \ell = 1, \ldots, r.$$

These intervals are pairwise disjoint and satisfy

$$S = \bigcup_{\ell=1}^{r} D_\ell,$$

so they form a finite partition of S.

The same construction applies to any finite collection of sets in Σ_0. Thus, in subsequent arguments, we may work without loss of generality with a common finite partition of S into intervals of this form, with each original set represented as a union of its pieces.

Moreover, every original nonempty interval $(a_i, b_i]$ is a union of consecutive D_ℓ. Indeed, since all its endpoints were selected, there exist $p < q$ such that $a_i = t_p$ and $b_i = t_q$. Hence,

$$(a_i, b_i] = \bigcup_{\ell=p+1}^{q} D_\ell.$$

The equality holds because consecutive half-open intervals include their right endpoints and exclude their left endpoints, so no points are lost or counted twice.

The same argument applies to every interval forming B. Consequently, $A \cup B$ is a finite union of some of the D_ℓ, which are pairwise disjoint intervals of the required form. Therefore,

$$A \cup B \in \Sigma_0.$$

Thus, Σ_0 contains S and is closed under complements and finite unions, so it is an algebra on $(0, 1]$.

HW 8 — Finite additivity of interval length

On the interval algebra Σ_0 from HW 7, define $\mu_0(A) = \sum_{i=1}^{n}(b_i - a_i)$ for a disjoint representation $A = \bigcup_{i=1}^{n}(a_i, b_i]$. Prove that μ_0 is well defined and finitely additive.

Proof.

First, for every $A \in \Sigma_0$ represented by pairwise disjoint intervals,

$$A = \bigcup_{i=1}^{n}(a_i, b_i], \qquad \mu_0(A) = \sum_{i=1}^{n}(b_i - a_i) \geq 0.$$

The inequality follows because $b_i \geq a_i$ for every i. Moreover,

$$\mu_0(\emptyset) = \mu_0((x, x]) = x - x = 0.$$

We also verify that μ_0 does not depend on the chosen disjoint interval representation. Given two representations of A, their common endpoint partition, constructed as in HW 7, yields

$$\sum_{i=1}^{n}(b_i - a_i) = \sum_{\ell:\, D_\ell \subseteq A} (t_\ell - t_{\ell-1}) = \sum_{j=1}^{m}(d_j - c_j).$$

Both equalities follow by splitting the intervals of each representation into consecutive partition intervals and adding their lengths.

Finally, let $A, B \in \Sigma_0$ be disjoint, with

$$A = \bigcup_{i=1}^{n}(a_i, b_i], \qquad B = \bigcup_{j=1}^{m}(c_j, d_j].$$

Since $A \cap B = \emptyset$, every interval of A is disjoint from every interval of B. Define

$$e_k = \begin{cases} a_k, & k = 1, \ldots, n, \\ c_{k-n}, & k = n+1, \ldots, n+m, \end{cases} \qquad f_k = \begin{cases} b_k, & k = 1, \ldots, n, \\ d_{k-n}, & k = n+1, \ldots, n+m. \end{cases}$$

Then

$$A \cup B = \bigcup_{k=1}^{n+m} (e_k, f_k],$$

where all intervals are pairwise disjoint. Hence,

$$\mu_0(A \cup B) = \sum_{k=1}^{n+m} (f_k - e_k) = \sum_{i=1}^{n} (b_i - a_i) + \sum_{j=1}^{m} (d_j - c_j) = \mu_0(A) + \mu_0(B).$$

The first equality uses the disjoint representation just constructed; the second separates the first n and last m terms; the last follows from the definition of μ_0. Repeating the argument proves additivity for any finite family of pairwise disjoint sets.

HW 9 — Borel singletons

Prove that $\{x\} \in \mathcal{B}(\mathbb{R})$ $\forall x \in \mathbb{R}$.

Proof.

The Borel sigma algebra contains all open intervals and is closed under countable intersections. Let $x \in \mathbb{R}$, we have that $\{x\} = \bigcap_{n=1}^{\infty} (x - \frac{1}{n}, x + \frac{1}{n})$. We know that for all $n \in \mathbb{N}$, $(x - \frac{1}{n}, x + \frac{1}{n}) \in \mathcal{B}(\mathbb{R})$ and countable intersection of measurable sets is a measurable set. For the set equality, x belongs to every interval. Conversely, if $y \neq x$, choose n such that $1/n < |y - x|$; then y is outside the corresponding interval. Thus the intersection is exactly $\{x\}$.

HW 10 — Lebesgue measure of a singleton

Prove that $\text{Leb}(\{x\}) = 0$ $\forall x \in \mathbb{R}$.

Proof.

Let $x \in \mathbb{R}$. From HW 9 we know that $\{x\} \in \mathcal{B}(\mathbb{R})$. Notice that for every $n \in \mathbb{N}$, $\{x\} \subseteq (x - \frac{1}{n}, x]$ and $(x - \frac{1}{n}, x] = (x - \frac{1}{n}, x) \cup \{x\}$ where the two sets are disjoint and belong to $\mathcal{B}(\mathbb{R})$. By additivity of measure we have

$$\frac{1}{n} = \text{Leb}((x - \tfrac{1}{n}, x]) = \text{Leb}((x - \tfrac{1}{n}, x)) + \text{Leb}(\{x\}) \geq \text{Leb}(\{x\}),$$

where the last inequality follows from the non-negativity of measures. Therefore $0 \leq \text{Leb}(\{x\}) \leq \frac{1}{n}$. Since this inequality holds for every $n \in \mathbb{N}$ by letting $n \to +\infty$ and applying the squeeze thm we obtain $\text{Leb}(\{x\}) = 0$. Moreover if a singleton had measure $a > 0$, choosing $n > \frac{1}{a}$ would contradict $a \leq \frac{1}{n}$ hence its measure is zero.

Lemma 1.4

Let (S, Σ, μ) be a measure space. Then

(a) $\mu(A \cup B) \leq \mu(A) + \mu(B) \qquad \forall A, B \in \Sigma,$

(b) $\mu\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} \mu(A_i) \qquad \forall (A_i)_{i=1}^{n} \subseteq \Sigma,$

(c) If $\mu(S) < \infty$ then $\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B) \quad \forall A, B \in \Sigma.$

HW 11 — Subadditivity and the union formula

Prove all three assertions of Lemma 1.4: binary subadditivity, finite subadditivity, and the union formula under $\mu(S) < \infty$.

Proof.

(a) Let $A, B \in \Sigma$, we have that $A \setminus B = A \cap B^c \in \Sigma$, $A \cup B = (A \setminus B) \cup B$ and $(A \setminus B) \cap B = \emptyset$. It follows that

$$\begin{aligned} \mu(A \cup B) &= \mu((A \setminus B) \cup B) = \mu(A \setminus B) + \mu(B) \leq \mu(A \setminus B) + \mu(A \cap B) + \mu(B) \\ &= \mu((A \setminus B) \cup (A \cap B)) + \mu(B) = \mu(A) + \mu(B). \end{aligned}$$

The first two equalities use the disjoint decomposition of $A \cup B$, the inequality uses $\mu(A \cap B) \geq 0$, and the last two equalities use the disjoint decomposition of A.

(b) Finite sub-additivity. We prove the statement by induction on n. **Base case:** $n = 1$. Statement is trivial $\mu(A_1) = \mu(A_1)$. **Inductive hypothesis:** $n \geq 1$, $\mu(\bigcup_{i=1}^{n} A_i) \leq \sum_{i=1}^{n} \mu(A_i)$. **Inductive step:** $n+1$. Notice that $\bigcup_{i=1}^{n+1} A_i = (\bigcup_{i=1}^{n} A_i) \cup A_{n+1}$. By prop (a) we have

$$\mu\left(\bigcup_{i=1}^{n+1} A_i\right) \leq \mu\left(\bigcup_{i=1}^{n} A_i\right) + \mu(A_{n+1}) \leq \sum_{i=1}^{n} \mu(A_i) + \mu(A_{n+1}) = \sum_{i=1}^{n+1} \mu(A_i),$$

where the first inequality uses part (a), the second the inductive hypothesis, and the equality the definition of the finite sum.

(c) $\mu(A \cup B) = \mu(A \setminus B) + \mu(B)$, since $A \setminus B$ and B are disjoint. $\mu(A) = \mu(A \setminus B) + \mu(A \cap B)$ since $A \setminus B$ and $A \cap B$ are disjoint. Adding $\mu(B)$ we obtain

$$\mu(A) + \mu(B) = \mu(A \setminus B) + \mu(A \cap B) + \mu(B) = \mu(A \cup B) + \mu(A \cap B),$$

where the first equality uses the disjoint decomposition of A, and the second uses the disjoint decomposition of $A \cup B$ from part (a). Since $\mu(S) < \infty$ for every $E \in \Sigma$, $\mu(E) < \infty$, in fact $\mu(S) = \mu(E) + \mu(E^c)$. Therefore subtracting $\mu(A \cap B)$ from from both sidess we obtain

$$\mu(A) + \mu(B) - \mu(A \cap B) = \mu(A \cup B).$$

HW 12 — Monotonicity of measures

Prove that if $A, B \in \Sigma$, $A \subseteq B$ then $\mu(A) \leq \mu(B)$.

Proof.

Notice that $B = (B \cap A) \cup (B \cap A^c)$ and that $(B \cap A) \cap (B \cap A^c) = \emptyset$. It follows

$$\mu(B) = \mu((B \cap A) \cup (B \cap A^c)) = \mu(B \cap A) + \mu(B \cap A^c) \geq \mu(B \cap A) = \mu(A),$$

because $A \subseteq B$. The second equality uses additivity on the two disjoint measurable pieces, the inequality their nonnegative measures, and the last equality $A \cap B = A$.

HW 13 — Measure of a difference

The operation of "subtracting" two sets is denoted by $A \setminus B$ and it means $A \cap B^c$. Show that for a finite measure μ, if $B \subseteq A$ then $\mu(A \setminus B) = \mu(A) - \mu(B)$.

Proof.

We obtain

$$\mu(A) = \mu((A \cap B) \cup (A \cap B^c)) = \mu(A \cap B) + \mu(A \cap B^c) = \mu(A \cap B) + \mu(A \setminus B).$$

Knowing that $A \cap B = B$ it follows $\mu(A) = \mu(B) + \mu(A \setminus B)$ and so $\mu(A \setminus B) = \mu(A) - \mu(B)$ if $\mu(B) < \infty$. The decomposition is disjoint and measurable. Finiteness of the measure ensures that the subtraction is between finite quantities; no $\infty - \infty$ expression is used.

Lemma 1.5. Continuity of measure from below

Let (S, Σ, μ) be a measure space. If $(A_n)_{n \in \mathbb{N}} \subseteq \Sigma$ such that $A_n \subseteq A_{n+1}$ and $A_n \uparrow A$ as $n \to +\infty$ (in the sense that $A = \bigcup_{n=1}^{\infty} A_n$). Then $\mu(A_n) \uparrow \mu(A)$ as $n \to +\infty$.

Proof.

We relabel sets to make them disjoint with: $B_1 = A_1$, $B_2 = A_2 \setminus B_1$, …, $B_n = A_n \setminus \bigcup_{i=1}^{n-1} B_i$. Each B_n is measurable, being a difference of measurable sets. As we will see in HW 14, the sets $(B_n)_{n \in \mathbb{N}}$ are pairwise disjoint. As we will see in HW 15, $\bigcup_{i=1}^{n} B_i = A_n$ and $\bigcup_{i=1}^{\infty} B_i = A$. We obtain

$$\mu(A_n) = \mu\left(\bigcup_{i=1}^{n} B_i\right) = \sum_{i=1}^{n} \mu(B_i) \leq \sum_{i=1}^{n+1} \mu(B_i) = \mu\left(\bigcup_{i=1}^{n+1} B_i\right) = \mu(A_{n+1}).$$

Here the union identities are supplied by HW 15, the finite-sum equalities use additivity on the disjoint sets from HW 14, and the inequality uses $\mu(B_{n+1}) \geq 0$. Therefore the sequence $\mu(A_n)$, $n \in \mathbb{N}$ is non decreasing hence it admits a limit which we can now compute

$$\lim_{n\to+\infty} \mu(A_n) = \lim_{n\to+\infty} \sum_{i=1}^{n} \mu(B_i) = \sum_{i=1}^{\infty} \mu(B_i) = \mu\left(\bigcup_{i=1}^{\infty} B_i\right) = \mu(A),$$

The first equality substitutes the finite-sum identity, the second is the definition of a nonnegative series, the third countable additivity for the disjoint measurable B_i, and the last $\bigcup_i B_i = A$ from HW 15.

HW 14 — Disjoint pieces

Prove that $B_i \cap B_j = \emptyset \ \forall i \neq j$ where $B_1 = A_1$, $B_n = A_n \setminus \bigcup_{k=1}^{n-1} B_k$, $n \geq 2$.

Proof.

We prove by induction on n that $B_1, \ldots, B_n$ are pairwise disjoint. **Base case:** $n = 1$ is trivial. **Inductive hypothesis:** $n \geq 1$. $B_1, \ldots, B_n$ are pairwise disjoint i.e. $B_i \cap B_j = \emptyset \ \forall 1 \leq i < j \leq n$. **Inductive step:** By definition $B_{n+1} = A_{n+1} \setminus \bigcup_{k=1}^{n} B_k$. Therefore $B_{n+1} \cap (\bigcup_{k=1}^{n} B_k) = \emptyset$. Since $B_i \subseteq \bigcup_{k=1}^{n} B_k \ \forall i = 1, \ldots, n$ we have that $B_{n+1} \cap B_i = \emptyset \ \forall i = 1, \ldots, n$. Moreover by inductive hyp $B_1, \ldots, B_n$ are already pairwise disjoint hence $B_1, \ldots, B_{n+1}$ are pairwise disjoint.

HW 15 — Unions of the disjoint pieces

Prove that $\bigcup_{i=1}^{n} B_i = A_n$ **(i)** and $\bigcup_{i=1}^{\infty} B_i = A$.

Proof.

Recall that $(A_n)_{n\in\mathbb{N}}$ is non decreasing sequence of sets such that $A_n \subseteq A_{n+1}$; $A = \bigcup_{i=1}^{\infty} A_n$. We define $B_1 = A_1$, $B_n = A_n \setminus \bigcup_{j=1}^{n-1} B_j$, $n \geq 2$. We prove **(i)** by induction: $N = 1$ is trivial. **Inductive hypothesis:** assume that statement hold for $n \geq 1$, i.e. $\bigcup_{i=1}^{n} B_i = A_n$. **Inductive step:** $B_{n+1} = A_{n+1} \setminus \bigcup_{i=1}^{n} B_i$, by inductive hyp we obtain $B_{n+1} = A_{n+1} \setminus A_n$. Therefore

$$\bigcup_{i=1}^{n+1} B_i = \left(\bigcup_{i=1}^{n} B_i\right) \cup B_{n+1} = A_n \cup (A_{n+1} \setminus A_n) = A_{n+1},$$

where the last equality follows from $A_n \subseteq A_{n+1}$. Therefore by induction $\bigcup_{i=1}^{n} B_i = A_n \ \forall n \in \mathbb{N}$ **(i)**. For **(ii)** we know that $A_n \uparrow A$ and $A = \bigcup_{i=1}^{\infty} A_n$ by ass. Therefore

$$\bigcup_{i=1}^{\infty} B_i = \bigcup_{n=1}^{\infty} \left(\bigcup_{i=1}^{n} B_i\right) = \bigcup_{n=1}^{\infty} A_n = A,$$

where the first equality follows the fact that B_i belongs to all finite unions with $n \geq i$ and the second follow the first part of HW 15 **(i)**. The last follow the assumption $A_n \uparrow A$.

Lemma 1.6. Continuity of measure from above

Let (S, Σ, μ) be a measure space. If $(A_n)_{n\in\mathbb{N}} \subseteq \Sigma$ such that $A_n \supseteq A_{n+1}$ and $A_n \downarrow A$ as $n \to +\infty$ (in the sense that $A = \bigcap_{n=1}^{\infty} A_n$). Then $\mu(A_n) \downarrow \mu(A)$ as $n \to +\infty$ provided that $\mu(A_m) < \infty$ for some $m \in \mathbb{N}$.

HW 16 — Continuity from above

Statement. Prove Lemma 1.6. **Hint:** take $B_n = A_k \setminus A_{k+n}$ so that $B_n \uparrow A_k \setminus A$ and use HW 13.

Proof.

Let A_m such that $\mu(A_m) < \infty$ and define a sequence $B_k = A_m \setminus A_{m+k}$. Since $A_{n+1} \subseteq A_n \ \forall n \in \mathbb{N}$ we have $B_{k+1} \supseteq B_k \ \forall k \in \mathbb{N}$. Moreover we have that $B_k \uparrow B$ with $B = A_m \setminus A = A_m \setminus \bigcap_{n=1}^{\infty} A_n$. By Lemma 1.5 we have

$$\lim_{k\to+\infty} \mu(B_k) = \mu(B) = \mu(A_m \setminus A) = \mu(A_m) - \mu(A).$$

By definition we have

$$\lim_{k\to+\infty}\mu(B_k)=\lim_{k\to+\infty}\mu(A_m\setminus A_{m+k})=\mu(A_m)-\lim_{k\to+\infty}\mu(A_{m+k}).$$

Combining these two equations we obtain $\mu(A_m)-\mu(A)=\mu(A_m)-\lim_{k\to+\infty}\mu(A_{m+k})$.
We need the condition $\mu(A_n)<\infty$, if we don't impose that we would have this counter-example: Take $S=[0,+\infty)$, $\Sigma=\mathcal{B}([0,+\infty))$ and $\mu=\text{Leb}$. Take $A_n=[n,+\infty)$ so that $A_n\downarrow\emptyset$, notice that $\mu(A_n)=\infty\ \forall n$ and so the lemma fails. All subtractions are legitimate because $A\subseteq A_{m+k}\subseteq A_m$ and $\mu(A_m)<\infty$, so the difference formula from HW 13 applies. Cancelling the common finite quantity gives $\lim_{k\to\infty}\mu(A_{m+k})=\mu(A)$. Omitting the finitely many initial indices does not change the limit; monotonicity of A_n and HW 12 show that $\mu(A_n)$ decreases to this value.

Lemma 1.7

Countable union of null set is a null set.

Proof.

Let $(N_n)_{n\in\mathbb{N}}\subseteq\Sigma$ be such that $\mu(N_n)=0$ for all n's. Set $\bar{N}=\bigcup_{n=1}^{\infty}N_n$ and $\bar{N}^k=\bigcup_{n=1}^{k}N_n$. Then by Lemma 1.5 and sub additivity of the measure

$$\mu(\bar{N})=\lim_{k\to+\infty}\mu(\bar{N}^k)\leq\lim_{k\to+\infty}\sum_{n=1}^{k}\mu(N_n)=0.$$

Here $\bar{N}^k\uparrow\bar{N}$, so the first equality uses continuity from below. The inequality uses finite subadditivity at each k, and the last equality uses that every summand is zero. Nonnegativity then gives $\mu(\bar{N})=0$.
For null sets N_n that are not assumed measurable, choose $B_n\in\Sigma$ such that $N_n\subseteq B_n$ and $\mu(B_n)=0$, as allowed by the definition of a null set. We obtain

$$\bigcup_{n=1}^{\infty}N_n\subseteq\bigcup_{n=1}^{\infty}B_n,\qquad\mu\left(\bigcup_{n=1}^{\infty}B_n\right)=0.$$

The inclusion follows from $N_n\subseteq B_n$ for every n, and the equality follows from the measurable case just proved. Thus $\bigcup_n N_n$ is a null set in the general sense, whether or not it is measurable; no measure is assigned to a non-measurable set.

HW 17 — Rational points form a null set

Take the measure space $([0,1],\mathcal{B}([0,1]),\text{Leb})$. Prove that $[0,1]\cap\mathbb{Q}\in\mathcal{B}([0,1])$ and $\text{Leb}([0,1]\cap\mathbb{Q})=0$.

Proof.

Recall that $\mathbb{Q}$ is countable and any singleton $\in\mathcal{B}(\mathbb{R})$. We can write $[0,1]\cap\mathbb{Q}=\{q_1,q_2,\ldots\}=\bigcup_{n=1}^{\infty}\{q_n\}$ where $q_n\in[0,1]\cap\mathbb{Q}$. Since $\{q_n\}\in\mathcal{B}([0,1])$ and $\mathcal{B}([0,1])$ is closed under countable unions then $[0,1]\cap\mathbb{Q}=\bigcup_{n=1}^{\infty}\{q_n\}\in\mathcal{B}([0,1])$. By HW 10 we know that $\text{Leb}(\{q_n\})=0\ \forall n\in\mathbb{N}$, moreover since q_n's are distinct are pairwise disjoint. Therefore by countable additivity

$$\text{Leb}([0,1]\cap\mathbb{Q})=\text{Leb}\left(\bigcup_{n=1}^{\infty}\{q_n\}\right)=\sum_{n=1}^{\infty}\text{Leb}(\{q_n\})=0.$$

2. Events

Probability spaces

A mathematical model for an experiment with uncertain (random) outcome is given by probability space $(\Omega, \mathcal{F}, \mathbb{P})$ where

Ω is the sample space (the set S from the previous section). A sample point is a point in Ω, $\omega \in \Omega$ and it represents (loosely speaking) a possible scenario in the experiment.

$\mathcal{F}$ is the collection of all events (the σ-algebra Σ from the previous section). An event $E \in \mathcal{F}$ is a collection of sample points, i.e. is a collection of all possible scenarios that may lead to the observed outcome in the experiment.

$\mathbb{P}$ is the probability measure that assigns to each event a probability of occurring. The probability $\mathbb{P}$ is a measure on $\mathcal{F}$ with $\mathbb{P}(\Omega) = 1$. (Notice that $\mathbb{P} : \mathcal{F} \to [0, 1]$.)

Example 1 (Simple coin tossing models)

(a) Toss a coin twice: $\Omega = \{HH, HT, TH, TT\}$ and $\mathcal{F} = \mathcal{P}(\Omega)$, the collection of all unions of singleton outcomes, closed under intersections and complements.

(b) Toss a coin infinitely many times: $\Omega = \{H, T\}^{\mathbb{N}} \Rightarrow \omega \in \Omega$ is a sequence $\omega = (\omega_1, \omega_2, \omega_3, \ldots)$ with $\omega_i = H$ or T. In principle one could consider $\mathcal{F} = \mathcal{P}(\Omega)$ where $\mathcal{P}(\Omega)$ is the collection of ALL subsets of Ω and it is called the power set. It turns out that this is not necessary and we can take the easier σ-algebra

$$\mathcal{F} = \sigma(\{\omega \in \Omega : \omega_n = W\},\ n \in \mathbb{N},\ W \in \{H, T\}).$$

To convince ourselves that this is a good choice we will work towards showing that the set

$$F := \left\{\omega \in \Omega : \lim_{n \to +\infty} \frac{\#(k \leq n : \omega_k = H)}{n} = \frac{1}{2}\right\}$$

belongs to $\mathcal{F}$ and, under the independent fair-coin probability model, $\mathbb{P}(F) = 1$. This is telling us that indeed the Law of Large Numbers (LLN) holds as it should in any sensible mathematical model for random experiments!

Now we prove a result that is in some sense a converse of Lemma 1.7. It tells us that countable intersection of sure events give us again a sure event. We need a small result first.

HW 18 — Sure events and null complements

For any $A \in \mathcal{F}$ show that $\mathbb{P}(A) = 1 \iff \mathbb{P}(A^c) = 0$.

Proof.

Let $A \in \mathcal{F}$, $A^c \in \mathcal{F}$. Notice that $A \cup A^c = \Omega$ and $A \cap A^c = \emptyset$. We obtain $1 = \mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(A^c)$, where the first equality is normalization and the second additivity on the two disjoint complementary events. **($\Rightarrow$)** If we assume that $\mathbb{P}(A) = 1$, then $\mathbb{P}(A^c) = 0$.

($\Leftarrow$) If we assume that $\mathbb{P}(A^c) = 0$ then $\mathbb{P}(A)$ has to be 1.

Proposition 2.1

If $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{F}$ are such that $\mathbb{P}(A_n) = 1$ for all $n \in \mathbb{N}$, then $\mathbb{P}(\bigcap_{n=1}^{\infty} A_n) = 1$.

Proof.

Since $A_n \in \mathcal{F} \Rightarrow A_n^c \in \mathcal{F}$ then $(\bigcap_{n=1}^{\infty} A_n)^c = \bigcup_{n=1}^{\infty} A_n^c \in \mathcal{F}$ and

$$\mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right) = 1 \iff \mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n^c\right) = 0.$$

Since $\mathbb{P}(A_n) = 1 \iff \mathbb{P}(A_n^c) = 0$ then $\mathbb{P}(\bigcup_{n=1}^{\infty} A_n^c) = 0$ by Lemma 1.7 which concludes the proof.

Def lim inf and lim sup

Given a sequence $(x_n)_{n\in\mathbb{N}} \subseteq \mathbb{R}$ we define

$$\liminf_{n\to+\infty} x_n = \sup_{n\geq 1} \inf_{m\geq n} x_m$$

(i.e. the largest minorant in the limit),

$$\limsup_{n\to+\infty} x_n = \inf_{n\geq 1} \sup_{m\geq n} x_m$$

(i.e. the smallest majorant in the limit). Notice that the sequence $y_n := \sup_{m\geq n} x_m$ is decreasing as $y_n = \sup_{m\geq n} x_m \geq \sup_{m\geq n+1} x_m = y_{n+1}$. Then $\inf_{n\geq 1} y_n = \lim_{n\to+\infty} y_n$ and

$$\limsup_{n\to+\infty} x_n = \inf_{n\geq 1} \sup_{m\geq n} x_m = \lim_{n\to+\infty} \sup_{m\geq n} x_m.$$

HW 19 — Tail infima

Show that $\liminf_{n\to+\infty} x_n = \lim_{n\to+\infty} \inf_{m\geq n} x_m$.

Proof.

Let $y_n = \inf_{m\geq n} x_m$. Notice that $\{x_m : m \geq n+1\} \subseteq \{x_m : m \geq n\}$. Therefore $\inf_{m\geq n} x_m \leq \inf_{m\geq n+1} x_m$ which implies $y_n \leq y_{n+1}$. Hence $(y_n)_{n\in\mathbb{N}}$ is a non-decreasing sequence, by monotonicity the limit existss in the extended real line and $\lim_{n\to+\infty} y_n = \sup_{n\geq 1} y_n$. Substituting the definition of y_n we conclude the proof.

Definition of convergence through liminf and limsup

Given $(x_n)_{n\in\mathbb{N}} \subseteq \mathbb{R}$, the values of $\limsup x_n$ and $\liminf x_n$ are always well-defined (not necessarily finite) whereas the value of $\lim x_n$ may not exist. We say that a limit exist if

$$\liminf_{n\to+\infty} x_n = \limsup_{n\to+\infty} x_n =: \lim_{n\to+\infty} x_n.$$

The common value may belong to $\overline{\mathbb{R}} = [-\infty, +\infty]$. Convergence to a finite real number additionally requires that this common value is finite.

HW 20 — An oscillating sequence

Consider the sequence $x_n = (-1)^n$, $n \in \mathbb{N}$, calculate its liminf and limsup. Does the limit exists?

Proof.

Notice that $x_n = \begin{cases} -1 & n \text{ odd}, \\ 1 & n \text{ even}. \end{cases}$ Therefore $\inf_{m\geq n} x_m = -1$ and $\sup_{m\geq n} x_m = 1$. Therefore $\liminf_{n\to+\infty} x_n = \sup_{n\geq 1} \inf_{m\geq n} x_m = \sup_{n\geq 1}(-1) = -1$ and in the same way $\limsup_{n\to+\infty} x_n = 1$. Therefore they don't coincide and the limit does not exist.

HW 21 — Threshold properties of limits

Prove the following properties of liminf and limsup.

(i) If $z > \limsup_{n\to+\infty} x_n$, then there exists $N \in \mathbb{N}$ such that $x_n < z \ \forall n \geq N$.

(ii) If $z < \liminf_{n\to+\infty} x_n$ then there exists $N \in \mathbb{N}$ such that $x_n > z \ \forall n \geq N$.

(iii) If $z < \limsup_{n\to+\infty} x_n$ then there are infinitely many $n \in \mathbb{N}$ such that $x_n > z$.

(iv) If $z > \liminf_{n\to+\infty} x_n$ then there are infinitely many $n \in \mathbb{N}$ such that $x_n < z$.

Proof.

(i) Assume that $z > \limsup_{n\to+\infty} x_n = \inf_{n\geq 1} \sup_{m\geq n} x_m := \inf_{n\geq 1} y_n$, where $y_n := \sup_{m\geq n} x_m$. By definition of infimum, if $\inf_{n\geq 1} y_n \in \mathbb{R}$ we have $\forall \varepsilon > 0 \ \exists N = N(\varepsilon) \in \mathbb{N}$ such that $y_N < \inf_{n\geq 1} y_n + \varepsilon$. Combining these two inequalities it exist $\varepsilon > 0$ sufficiently small so that $y_N \leq \inf_{n\geq 1} y_n + \varepsilon < z$. Restoring the definition of y_N we obtain

$$x_m \leq \sup_{j\geq N} x_j = y_N < z \quad \forall m \geq N \quad \Rightarrow \quad x_m < z \quad \forall m \geq N,$$

where the first inequality follows from the definition of supremum.

(ii) Assume that $z < \liminf_{n\to+\infty} x_n = \sup_{n\geq 1} \inf_{m\geq n} x_m := \sup_{n\geq 1} y_n$, where $y_n := \inf_{m\geq n} x_m$. By definition of supremum if $\sup_{n\geq 1} y_n \in \mathbb{R}$ we have $\forall \varepsilon > 0\ \exists N = N(\varepsilon) \in \mathbb{N}$ such that $y_N > \sup_{n\geq 1} y_n - \varepsilon$. Since $z < \sup_{n\geq 1} y_n$ we can choose $\varepsilon > 0$ sufficiently small such that $z < \sup_{n\geq 1} y_n - \varepsilon$. Combining these two inequalities we obtain $z \leq \sup_{n\geq 1} y_n - \varepsilon < y_N$. Restoring the definition of y_N we obtain

$$x_m \geq \inf_{j\geq N} x_j = y_N > z \quad \forall m \geq N.$$

Hence $x_m > z\ \forall m \geq N$.

For the tail infima in **(ii)**, if $\sup_{n\geq 1} y_n = +\infty$ its definition directly guarantees an N such that $y_N > z$ and the conclusion follows in the same way.

For the tail suprema $y_n = \sup_{m\geq n} x_m$ in **(i)**, if $\inf_{n\geq 1} y_n = -\infty$ then by definition of infimum $\forall z \in \mathbb{R}$ there exists $N \in \mathbb{N}$ such that $y_N < z$ and restoring def of y_N we obtain $x_m < z\ \forall m \geq N$.

(iii) Assume that $z < \limsup_{n\to+\infty} x_n = \inf_{n\geq 1} \sup_{m\geq n} x_m = \inf_{n\geq 1} y_n$. By the definition of infimum we have $z < \inf_{n\geq 1} y_n \leq y_n\ \forall n \in \mathbb{N}$. Restoring the definition of y_n we obtain $z < \sup_{m\geq n} x_m\ \forall n \in \mathbb{N}$. Taking $n = 1$ by the definition of supremum there exists $n_1 \in \mathbb{N}$ such that $x_{n_1} > z$. Now taking $n = n_1 + 1$ we have $z < \sup_{m\geq n_1+1} x_m$. Hence by definition of supremum there exists $n_2 > n_1$ such that $x_{n_2} > z$. Repeating this argument we obtain a strictly increasing sequence of indices $n_1 < n_2 < n_3 < \cdots$ such that $x_{n_j} > z\ \forall j \in \mathbb{N}$. Therefore $x_n > z$ for infinitely many $n \in \mathbb{N}$.

(iv) Assume that $z > \liminf_{n\to+\infty} x_n = \sup_{n\geq 1} \inf_{m\geq n} x_m := \sup_{n\geq 1} y_n$. By the definition of supremum we have $z > \sup_{n\geq 1} y_n \geq y_n\ \forall n \in \mathbb{N}$. Restoring the definition of y_n we obtain $z > \inf_{m\geq n} x_m\ \forall n \in \mathbb{N}$. Taking $n = 1$ by the definition of infimum there exists $n_1 \in \mathbb{N}$ such that $x_{n_1} < z$. Now taking $n = n_1 + 1$ we have $z > \inf_{m\geq n_1+1} x_m$. Hence by the definition of infimum there exists $n_2 > n_1$ such that $x_{n_2} < z$. Repeating this argument we obtain a strictly increasing sequence of indices $n_1 < n_2 < n_3 < \cdots$ such that $x_{n_j} < z\ \forall j \in \mathbb{N}$. Therefore $x_n < z$ for infinitely many $n \in \mathbb{N}$.

HW 22 — Negation, suprema and infima

Let $(x_\alpha)_{\alpha\in\mathcal{I}}$ be a family of real numbers ($\mathcal{I} \neq \emptyset$ is not necessarily countable, e.g., $\mathcal{I} = [0, 1]$). Prove that $\sup_{\alpha\in\mathcal{I}} x_\alpha = -\inf_{\alpha\in\mathcal{I}}(-x_\alpha)$ **(i)**. Deduce that $\liminf_{n\to+\infty} x_n = -\limsup_{n\to+\infty}(-x_n)$ **(ii)**.

Proof.

(i) Let $M = \inf_{\alpha\in\mathcal{I}}(-x_\alpha)$. We distinguish two cases.

$M \in \mathbb{R}$. By definition of infimum we have $M \leq -x_\alpha\ \forall \alpha \in \mathcal{I}$ and for every $\varepsilon > 0$ there exists $\beta \in \mathcal{I}$ such that $-x_\beta \leq M + \varepsilon$. Multiplying both inequalities by (-1) we obtain $-M \geq x_\alpha\ \forall \alpha \in \mathcal{I}$ and $x_\beta \geq -M - \varepsilon$. Therefore $-M$ is an upper bound of $(x_\alpha)_{\alpha\in\mathcal{I}}$ and for every $\varepsilon > 0$ there exists $\beta \in \mathcal{I}$ such that $-M - \varepsilon \leq x_\beta$. Hence by definition of supremum $\sup_{\alpha\in\mathcal{I}} x_\alpha = -M = -\inf_{\alpha\in\mathcal{I}}(-x_\alpha)$.

$M = -\infty$. By definition of infimum, for every $K \in \mathbb{R}$ there exists $\beta \in \mathcal{I}$ such that $-x_\beta < -K$. Multiplying by (-1) we obtain $x_\beta > K$. Therefore $(x_\alpha)_{\alpha\in\mathcal{I}}$ is not bounded above and hence $\sup_{\alpha\in\mathcal{I}} x_\alpha = +\infty$. Since $M = -\infty$ we conclude that $\sup_{\alpha\in\mathcal{I}} x_\alpha = +\infty = -M = -\inf_{\alpha\in\mathcal{I}}(-x_\alpha)$.

$M = +\infty$ is impossible since $\mathcal{I} \neq \emptyset$ and every $x_\alpha \in \mathbb{R}$.

(ii) By definition $\liminf_{n\to+\infty} x_n = \sup_{n\geq 1} \inf_{m\geq n} x_m$. Notice that

$$\liminf_{n\to+\infty} x_n = \sup_{n\geq 1} \inf_{m\geq n} x_m = \sup_{n\geq 1} \left(-\sup_{m\geq n} (-x_m) \right) = -\inf_{n\geq 1} \sup_{m\geq n} (-x_m) = -\limsup_{n\to+\infty} (-x_n),$$

where the second equality follows by **(i)** taking $\inf_{m\geq n} x_m = -\sup_{m\geq n}(-x_m)$ and the third equality follows by **(i)** taking $\sup_{n\geq 1}(-a_n) = -\inf_{n\geq 1} a_n$ with $a_n := \sup_{m\geq n}(-x_m)$. The same reversal of order exchanges arbitrary upper bounds with lower bounds even when the bounds are infinite. Thus both applications of the sup/inf identity remain valid for the extended-real tail values.

Def limsup of collection of sets

Let $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$ and define

$$\begin{aligned}\limsup_{n\to+\infty} E_n &:= \{\omega \in \Omega : \omega \in E_n \text{ for infinitely many } n\text{'s}\} \\ &= \{\omega \in \Omega : \text{for any } n \in \mathbb{N} \text{ there is } m \geq n \text{ such that } \omega \in E_m\} = \bigcap_{n\geq 1} \bigcup_{m\geq n} E_m.\end{aligned}$$

For fixed n, $\bigcup_{m\geq n} E_m$ contains exactly those ω that belong to at least one E_m with $m \geq n$. Intersecting over all n requires such occurrences arbitrarily far out in the sequence. Notice also that $\limsup_{n\to+\infty} E_n \in \mathcal{F}$ because it is countable intersection of countable unions of sets in $\mathcal{F}$.

HW 23 — Measurability of the limsup event

Prove that $\limsup_{n\to+\infty} E_n \in \mathcal{F}$.

Proof.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$. By definition $\limsup_{n\to+\infty} E_n = \bigcap_{n\geq 1} \bigcup_{m\geq n} E_m$. Since $\mathcal{F}$ is a σ-algebra is closed under countable unions therefore $\bigcup_{m\geq n} E_m \in \mathcal{F}$ $\forall n \in \mathbb{N}$. Moreover by HW 3, a σ-algebra is also closed under countable intersections hence $\bigcap_{n\geq 1}(\bigcup_{m\geq n} E_m) \in \mathcal{F}$.

Lemma 2.2 (Reverse Fatou's lemma for sets)

Let $(\Omega, \mathcal{F}, \mathbb{P})$. Given $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$ we have

$$\mathbb{P}\left(\limsup_{n\to+\infty} E_n\right) \geq \limsup_{n\to+\infty} \mathbb{P}(E_n).$$

Proof.

Define $G_m := \bigcup_{n\geq m} E_n$. Notice that $G_{m+1} \subseteq G_m$ $\forall m \in \mathbb{N}$. Indeed by definition $G_m = \bigcup_{n\geq m} E_n = E_m \cup (\bigcup_{n\geq m+1} E_n) = E_m \cup G_{m+1} \supseteq G_{m+1}$. Therefore $(G_m)_{m\in\mathbb{N}}$ is a non-increasing sequence of sets and $G_m \downarrow G$ where $G = \bigcap_{m=1}^{\infty} G_m = \bigcap_{m=1}^{\infty} \bigcup_{n\geq m} E_n = \limsup_{n\to+\infty} E_n$. Since $\mathbb{P}$ is a probability measure we have $\mathbb{P}(G_m) \leq 1 < \infty$ $\forall m \in \mathbb{N}$. Hence by Lemma 1.6 (continuity from above) $\mathbb{P}(G) = \lim_{m\to+\infty} \mathbb{P}(G_m)$. Moreover by definition of G_m we have $E_n \subseteq G_m$ $\forall n \geq m$. By the monotonicity of probability measures (HW 12), $\mathbb{P}(E_n) \leq \mathbb{P}(G_m)$ $\forall n \geq m$. Taking the limit as $m \to +\infty$ we have $\lim_{m\to+\infty} \sup_{n\geq m} \mathbb{P}(E_n) \leq \lim_{m\to+\infty} \mathbb{P}(G_m)$.

Recalling the definition of limsup for numerical sequences and using Lemma 1.6 we conclude that

$$\mathbb{P}\left(\limsup_{n\to+\infty} E_n\right) = \mathbb{P}(G) = \lim_{m\to+\infty} \mathbb{P}(G_m) \geq \lim_{m\to+\infty} \sup_{n\geq m} \mathbb{P}(E_n) = \limsup_{n\to+\infty} \mathbb{P}(E_n).$$

Hence $\mathbb{P}(\limsup_{n\to+\infty} E_n) \geq \limsup_{n\to+\infty} \mathbb{P}(E_n)$.

Lemma 2.3 (First Borel–Cantelli's lemma)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$ be such that $\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty$. Then $\mathbb{P}(\limsup_{n\to+\infty} E_n) = 0$ (i.e. almost every ω belongs to only finitely many E_n's).

Proof.

Define $G_m := \bigcup_{n\geq m} E_n$. Notice that $G_{m+1} \subseteq G_m$ $\forall m \in \mathbb{N}$. Indeed by definition $G_m = E_m \cup G_{m+1}$. Therefore (G_m) is a non-increasing sequence of sets and $G_m \downarrow G$ where $G = \bigcap_{m=1}^{\infty} G_m = \limsup_{n\to+\infty} E_n$. Since $\mathbb{P}(G_m) \leq 1 < \infty$ by Lemma 1.6 (continuity from above) we obtain $\mathbb{P}(G) = \lim_{m\to+\infty} \mathbb{P}(G_m)$. Now for each fixed $m \in \mathbb{N}$ we define $G_m^k := \bigcup_{n=m}^{m+k} E_n$. Notice that $G_m^k \subseteq G_m^{k+1}$ since $G_m^{k+1} = G_m^k \cup E_{m+k+1}$. Therefore $(G_m^k)_{k\in\mathbb{N}}$ is a non-decreasing and $G_m^k \uparrow G_m = \bigcup_{n\geq m} E_n$. Hence by Lemma 1.5 (continuity from below) $\mathbb{P}(G_m) = \lim_{k\to+\infty} \mathbb{P}(G_m^k)$. Moreover by finite sub-

additivity of probability measures (Lemma 1.4) $\mathbb{P}(G_m^k) = \mathbb{P}(\bigcup_{n=m}^{m+k} E_n) \leq \sum_{n=m}^{m+k} \mathbb{P}(E_n)$. Taking the limit as $k \to +\infty$ we obtain

$$\mathbb{P}(G_m) = \lim_{k\to+\infty} \mathbb{P}(G_m^k) \leq \lim_{k\to+\infty} \sum_{n=m}^{m+k} \mathbb{P}(E_n) = \sum_{n=m}^{\infty} \mathbb{P}(E_n).$$

Combining this inequality with the previous result we have

$$\mathbb{P}\left(\limsup_{n\to+\infty} E_n\right) = \mathbb{P}(G) = \lim_{m\to+\infty} \mathbb{P}(G_m) \leq \lim_{m\to+\infty} \sum_{n=m}^{\infty} \mathbb{P}(E_n) = 0,$$

where the last equality follows from the ass. $\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty$. Finally, since probabilities are non-negative $0 \leq \mathbb{P}(\limsup_{n\to+\infty} E_n) \leq 0$. Therefore $\mathbb{P}(\limsup_{n\to+\infty} E_n) = 0$.

Def liminf of collection of sets

Let $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$ and define

$$\begin{aligned}\liminf_{n\to+\infty} E_n &:= \{\omega \in \Omega : \omega \in E_n \text{ for all but finitely many } n\text{'s}\} \\ &= \{\omega \in \Omega : \exists N_\omega \in \mathbb{N} \text{ such that } \omega \in E_n \ \forall n \geq N_\omega\} = \bigcup_{n\geq 1} \bigcap_{m\geq n} E_m.\end{aligned}$$

Notice that $\bigcap_{m\geq n} E_m$ contains those ω's that belong to all sets E_m for $m \geq n$. Taking the union allows to include all those ω's that belong to all but finitely many sets E_n. Notice that $\liminf_{n\to+\infty} E_n \in \mathcal{F}$ because it is countable union of countable intersections of sets in $\mathcal{F}$.

HW 24 — Measurability of the liminf event

Prove that $\liminf_{n\to+\infty} E_n \in \mathcal{F}$.

Proof.

Define $Y_n := \bigcap_{m\geq n} E_m = E_n \cap (\bigcap_{m\geq n+1} E_m) = E_n \cap Y_{n+1}$. Therefore $Y_n \subseteq Y_{n+1}$ $\forall n \in \mathbb{N}$, hence $(Y_n)_{n\in\mathbb{N}}$ is a non-decreasing sequence of sets. Moreover since $E_m \in \mathcal{F}$ $\forall m \in \mathbb{N}$ and $\mathcal{F}$ is a σ-algebra, by HW 3 it follows that $Y_n = \bigcap_{m\geq n} E_m \in \mathcal{F}$ $\forall n \in \mathbb{N}$. Finally since $\mathcal{F}$ is closed under countable unions, we obtain

$$\liminf_{n\to+\infty} E_n = \bigcup_{n=1}^{\infty} Y_n = \bigcup_{n=1}^{\infty} \left(\bigcap_{m\geq n} E_m\right) \in \mathcal{F}.$$

HW 25 — Complement relation of event limits

Prove that $\liminf_{n\to+\infty} E_n = (\limsup_{n\to+\infty} E_n^c)^c$.

Proof.

We obtain

$$\begin{aligned}\left(\limsup_{n\to+\infty} E_n^c\right)^c &= \left(\bigcap_{n=1}^{\infty} \bigcup_{m\geq n} E_m^c\right)^c = \bigcup_{n=1}^{\infty} \left(\bigcup_{m\geq n} E_m^c\right)^c \\ &= \bigcup_{n=1}^{\infty} \bigcap_{m\geq n} (E_m^c)^c = \bigcup_{n=1}^{\infty} \bigcap_{m\geq n} E_m = \liminf_{n\to+\infty} E_n,\end{aligned}$$

where the second and third equalities follow De Morgan's laws, the fourth from double complement property and the last from the definition of liminf.

Lemma 2.4 (Fatou's lemma for sets)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$. Prove that

$$\mathbb{P}\left(\liminf_{n\to+\infty} E_n\right) \leq \liminf_{n\to+\infty} \mathbb{P}(E_n).$$

HW 26 — Fatou inequality by complements

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$. Prove Lemma 2.4, namely that

$$\mathbb{P}\left(\liminf_{n\to+\infty} E_n\right) \leq \liminf_{n\to+\infty} \mathbb{P}(E_n).$$

Hint: follow the steps in the proof of Lemma 2.2.

Proof.

We obtain

$$\mathbb{P}\left(\liminf_{n\to+\infty} E_n\right) = \mathbb{P}\left(\left(\limsup_{n\to+\infty} E_n^c\right)^c\right) = 1 - \mathbb{P}\left(\limsup_{n\to+\infty} E_n^c\right) \leq 1 - \limsup_{n\to+\infty} \mathbb{P}(E_n^c),$$

The first equality is HW 25, the second is the complement identity from HW 18, and the inequality is Lemma 2.2 applied to (E_n^c), after multiplying by -1 and adding 1. Hence

$$\begin{aligned}\mathbb{P}\left(\liminf_{n\to+\infty} E_n\right) &\leq 1 - \limsup_{n\to+\infty} \mathbb{P}(E_n^c) = 1 - \limsup_{n\to+\infty}(1 - \mathbb{P}(E_n)) \\ &= 1 - \left(1 + \limsup_{n\to+\infty}(-\mathbb{P}(E_n))\right) = -\limsup_{n\to+\infty}(-\mathbb{P}(E_n)) = \liminf_{n\to+\infty} \mathbb{P}(E_n),\end{aligned}$$

The first equality again uses HW 18; the next two use the constant-shift rule for limsup and cancel 1, and the last is HW 22.

Definition of indicator function

The indicator function of a set $E \in \mathcal{F}$ is defined as

$$\mathbf{1}_E(\omega) = \begin{cases} 1 & \text{if } \omega \in E, \\ 0 & \text{if } \omega \notin E. \end{cases}$$

For example, if we take $\Omega = [0,1]$ (i.e. ω's are numbers in $[0,1]$) and $\mathcal{F} = \mathcal{B}([0,1])$, then $\mathbf{1}_{[\frac{1}{2},1]}(\omega)$ is a function which is zero when $\omega \in [0, \frac{1}{2})$ and it is equal to one when $\omega \in [\frac{1}{2}, 1]$.

HW 27 — Indicators and limsup

(Ex. 2.9 from the book.) Given $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$. Prove that, for every $\omega \in \Omega$

$$\mathbf{1}_{\limsup_{n\to+\infty} E_n}(\omega) = \limsup_{n\to+\infty} \mathbf{1}_{E_n}(\omega).$$

Hint: follow the intermediate steps set out below. **(i)** prove that $\mathbf{1}_{E_i\cap E_j} = \mathbf{1}_{E_i} \cdot \mathbf{1}_{E_j} = \min\{\mathbf{1}_{E_i}, \mathbf{1}_{E_j}\}$; **(ii)** prove that $\mathbf{1}_{E_i\cup E_j} = \mathbf{1}_{E_i} + \mathbf{1}_{E_j} - \mathbf{1}_{E_i\cap E_j} = \max\{\mathbf{1}_{E_i}, \mathbf{1}_{E_j}\}$; **(iii)** use **(i)** and **(ii)** to conclude.

Proof.

(i) *We prove the intersection identity by its two possible values.*

$$\mathbf{1}_{A\cap B}(\omega) = \begin{cases} 0 & \text{if } \omega \notin A\cap B \iff \omega \notin A \text{ or } \omega \notin B \\ & \quad \iff \mathbf{1}_A(\omega)\mathbf{1}_B(\omega) = 0 = \min\{\mathbf{1}_A(\omega), \mathbf{1}_B(\omega)\}, \\ 1 & \text{if } \omega \in A\cap B \iff \omega \in A \text{ and } \omega \in B \\ & \quad \iff \mathbf{1}_A(\omega)\mathbf{1}_B(\omega) = 1 = \min\{\mathbf{1}_A(\omega), \mathbf{1}_B(\omega)\}. \end{cases}$$

(ii) *We prove the union identity by its two possible values.*

$$\mathbf{1}_{A\cup B}(\omega) = \begin{cases} 0 & \text{if } \omega \notin A \text{ and } \omega \notin B \\ & \quad \iff \mathbf{1}_A(\omega) + \mathbf{1}_B(\omega) - \mathbf{1}_{A\cap B}(\omega) = 0 = \max\{\mathbf{1}_A(\omega), \mathbf{1}_B(\omega)\}, \\ 1 & \text{if } \omega \in A \text{ or } \omega \in B \\ & \quad \iff \mathbf{1}_A(\omega) + \mathbf{1}_B(\omega) - \mathbf{1}_{A\cap B}(\omega) = 1 = \max\{\mathbf{1}_A(\omega), \mathbf{1}_B(\omega)\}. \end{cases}$$

The same pointwise reasoning applies to countably many indicators. An intersection indicator equals one exactly when all the indicators equal one; a union indicator equals one exactly when at least one equals one. Since every value is either zero or one, infimum and supremum are attained. Thus

$$\mathbf{1}_{\bigcap_{n=1}^{\infty} A_n}(\omega) = \inf_{n\geq 1} \mathbf{1}_{A_n}(\omega) = \min_{n\geq 1} \mathbf{1}_{A_n}(\omega), \qquad \mathbf{1}_{\bigcup_{n=1}^{\infty} A_n}(\omega) = \sup_{n\geq 1} \mathbf{1}_{A_n}(\omega) = \max_{n\geq 1} \mathbf{1}_{A_n}(\omega).$$

For simplicity define $F_n := \bigcup_{m\geq n} E_m$ and let $\omega \in \Omega$, we have

$$\begin{aligned}\mathbf{1}_{\limsup_{n\to+\infty} E_n}(\omega) &= \mathbf{1}_{\bigcap_{n\geq 1} F_n}(\omega) = \inf_{n\geq 1} \mathbf{1}_{F_n}(\omega) \\ &= \inf_{n\geq 1} \mathbf{1}_{\bigcup_{m\geq n} E_m}(\omega) = \inf_{n\geq 1} \sup_{m\geq n} \mathbf{1}_{E_m}(\omega) = \limsup_{n\to+\infty} \mathbf{1}_{E_n}(\omega).\end{aligned}$$

In the chain, the first equality substitutes the limsup event, the second uses the intersection identity, the third substitutes F_n, the fourth uses the union identity, and the last is the numerical definition of limsup. Each step holds for the same arbitrary ω.

3. Random Variables

Random variables and pre-images

Random variables are measurable functions.

Def — pre-image of a set

Let (S, Σ) be a measurable space and let $h : S \to \mathbb{R}$ be a function. We define the pre-image of a set $A \subseteq \mathbb{R}$ as $h^{-1}(A) = \{s \in S : h(s) \in A\}$.

Def 3.1 (Measurable functions)

A function $h : S \to \mathbb{R}$ is Σ-measurable (also denoted $h \in \mathrm{m}\Sigma$) if $h^{-1}(A) \in \Sigma \ \forall A \in \mathcal{B}(\mathbb{R})$. Equivalently we say that $h \in \mathrm{m}\Sigma$ if $h^{-1} : \mathcal{B}(\mathbb{R}) \to \Sigma$.

Notation: we use $(\mathrm{m}\Sigma)^+$ for the class of non-negative Σ-measurable functions. We use $\mathrm{b}\Sigma$ for the class of bounded Σ-measurable functions. **Borel-measurable functions:** If S is a topological space and $\Sigma = \mathcal{B}(S)$, then $\mathrm{m}\Sigma$ is the class of Borel-measurable functions. For example if $S = \mathbb{R}$, $\Sigma = \mathcal{B}(\mathbb{R})$ and $h : \mathbb{R} \to \mathbb{R}$ is such $h \in \mathrm{m}\Sigma$, then h is Borel-measurable.

Properties of pre-image mapping

Let $h : S \to \mathbb{R}$ and let $(A_\alpha)_{\alpha \in \mathcal{I}}$ be a family of subsets of $\mathbb{R}$ with $\mathcal{I}$ possibly uncountable (i.e. $\mathcal{I} = [0, 1]$). Then

$$\textbf{(i)} \quad h^{-1}\left(\bigcup_{\alpha \in \mathcal{I}} A_\alpha\right) = \bigcup_{\alpha \in \mathcal{I}} h^{-1}(A_\alpha),$$

$$\textbf{(ii)} \quad h^{-1}(A^c) = (h^{-1}(A))^c,$$

$$\textbf{(iii)} \quad h^{-1}\left(\bigcap_{\alpha \in \mathcal{I}} A_\alpha\right) = \bigcap_{\alpha \in \mathcal{I}} h^{-1}(A_\alpha).$$

Proof (i).

We first prove that $h^{-1}(\bigcup_{\alpha \in \mathcal{I}} A_\alpha) \subseteq \bigcup_{\alpha \in \mathcal{I}} h^{-1}(A_\alpha)$. Let $s_0 \in h^{-1}(\bigcup_{\alpha \in \mathcal{I}} A_\alpha)$. By definition of pre-image we have $h(s_0) \in \bigcup_{\alpha \in \mathcal{I}} A_\alpha$. By definition of union there exists $\alpha_0 \in \mathcal{I}$ such that $h(s_0) \in A_{\alpha_0}$. Therefore $s_0 \in h^{-1}(A_{\alpha_0}) \subseteq \bigcup_{\alpha \in \mathcal{I}} h^{-1}(A_\alpha)$, where the membership $(\in)$ follows from the definition of pre-image and the inclusion follows the fact that $h^{-1}(A_{\alpha_0})$ is one of the sets in the union. Since s_0 was arbitrary it follows that $h^{-1}(\bigcup_{\alpha \in \mathcal{I}} A_\alpha) \subseteq \bigcup_{\alpha \in \mathcal{I}} h^{-1}(A_\alpha)$. For the reverse inclusion, let $s_0 \in \bigcup_{\alpha \in \mathcal{I}} h^{-1}(A_\alpha)$. By definition of union, there exists $\alpha_0 \in \mathcal{I}$ such that $s_0 \in h^{-1}(A_{\alpha_0})$. By definition of pre-image we obtain $h(s_0) \in A_{\alpha_0}$. Therefore $h(s_0) \in A_{\alpha_0} \subseteq \bigcup_{\alpha \in \mathcal{I}} A_\alpha$
where the inclusion follows from the definition of union. Hence by definition of pre-image $s_0 \in h^{-1}(\bigcup_{\alpha \in \mathcal{I}} A_\alpha)$. Since s_0 is arbitrary we obtain $\bigcup_{\alpha \in \mathcal{I}} h^{-1}(A_\alpha) \subseteq h^{-1}(\bigcup_{\alpha \in \mathcal{I}} A_\alpha)$. Combining the two inclusions we conclude the proof.

HW 28 — Pre-images of complements and intersections

For $h : S \to \mathbb{R}$, prove properties **(ii)** and **(iii)**:

$$h^{-1}(A^c) = (h^{-1}(A))^c, \qquad h^{-1}\left(\bigcap_{\alpha \in \mathcal{I}} A_\alpha\right) = \bigcap_{\alpha \in \mathcal{I}} h^{-1}(A_\alpha).$$

Complements on the left are taken in $\mathbb{R}$ and those on the right in S.

Proof.

(ii) Let $A \subseteq \mathbb{R}$, *we first prove that* $h^{-1}(A^c) \subseteq (h^{-1}(A))^c$. Let $s_0 \in h^{-1}(A^c)$. By definition of pre-image we have $h(s_0) \in A^c \Rightarrow h(s_0) \notin A$. Therefore $s_0 \notin h^{-1}(A) \Rightarrow s_0 \in (h^{-1}(A))^c$ where the first statement follows from the definition of pre-image and the second from the definition of complement. Since s_0 is arbitrary it follows that $h^{-1}(A^c) \subseteq (h^{-1}(A))^c$. For the reverse inclusion

let $s_0 \in (h^{-1}(A))^c \Rightarrow s_0 \notin h^{-1}(A)$. By the definition of pre-image this implies that $h(s_0) \notin A$. Therefore $h(s_0) \in A^c \Rightarrow s_0 \in h^{-1}(A^c)$ where the first membership follows from the definition of complement and the second from the definition of pre-image. Since s_0 was arbitrary we obtain $(h^{-1}(A))^c \subseteq h^{-1}(A^c)$. Combining the two inclusionss we conclude the proof.

(iii) Let $(A_\alpha)_{\alpha\in\mathcal{I}}$ be a family of subsets of $\mathbb{R}$ where $\mathcal{I}$ is possibly uncountable. *We first prove that* $h^{-1}(\bigcap_{\alpha\in\mathcal{I}} A_\alpha) \subseteq \bigcap_{\alpha\in\mathcal{I}} h^{-1}(A_\alpha)$. Let $s_0 \in h^{-1}(\bigcap_{\alpha\in\mathcal{I}} A_\alpha)$. By definition of pre-image we have $h(s_0) \in \bigcap_{\alpha\in\mathcal{I}} A_\alpha$. By definition of intersection this means that $h(s_0) \in A_\alpha \ \forall\alpha \in \mathcal{I}$. Therefore $s_0 \in h^{-1}(A_\alpha) \ \forall\alpha \in \mathcal{I}$ where the membership follows the def of pre-image. Hence $s_0 \in \bigcap_{\alpha\in\mathcal{I}} h^{-1}(A_\alpha)$ by def of intersection. Since s_0 is arbitrary it follows that $h^{-1}(\bigcap_{\alpha\in\mathcal{I}} A_\alpha) \subseteq \bigcap_{\alpha\in\mathcal{I}} h^{-1}(A_\alpha)$. For the reverse inclusion, let $s_0 \in \bigcap_{\alpha\in\mathcal{I}} h^{-1}(A_\alpha)$. By definition of intersection we have $s_0 \in h^{-1}(A_\alpha) \ \forall\alpha \in \mathcal{I}$. By definition of pre-image it follows that $h(s_0) \in A_\alpha \ \forall\alpha \in \mathcal{I}$. Therefore by def of intersection $h(s_0) \in \bigcap_{\alpha\in\mathcal{I}} A_\alpha$ hence $s_0 \in h^{-1}(\bigcap_{\alpha\in\mathcal{I}} A_\alpha)$. Since s_0 is arbitrary we obtain $\bigcap_{\alpha\in\mathcal{I}} h^{-1}(A_\alpha) \subseteq h^{-1}(\bigcap_{\alpha\in\mathcal{I}} A_\alpha)$. Combining the two inclusions we conclude the proof.

Lemma 3.1 — The pre-image of Borel sets form a σ-algebra

Let $h : S \to \mathbb{R}$, we want to prove that $h^{-1}(\mathcal{B}(\mathbb{R})) := \{h^{-1}(A) : A \in \mathcal{B}(\mathbb{R})\}$ is a σ-algebra on S.

Proof.

By definition of σ-algebra we must verify the following three prop.

(i) $\emptyset \in h^{-1}(\mathcal{B}(\mathbb{R}))$. Notice that $\emptyset, \mathbb{R} \in \mathcal{B}(\mathbb{R})$ since $\mathcal{B}(\mathbb{R})$ is a σ-algebra on $\mathbb{R}$. Moreover by definition of pre-image $h^{-1}(\emptyset) = \emptyset$ and $h^{-1}(\mathbb{R}) = S$. Therefore $\emptyset, S \in h^{-1}(\mathcal{B}(\mathbb{R}))$.

(ii) Closure under complements. Let $H \in h^{-1}(\mathcal{B}(\mathbb{R}))$. By definition of this class, there exists $A \in \mathcal{B}(\mathbb{R})$ such that $H = h^{-1}(A)$. Since $\mathcal{B}(\mathbb{R})$ is a σ-algebra, it is closed under complements. Therefore $A^c \in \mathcal{B}(\mathbb{R})$. Now consider the complement of H:

$$H^c = (h^{-1}(A))^c = h^{-1}(A^c) \in h^{-1}(\mathcal{B}(\mathbb{R})),$$

where the first equality follows from the definition of H, the second from prop **(ii)** HW 28 and the last membership follows from $A^c \in \mathcal{B}(\mathbb{R})$. Therefore $H \in h^{-1}(\mathcal{B}(\mathbb{R})) \Rightarrow H^c \in h^{-1}(\mathcal{B}(\mathbb{R}))$.

(iii) Closure under countable unions. Let $(H_n)_{n\in\mathbb{N}} \subseteq h^{-1}(\mathcal{B}(\mathbb{R}))$. By definition of this class $\forall n \in \mathbb{N}$ $\exists A_n \in \mathcal{B}(\mathbb{R})$ such that $H_n = h^{-1}(A_n)$. Since $\mathcal{B}(\mathbb{R})$ is a σ-alg. it is closed under countable unions therefore $\bigcup_{n=1}^{\infty} A_n \in \mathcal{B}(\mathbb{R})$. Now consider the union of the sets H_n:

$$\bigcup_{n=1}^{\infty} H_n = \bigcup_{n=1}^{\infty} h^{-1}(A_n) = h^{-1}\left(\bigcup_{n=1}^{\infty} A_n\right) \in h^{-1}(\mathcal{B}(\mathbb{R})),$$

where the first equality follows from the def of H_n, the second from prop **(i)** of the pre-image mapping and the last membership follows from $\bigcup_{n=1}^{\infty} A_n \in \mathcal{B}(\mathbb{R})$. Therefore $(H_n)_{n\in\mathbb{N}} \subseteq h^{-1}(\mathcal{B}(\mathbb{R})) \Rightarrow \bigcup_{n=1}^{\infty} H_n \in h^{-1}(\mathcal{B}(\mathbb{R}))$. Since all three prop are satisfied $h^{-1}(\mathcal{B}(\mathbb{R}))$ is a σ-alg on S.

Rmk: It should be clear that h is measurable respect to $h^{-1}(\mathcal{B}(\mathbb{R}))$. Moreover for any $h : S \to \mathbb{R}$ we can construct a σ-alg on S with respect to which h is measurable. The smallest of such σ-algebras is exactly $h^{-1}(\mathcal{B}(\mathbb{R}))$.

Problem: According to Def 3.1 in order to check whether a function is Σ-measurable or not one need to verify the pre-images of ALL Borel sets are in Σ. Checking every Borel set directly is generally impractical; the next lemma provides a generating-class test.

Lemma 3.2. Measurability through a generating class

Let (S, Σ) be a measurable space, $h : S \to \mathbb{R}$ and let $\mathcal{C} \subseteq \mathcal{B}(\mathbb{R})$ be such that $\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R})$. If $h^{-1}(A) \in \Sigma \ \forall A \in \mathcal{C}$ then h is Σ-measurable i.e. $h^{-1}(A) \in \Sigma \ \forall A \in \mathcal{B}(\mathbb{R})$.

Proof.

Define $\mathcal{E} := \{A \in \mathcal{B}(\mathbb{R}) : h^{-1}(A) \in \Sigma\}$. By definition of $\mathcal{E}$ and by our assumption we have $\mathcal{C} \subseteq \mathcal{E} \subseteq \mathcal{B}(\mathbb{R})$. *We want to prove that* $\mathcal{E}$ *is a* σ*-algebra on* $\mathbb{R}$. By definition of σ-alg we must verify the three prop.

(i) $\emptyset$ and $\mathbb{R}$ belong to $\mathcal{E}$. Notice that by definition of pre-image $h^{-1}(\emptyset) = \emptyset$ and $h^{-1}(\mathbb{R}) = S$. Since Σ is a σ-algebra on S we have $\emptyset, S \in \Sigma$. Therefore $h^{-1}(\emptyset) \in \Sigma$ and $h^{-1}(\mathbb{R}) \in \Sigma$ hence by definition of $\mathcal{E}$, $\emptyset, \mathbb{R} \in \mathcal{E}$.

(ii) Closure under complements. Let $A \in \mathcal{E}$, by def of $\mathcal{E}$ we have $A \in \mathcal{B}(\mathbb{R})$ and $h^{-1}(A) \in \Sigma$. Since both $\mathcal{B}(\mathbb{R})$ and Σ are σ-algebras they are closed under complements. Therefore $A^c \in \mathcal{B}(\mathbb{R})$ and $(h^{-1}(A))^c \in \Sigma$. Now by prop **(ii)** of HW 28 $h^{-1}(A^c) = (h^{-1}(A))^c \in \Sigma$ where the equality follows from the property of pre-images of complements and the last membership follows from the closure of Σ under complements. Since $A^c \in \mathcal{B}(\mathbb{R})$ and $h^{-1}(A^c) \in \Sigma$ by definition of $\mathcal{E}$, $A^c \in \mathcal{E}$. Hence $\mathcal{E}$ is closed under complements.

(iii) Closure under countable unions. Let $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{E}$, by definition of $\mathcal{E}$, $A_n \in \mathcal{B}(\mathbb{R})$ and $h^{-1}(A_n) \in \Sigma\ \forall n \in \mathbb{N}$. Since $\mathcal{B}(\mathbb{R})$ is a σ-algebra, it is closed under countable unions. Therefore $\bigcup_{n=1}^{\infty} A_n \in \mathcal{B}(\mathbb{R})$. Moreover by property **(i)** of the pre-image mapping

$$h^{-1}\left(\bigcup_{n=1}^{\infty} A_n\right) = \bigcup_{n=1}^{\infty} h^{-1}(A_n) \in \Sigma$$

where the equality follows from property **(i)** of pre-images and the last membership follows from the closure of Σ under countable unions, since $h^{-1}(A_n) \in \Sigma$ for every $n \in \mathbb{N}$. Hence by definition of $\mathcal{E}$, $\bigcup_{n=1}^{\infty} A_n \in \mathcal{E}$, thus $\mathcal{E}$ is closed under countable unions. Since all three prop are satisfied $\mathcal{E}$ is a σ-alg on $\mathbb{R}$. Moreover since $\mathcal{C} \subseteq \mathcal{E}$ and $\mathcal{E}$ is a σ-algebra, by the definition of the generated σ-algebra $\sigma(\mathcal{C}) \subseteq \mathcal{E}$. Therefore $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{C}) \subseteq \mathcal{E} \subseteq \mathcal{B}(\mathbb{R})$ hence $\mathcal{E} = \mathcal{B}(\mathbb{R})$. By def of $\mathcal{E}$ this implies $h^{-1}(A) \in \Sigma\ \forall A \in \mathcal{B}(\mathbb{R})$ therefore $h \in \mathrm{m}\Sigma$.

Remarks

There are some important consequences of the above lemma:

Proof.

(i) If S is a topological space and $h : S \to \mathbb{R}$ is continuous then h is Borel measurable (i.e. $h^{-1}(A) \in \mathcal{B}(S)\ \forall A \in \mathcal{B}(\mathbb{R})$). Indeed, $h^{-1}(A)$ is an open set in S whenever A is an open set in $\mathbb{R}$, due to continuity. Take the generating class of open sets in $\mathbb{R}$. Their pre-images are open in S, hence in $\mathcal{B}(S)$, and they generate $\mathcal{B}(\mathbb{R})$. Lemma 3.2 therefore applies with domain sigma-algebra $\mathcal{B}(S)$.

(ii) Given a measurable space (S, Σ) a function $h : S \to \mathbb{R}$ is Σ-measurable if and only if $\{h \leq c\} := \{s \in S : h(s) \in (-\infty, c]\} = h^{-1}((-\infty, c]) \in \Sigma$ for all $c \in \mathbb{R}$. Indeed, if $h \in \mathrm{m}\Sigma$, then $h^{-1}((-\infty, c]) \in \Sigma$ for all $c \in \mathbb{R}$ because $(-\infty, c] \in \mathcal{B}(\mathbb{R})$. If instead $h^{-1}((-\infty, c]) \in \Sigma$ for all $c \in \mathbb{R}$, then we apply the lemma upon recalling the π-system $\pi(\mathbb{R})$ and the fact that $\sigma(\pi(\mathbb{R})) = \mathcal{B}(\mathbb{R})$. This is a clever way to test measurability.

HW 29 — A Borel indicator

Let $S = \mathbb{R}$, $\Sigma = \mathcal{B}(\mathbb{R})$ and $h(s) = \mathbf{1}_{[0,1]}(s)$. Prove that h is Borel-measurable. Hint: first you need to show that $[0, 1] \in \mathcal{B}(\mathbb{R})$.

Proof.

First, $[0, 1] = \mathbb{R} \setminus ((-\infty, 0) \cup (1, +\infty)) \in \mathcal{B}(\mathbb{R})$, since the set being complemented is open. Recall that by consequence of **(ii)** of Lemma 3.2 a function $h : \mathbb{R} \to \mathbb{R}$ is Borel measurable if and only if $h^{-1}((-\infty, c]) \in \mathcal{B}(\mathbb{R})\ \forall c \in \mathbb{R}$. By definition of indicator function $h(s) = \begin{cases} 0 & s \notin [0, 1], \\ 1 & s \in [0, 1]. \end{cases}$ Therefore by definition of pre-image

$$h^{-1}((-\infty, c]) = \{s \in \mathbb{R} : h(s) \leq c\} = \begin{cases} \emptyset & c < 0, \\ (-\infty, 0) \cup (1, +\infty) & 0 \leq c < 1, \\ \mathbb{R} & c \geq 1. \end{cases}$$

Indeed if $c < 0$ neither 0 nor 1 satisfies $h(s) \leq c$ so the pre-image is empty. If $0 \leq c < 1$ only $h(s) = 0$ satisfies the inequality which occurs when $s \notin [0, 1]$. Finally if $c \geq 1$, both value satisfies the inequality so the pre-image is $\mathbb{R}$. Now since $\mathcal{B}(\mathbb{R})$ is a σ-algebra we have $\emptyset, \mathbb{R} \in \mathcal{B}(\mathbb{R})$. Moreover since $(-\infty, 0) \cup (1, +\infty)$ are open sets, their union is also open. Therefore by definition of σ-algebra $(-\infty, 0) \cup (1, +\infty) \in \mathcal{B}(\mathbb{R})$. Hence in all three cases $h^{-1}((-\infty, c]) \in \mathcal{B}(\mathbb{R})$ $\forall c \in \mathbb{R}$, therefore by consequence **(ii)** of Lemma 3.2 $h \in \mathrm{m}\mathcal{B}(\mathbb{R})$.

HW 30 — Equivalent threshold tests

Prove that $h : S \to \mathbb{R}$ is Σ-measurable if and only if each of the following holds:

$$\begin{aligned} &\textbf{(i)} \quad \{h < c\} := h^{-1}((-\infty, c)) \in \Sigma \quad \forall c \in \mathbb{R}, \\ &\textbf{(ii)} \quad \{h > c\} := h^{-1}((c, +\infty)) \in \Sigma \quad \forall c \in \mathbb{R}, \\ &\textbf{(iii)} \quad \{h \geq c\} := h^{-1}([c, +\infty)) \in \Sigma \quad \forall c \in \mathbb{R}. \end{aligned}$$

Proof.

(⇒) We assume that h is Σ-measurable then by Definition 3.1 $\forall A \in \mathcal{B}(\mathbb{R})$, $h^{-1}(A) \in \Sigma$. Since $(-\infty, c)$ and $(c, +\infty)$ are open sets they are Borel and $[c, +\infty) = (-\infty, c)^c$ hence is Borel so their pre-image belong to Σ.

(⇐)

(i) Assume that $\{h < c\} \in \Sigma$ $\forall c \in \mathbb{R}$. Notice that $(-\infty, c] = \bigcap_{n=1}^{\infty}(-\infty, c + \frac{1}{n})$. Therefore

$$\begin{aligned} \{h \leq c\} = h^{-1}((-\infty, c]) &= h^{-1}\left(\bigcap_{n=1}^{\infty}(-\infty, c + \tfrac{1}{n})\right) \\ &= \bigcap_{n=1}^{\infty} h^{-1}((-\infty, c + \tfrac{1}{n})) = \bigcap_{n=1}^{\infty}\{h < c + \tfrac{1}{n}\} \in \Sigma, \end{aligned}$$

where the first equality follows from the definition of pre-image, the second from the identity above, the third from property **(iii)** of HW 28 and the fourth from the definition of $\{h < c\}$. The last membership follows from the closure under countable intersections since $\{h < c + \frac{1}{n}\} \in \Sigma$ $\forall n \in \mathbb{N}$ by assumption. Hence $\{h \leq c\} \in \Sigma$ $\forall c \in \mathbb{R}$ and by Lemma 3.2 h is Σ-measurable.

(ii) Assume that $\{h > c\} \in \Sigma$ $\forall c \in \mathbb{R}$. Notice that $(-\infty, c] = (c, +\infty)^c$. Therefore

$$\{h \leq c\} = h^{-1}((-\infty, c]) = h^{-1}((c, +\infty)^c) = (h^{-1}((c, +\infty)))^c = \{h > c\}^c \in \Sigma,$$

where the first equality follows from the definition of pre-image, the second from the complement identity, the third from prop **(ii)** of HW 28, the fourth from the definition of $\{h > c\}$. The last membership follows closure of Σ under complements since $\{h > c\} \in \Sigma$ by assumption. Hence by Lemma 3.2 h is Σ-measurable.

(iii) Assume that $\{h \geq c\} \in \Sigma$ $\forall c \in \mathbb{R}$. Notice that $(-\infty, c) = [c, +\infty)^c$. Therefore

$$\{h < c\} = h^{-1}((-\infty, c)) = h^{-1}([c, +\infty)^c) = (h^{-1}([c, +\infty)))^c = \{h \geq c\}^c \in \Sigma,$$

where the first equality follows from the definition of pre-image, the second from the complement identity, the third from property **(ii)** of HW 28 and the fourth from the definition of $\{h \geq c\}$. The last membership follows from the closure of Σ under complements since $\{h \geq c\} \in \Sigma$ by assumption. Hence condition **(i)** holds and by the first part of the proof h is Σ-measurable. Therefore each of the three condition is equivalent to Σ-measurability.

Lemma 3.3. Composition of functions

Let $h : S \to \mathbb{R}$ and $f : \mathbb{R} \to \mathbb{R}$ be such that $h \in \mathrm{m}\Sigma$ and $f \in \mathrm{m}\mathcal{B}$. Then $g := f \circ h : S \to \mathbb{R}$ (i.e. $g(s) = f(h(s))$, $s \in S$) is a Σ-measurable functions.

Proof.

For any $A \in \mathcal{B}(\mathbb{R})$ we have

$$g^{-1}(A) = \{s \in S : f(h(s)) \in A\} = \{s \in S : h(s) \in f^{-1}(A)\} = h^{-1}(f^{-1}(A)) \in \Sigma,$$

The first equality uses $g(s) = f(h(s))$, the second the definition of the pre-image under f, and the third that under h. Since $f^{-1}(A)$ is Borel by measurability of f, measurability of h gives the final membership.

Lemma 3.4

If $h, h_1, h_2 \in \mathrm{m}\Sigma$ then **(i)** $h_1 + h_2 \in \mathrm{m}\Sigma$, **(ii)** $h_1 \cdot h_2 \in \mathrm{m}\Sigma$, **(iii)** $\lambda h \in \mathrm{m}\Sigma$ for any $\lambda \in \mathbb{R}$.

Proof.

(i) Recall by HW 30 it is sufficient to show that $\{h_1 + h_2 > c\} \in \Sigma \ \forall c \in \mathbb{R}$. Notice that if $h_1(s) + h_2(s) > c$ then $c - h_2(s) < h_1(s)$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$ there exists $q \in \mathbb{Q}$ such that $h_1(s) > q > c - h_2(s)$. Therefore

$$\{h_1 + h_2 > c\} = \bigcup_{q\in\mathbb{Q}} \{s \in S : h_1(s) > q > c - h_2(s)\} = \bigcup_{q\in\mathbb{Q}} (\{h_1 > q\} \cap \{h_2 > c - q\}) \in \Sigma.$$

Since h_1, h_2 are Σ-measurable by HW 30 we have $\{h_1 > q\} \in \Sigma$ and $\{h_2 > c-q\} \in \Sigma$. The second equality rewrites $q > c - h_2(s)$ as $h_2(s) > c - q$. Finite-intersection closure and then countable-union closure of Σ, using countability of $\mathbb{Q}$, give the final membership. The rational-union identity in **(i)** also has the reverse implication: if $h_1(s) > q$ and $h_2(s) > c-q$, adding gives $h_1(s)+h_2(s) > c$. Thus no point is added by that union. Since $c \in \mathbb{R}$ was arbitrary, HW 30 gives $h_1 + h_2 \in \mathrm{m}\Sigma$.

(iii) By HW 30 it is sufficient to prove that $\{\lambda h > c\} \in \Sigma \ \forall c \in \mathbb{R}$. We distinguish three cases. Case 1. $\lambda = 0$. Notice that $\{\lambda h > c\} = \{s \in S : 0 > c\} = \begin{cases} \emptyset & c \geq 0, \\ S & c < 0. \end{cases}$ Since $\emptyset, S \in \Sigma$ it follows that $\{\lambda h > c\} \in \Sigma$. Case 2. $\lambda > 0$. Notice that $\{\lambda h > c\} = \{s \in S : \lambda h(s) > c\} = \{s \in S : h(s) > \frac{c}{\lambda}\} \in \Sigma$ since $h \in \mathrm{m}\Sigma$. Case 3. $\lambda < 0$. Notice that $\{\lambda h > c\} = \{s \in S : h(s) < \frac{c}{\lambda}\} \in \Sigma$ by HW 30. Hence in all three cases $\{\lambda h > c\} \in \Sigma$ therefore by HW 30 $\lambda h \in \mathrm{m}\Sigma$.

(ii) Let $h_1, h_2 \in \mathrm{m}\Sigma$. Notice that the function $f(x) = x^2$ is continuous and therefore is Borel-measurable by consequence **(i)** of Lemma 3.2. Hence by Lemma 3.3 $h_1^2, h_2^2 \in \mathrm{m}\Sigma$ and, applying the same composition lemma to the measurable sum from **(i)**, $(h_1 + h_2)^2 \in \mathrm{m}\Sigma$. Now notice that $h_1 h_2 = \frac{1}{2}(h_1 + h_2)^2 - \frac{1}{2}(h_1^2 + h_2^2)$, since sum, difference and scalar multiples are Σ-measurable also $h_1 h_2 \in \mathrm{m}\Sigma$.

Lemma 3.5. Measurability of infimum, supremum, liminf and limsup

Let (S, Σ) be a measurable space and let $(h_n)_{n\in\mathbb{N}} \subseteq \mathrm{m}\Sigma$. The infimum, supremum and limits below may take values in $\overline{\mathbb{R}} = [-\infty, +\infty]$. For a map $u : S \to \overline{\mathbb{R}}$, measurability means that $\{u \leq c\} \in \Sigma$ for every $c \in \mathbb{R}$; the equivalent threshold tests from HW 30 apply in this extended setting as well. Memberships in $\mathrm{m}\Sigma$ within this lemma are understood in this extended sense. Whenever one of these functions is finite at every point, it belongs to the real-valued class of Definition 3.1. Then **(i)** $\inf_{n\in\mathbb{N}} h_n \in \mathrm{m}\Sigma$, $\sup_{n\in\mathbb{N}} h_n \in \mathrm{m}\Sigma$; **(ii)** $\liminf_{n\to+\infty} h_n \in \mathrm{m}\Sigma$, $\limsup_{n\to+\infty} h_n \in \mathrm{m}\Sigma$.

Proof.

(i) Let $h(s) := \inf_{n\in\mathbb{N}} h_n(s)$. Recall that by HW 30 to prove that h is Σ-measurable it is sufficient to show that $\{h \geq c\} \in \Sigma \ \forall c \in \mathbb{R}$. As we will see in HW 31, $\{h \geq c\} = \bigcap_{n=1}^{\infty} \{h_n \geq c\}$. Each set in the intersection is measurable by HW 30, so closure under countable intersections gives $\{h \geq c\} \in \Sigma$. By HW 22 $\sup_{n\in\mathbb{N}} h_n = -\inf_{n\in\mathbb{N}}(-h_n)$, since $h_n \in \mathrm{m}\Sigma$ by Lemma 3.4 **(iii)**, $-h_n \in \mathrm{m}\Sigma$ and by the first part of the proof $\inf_{n\in\mathbb{N}}(-h_n) \in \mathrm{m}\Sigma$. Negation also preserves extended-real measurability, since $\{-u > c\} = \{u < -c\}$ for every real threshold c. Consequently $-\inf_{n\in\mathbb{N}}(-h_n) = \sup_{n\in\mathbb{N}} h_n$ is measurable.

(ii) By definition of liminf $\liminf_{n\to+\infty} h_n(s) = \sup_{n\geq 1} \inf_{m\geq n} h_m(s)$. For every $n \in \mathbb{N}$ define $\ell_n(s) := \inf_{m\geq n} h_m(s)$. Notice that ℓ_n is the infimum of a countable family of Σ-measurable functions therefore by prop **(i)** $\ell_n \in \mathrm{m}\Sigma \ \forall n \in \mathbb{N}$. Therefore also $\sup_{n\geq 1} \ell_n(s) \in \mathrm{m}\Sigma$ thanks to **(i)** hence $\liminf h_n(s) \in \mathrm{m}\Sigma$. By HW 22 $\limsup_{n\to+\infty} h_n(s) = -\liminf_{n\to+\infty}(-h_n(s))$ and the same threshold test for negation shows that $\limsup_{n\to+\infty} h_n$ is measurable.

HW 31 — Superlevel sets of an infimum

For $h = \inf_n h_n$, prove that $\{h \geq c\} = \bigcap_{n=1}^{\infty} \{h_n \geq c\}$.

Proof.

We first prove that $\{h \geq c\} \subseteq \bigcap_{n=1}^{\infty} \{h_n \geq c\}$. Let $s_0 \in \{h \geq c\}$. By definition of this set we have $h(s_0) = \inf_{n\in\mathbb{N}} h_n(s_0) \geq c$. By definition of infimum $\inf_{n\in\mathbb{N}} h_n(s_0) \leq h_n(s_0)\ \forall n \in \mathbb{N}$. Therefore $h_n(s_0) \geq \inf_{n\in\mathbb{N}} h_n(s_0) \geq c\ \forall n \in \mathbb{N}$ hence $s_0 \in \{h_n \geq c\}$ which implies $s_0 \in \bigcap_{n=1}^{\infty} \{h_n \geq c\}$. Since s_0 was arbitrary $\bigcap_{n=1}^{\infty} \{h_n \geq c\} \supseteq \{h \geq c\}$. For the reverse inclusion let $s_0 \in \bigcap_{n=1}^{\infty} \{h_n \geq c\}$. By definition of intersection we have $s_0 \in \{h_n \geq c\}\ \forall n \in \mathbb{N}$. Therefore by definition of these sets $h_n(s_0) \geq c\ \forall n \in \mathbb{N}$. Notice that c is a lower bound for the family $\{h_n(s_0), n \in \mathbb{N}\}$. Since the infimum is the greatest lower bound we obtain $h(s_0) = \inf_{n\in\mathbb{N}} h_n(s_0) \geq c$ hence $s_0 \in \{h \geq c\}$. Since s_0 was arbitrary $\bigcap_{n=1}^{\infty} \{h_n \geq c\} \subseteq \{h \geq c\}$. Combining the two inclusions we conclude the proof.

Corollary 3.6 — Measurability of the Convergence Set

Let $(h_n)_{n\in\mathbb{N}} \subset m\Sigma$. Then $\{s \in \mathcal{S} : \lim_{n\to\infty} h_n(s) \text{exists}\} \in \Sigma$.

Expanded proof.

Let

$$\underline{h} = \liminf_{n\to\infty} h_n,\ \overline{h} = \limsup_{n\to\infty} h_n.$$

By Lemma 3.5 these are extended-real measurable, and $\underline{h} \leq \overline{h}$ pointwise. We follow the lecturer's idea of testing their difference at zero, first restricting to where the subtraction is defined.

Part 1 — Both liminf and limsup are finite real numbers. Let $C = \{s \in \mathcal{S} : \underline{h}(s), \overline{h}(s) \in \mathbb{R}\}$. This set is measurable: since $\underline{h} \leq \overline{h}$, $C = \bigcup_{k\geq 1}(\{\underline{h} \geq -k\} \cap \{\overline{h} \leq k\}) \in \Sigma$. On C both functions are real-valued, so we can use the lecturer's subtraction argument. Define $g(s) := \underline{h}(s) - \overline{h}(s)$ for $s \in C$. By Lemma 3.5 the two functions are measurable, and Lemma 3.4**(i)**,**(iii)** makes their difference g measurable on C.

We show that finite real convergence is equivalent to $g = 0$. A real sequence converges to a finite limit if and only if its liminf and limsup coincide in $\mathbb{R}$. Since both values are finite on C, this is equivalent to their difference being zero. Therefore

$$E_{\text{fin}} = \{s \in C : \underline{h}(s) - \overline{h}(s) = 0\} = \{s \in C : g(s) = 0\} = g^{-1}(\{0\}) \in \Sigma.$$

The first equality is the finite liminf/limsup convergence criterion, the second substitutes the definition of g, and the third is the definition of pre-image. Since $\{0\} \in \mathcal{B}(\mathbb{R})$ and g is measurable on the measurable set C, its pre-image belongs to Σ.

Part 2 — Both liminf and limsup equal $+\infty$. *We show that convergence to* $+\infty$ *is a measurable event.* The inequality $\underline{h} \leq \overline{h}$ implies that $\underline{h}(s) = +\infty$ forces $\overline{h}(s) = +\infty$. Consequently,

$$E_+ := \{s : \underline{h}(s) = \overline{h}(s) = +\infty\} = \{\underline{h} = +\infty\} = \bigcap_{k\geq 1} \{\underline{h} > k\} \in \Sigma.$$

The first equality uses the inequality between liminf and limsup. For the second, an extended real number equals $+\infty$ exactly when it exceeds every positive integer. Each level set is measurable by Lemma 3.5; closure under countable intersections gives the final membership. If $\underline{h}(s) = +\infty$, the identity $\underline{h}(s) = \sup_N \inf_{n\geq N} h_n(s)$ shows that, for every real M, some tail infimum exceeds M. Hence $h_n(s) > M$ throughout that tail, which is convergence to $+\infty$. Conversely, convergence to $+\infty$ puts $h_n(s) > M + 1$ eventually for every M. The corresponding tail infimum is at least $M + 1$, so $\underline{h}(s) > M$ for every M, and therefore $\underline{h}(s) = +\infty$.

Part 3 — Both liminf and limsup equal $-\infty$. *We show that convergence to* $-\infty$ *is a measurable event.* Again $\underline{h} \leq \overline{h}$, so $\overline{h}(s) = -\infty$ forces $\underline{h}(s) = -\infty$. Thus

$$E_- := \{s : \underline{h}(s) = \overline{h}(s) = -\infty\} = \{\overline{h} = -\infty\} = \bigcap_{k\geq 1} \{\overline{h} < -k\} \in \Sigma.$$

The first equality uses the same order inequality. The second characterizes $-\infty$ by all negative integer thresholds. Each level set is measurable by Lemma 3.5, so its countable intersection belongs to Σ. If $\overline{h}(s) = -\infty$, the identity $\overline{h}(s) = \inf_N \sup_{n \geq N} h_n(s)$ gives, for every real M, a tail supremum below M. Every term of that tail is then below M, which is convergence to $-\infty$. Conversely, that convergence gives $h_n(s) < M - 1$ eventually for every M. The tail supremum is at most $M - 1$, so $\overline{h}(s) < M$ for every M, and therefore $\overline{h}(s) = -\infty$.

Combining the three parts, the event of convergence in $\overline{\mathbb{R}}$ is

$$\{s : \lim_{n\to\infty} h_n(s) \text{ exists in } \overline{\mathbb{R}}\} = E_{\text{fin}} \cup E_+ \cup E_- \in \Sigma.$$

For convergence to a finite real number, the event is E_{fin} alone. No difference of two infinite values is used. QED

Def: Random variable

Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ a r.v. X is a $\mathcal{F}$-measurable function i.e. $X : \Omega \to \mathbb{R}$ and $X^{-1} : \mathcal{B}(\mathbb{R}) \to \mathcal{F}$.

Ex 1b proof

Consider the infinite coin tossing where $\Omega = \{H, T\}^{\mathbb{N}}$ and $\mathcal{F} = \sigma(\{\omega \in \Omega : \omega_n = W\},\ n \in \mathbb{N},\ W \in \{H, T\})$. For every $n \in \mathbb{N}$ define $X_n(\omega) = \begin{cases} 1 & \text{if } \omega_n = H, \\ 0 & \text{if } \omega_n = T, \end{cases}$ and $S_n = X_1 + \cdots + X_n$. Prove that $F := \{\omega \in \Omega : \lim_{n\to+\infty} \frac{S_n(\omega)}{n} = \frac{1}{2}\} \in \mathcal{F}$.

Proof.

We first prove that X_n is $\mathcal{F}$-measurable $\forall n \in \mathbb{N}$. By HW 30 it is sufficient to show that $\{X_n \leq c\} \in \mathcal{F}\ \forall c \in \mathbb{R}$. By definition of X_n we have

$$\{X_n \leq c\} = \begin{cases} \emptyset & c < 0, \\ \{\omega \in \Omega : \omega_n = T\} & 0 \leq c < 1, \\ \Omega & c \geq 1. \end{cases}$$

Notice that $\emptyset, \Omega \in \mathcal{F}$ and $\{\omega \in \Omega : \omega_n = T\} \in \mathcal{F}$ by def of $\mathcal{F}$ in example 1b. Therefore $\{X_n \leq c\} \in \mathcal{F}\ \forall c \in \mathbb{R}$ hence $X_n \in \mathrm{m}\mathcal{F}\ \forall n \in \mathbb{N}$. Now by Lemma 3.4 **(i)** $S_n := \sum_{k=1}^n X_k \in \mathrm{m}\mathcal{F}$ $\forall n \in \mathbb{N}$. Moreover since $\frac{1}{n} \in \mathbb{R}$, by Lemma 3.4 **(iii)** $\frac{1}{n} S_n \in \mathrm{m}\mathcal{F}\ \forall n \in \mathbb{N}$. *We prove now that $F \in \mathcal{F}$.* Recall that a sequence of real numbers converges to $a \in \mathbb{R}$ if and only if its limsup and liminf are both equal to a. Therefore

$$F = \{\omega \in \Omega : \lim_{n\to+\infty} \tfrac{S_n(\omega)}{n} = \tfrac{1}{2}\} = \{\omega \in \Omega : \limsup_{n\to+\infty} \tfrac{S_n(\omega)}{n} = \tfrac{1}{2}\} \cap \{\omega \in \Omega : \liminf_{n\to+\infty} \tfrac{S_n(\omega)}{n} = \tfrac{1}{2}\}.$$

Define $U(\omega) := \limsup_{n\to+\infty} \frac{S_n(\omega)}{n}$ and $L(\omega) := \liminf_{n\to+\infty} \frac{S_n(\omega)}{n}$. Since $\frac{S_n}{n} \in \mathrm{m}\mathcal{F}\ \forall n \in \mathbb{N}$ by Lemma 3.5 **(ii)** $U, L \in \mathrm{m}\mathcal{F}$. Notice that both functions are real valued since $0 \leq \frac{S_n(\omega)}{n} \leq 1\ \forall \omega$. Moreover $\{\frac{1}{2}\} \in \mathcal{B}(\mathbb{R})$ since it is a singleton. In particular

$$A := \{\omega \in \Omega : \limsup_{n\to+\infty} \tfrac{S_n(\omega)}{n} = \tfrac{1}{2}\} = U^{-1}(\{\tfrac{1}{2}\}) \in \mathcal{F},$$
$$B := \{\omega \in \Omega : \liminf_{n\to+\infty} \tfrac{S_n(\omega)}{n} = \tfrac{1}{2}\} = L^{-1}(\{\tfrac{1}{2}\}) \in \mathcal{F}.$$

Hence recalling $F = A \cap B$ since $\mathcal{F}$ is closed under intersection we conclude that $F \in \mathcal{F}$. As a consequence, since $\mathbb{P} : \mathcal{F} \to [0, 1]$ is a probability measure and $F \in \mathcal{F}$, the probability $\mathbb{P}(F)$ is well-defined.

Def Law of a random variable

On a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ let X be a real-valued random variable. The law of X is a probability measure on $\mathcal{B}(\mathbb{R})$ i.e. a mapping $\mathcal{L}_X : \mathcal{B}(\mathbb{R}) \to [0, 1]$ defined as $\mathcal{L}_X := \mathbb{P} \circ X^{-1}$.

That is, for any $A \in \mathcal{B}(\mathbb{R})$ $\mathcal{L}_X(A) := \mathbb{P}(X \in A)$. The diagram below should help your intuition: $\mathcal{B}(\mathbb{R}) \xrightarrow{X^{-1}} \mathcal{F} \xrightarrow{\mathbb{P}} [0,1]$.

Proof.

Notice that $\mathcal{L}_X$ is well defined. Indeed since X is $\mathcal{F}$-measurable by definition 3.1, $X^{-1}(A) \in \mathcal{F}$ $\forall A \in \mathcal{B}(\mathbb{R})$. Therefore $\mathcal{L}_X(A) = \mathbb{P}(X^{-1}(A)) \in [0,1]$. *We now verify the properties of a probability measure.*

(i) *We first prove that* $\mathcal{L}_X(\emptyset) = 0$. We obtain

$$\mathcal{L}_X(\emptyset) = \mathbb{P}(X^{-1}(\emptyset)) = \mathbb{P}(\emptyset) = 0$$

where the first equality follows from the definition of the law X, the second from the fact that pre-image of the empty set is empty, the last from properties of the probability measure $\mathbb{P}$.

(ii) *We now prove that* $\mathcal{L}_X(\mathbb{R}) = 1$. We obtain

$$\mathcal{L}_X(\mathbb{R}) = \mathbb{P}(X^{-1}(\mathbb{R})) = \mathbb{P}(\Omega) = 1$$

where the first equality is the definition of $\mathcal{L}_X$, the second follows from the fact that $X : \Omega \to \mathbb{R}$ so $X(\omega) \in \mathbb{R}$ $\forall \omega \in \Omega$ and the last equality follows from the fact that $\mathbb{P}$ is a probability measure.

(iii) *Finally we prove countable additivity.* Let $(A_n)_{n\in\mathbb{N}} \subset \mathcal{B}(\mathbb{R})$ be a sequence of pairwise disjoint sets namely $A_i \cap A_j = \emptyset$ $\forall i \neq j$. *We want to prove that*

$$\mathcal{L}_X\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathcal{L}_X(A_n).$$

As we will see in HW 32, $X^{-1}(\bigcup_{n=1}^{\infty} A_n) = \bigcup_{n=1}^{\infty} X^{-1}(A_n)$. We use that identity without proving it here. First notice that the pre-images $X^{-1}(A_n)$ are also pairwise disjoint. Indeed for $i \neq j$ by property **(iii)** of HW 28

$$X^{-1}(A_i) \cap X^{-1}(A_j) = X^{-1}(A_i \cap A_j) = X^{-1}(\emptyset) = \emptyset$$

Moreover since X is $\mathcal{F}$-measurable $X^{-1}(A_n) \in \mathcal{F}$ $\forall n \in \mathbb{N}$. Therefore we can apply the countable additivity of $\mathbb{P}$ to the sequence $(X^{-1}(A_n))_{n\in\mathbb{N}}$. We obtain

$$\mathcal{L}_X\left(\bigcup_{n=1}^{\infty} A_n\right) = \mathbb{P}\left(X^{-1}\left(\bigcup_{n=1}^{\infty} A_n\right)\right) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} X^{-1}(A_n)\right) = \sum_{n=1}^{\infty} \mathbb{P}(X^{-1}(A_n)) = \sum_{n=1}^{\infty} \mathcal{L}_X(A_n)$$

where the first equality follows from the definition of $\mathcal{L}_X$, the second from HW 32, the third from the countable additivity of $\mathbb{P}$ since the pre-images are pairwise disjoint, and the last from the definition of $\mathcal{L}_X$. Hence $\mathcal{L}_X(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mathcal{L}_X(A_n)$. Therefore $\mathcal{L}_X$ is countably additive. Since $\mathcal{L}_X(\emptyset) = 0$, $\mathcal{L}_X(\mathbb{R}) = 1$ and $\mathcal{L}_X$ is countably additive on $\mathcal{B}(\mathbb{R})$ we conclude that $\mathcal{L}_X$ is a probability measure on $\mathcal{B}(\mathbb{R})$.

HW 32 — Pre-image of a countable union

Prove that if $(A_n)_{n\in\mathbb{N}}$ is a family of pairwise disjoint subsets of $\mathbb{R}$ then

$$\{\omega \in \Omega : X(\omega) \in \bigcup_{n=1}^{\infty} A_n\} = \bigcup_{n=1}^{\infty} \{\omega \in \Omega : X(\omega) \in A_n\}.$$

Proof.

We first prove that $\{\omega \in \Omega : X(\omega) \in \bigcup_{n=1}^{\infty} A_n\} \subseteq \bigcup_{n=1}^{\infty} \{\omega \in \Omega : X(\omega) \in A_n\}$. Let $\omega_0 \in \{\omega \in \Omega : X(\omega) \in \bigcup_{n=1}^{\infty} A_n\}$. By definition of this set we have $X(\omega_0) \in \bigcup_{n=1}^{\infty} A_n$. By definition of union there exists $n_0 \in \mathbb{N}$ such that $X(\omega_0) \in A_{n_0}$. n_0 is unique since A_n are pairwise disjoint. Therefore $\omega_0 \in \{\omega \in \Omega : X(\omega) \in A_{n_0}\} \subseteq \bigcup_{n=1}^{\infty} \{\omega \in \Omega : X(\omega) \in A_n\}$ where the first membership follows from the definition of the set and the inclusion follows from the fact that $\{\omega \in \Omega : X(\omega) \in A_{n_0}\}$ is one of the sets of the union. Since ω_0 was arbitrary we obtain $\{\omega \in \Omega : X(\omega) \in \bigcup_{n=1}^{\infty} A_n\} \subseteq \bigcup_{n=1}^{\infty} \{\omega \in \Omega : X(\omega) \in A_n\}$. For the reverse inclusion we let $\omega_0 \in \bigcup_{n=1}^{\infty} \{\omega \in \Omega : X(\omega) \in A_n\}$.

By definition of union there exists $n_0 \in \mathbb{N}$ such that $\omega_0 \in \{\omega \in \Omega : X(\omega) \in A_{n_0}\}$. By definition of this set we have $X(\omega_0) \in A_{n_0}$. Therefore $X(\omega_0) \in A_{n_0} \subseteq \bigcup_{n=1}^{\infty} A_n$. Hence $\omega_0 \in \{\omega \in \Omega : X(\omega) \in \bigcup_{n=1}^{\infty} A_n\}$. Since ω_0 was arbitrary it follows that $\bigcup_{n=1}^{\infty}\{\omega \in \Omega : X(\omega) \in A_n\} \subseteq \{\omega \in \Omega : X(\omega) \in \bigcup_{n=1}^{\infty} A_n\}$. Combining the two inclusions we conclude the proof.

Why is it sufficient to specify the law on a π-system

Since $\mathcal{L}_X$ is a probability measure on $\mathcal{B}(\mathbb{R})$ we may ask whether it is necessary to specify $\mathcal{L}_X(A)$ $\forall$ Borel set A in order to uniquely determine the law of X. The answer is no. By Lemma 1.2 it is sufficient to specify the values of $\mathcal{L}_X$ on a π-system generating $\mathcal{B}(\mathbb{R})$. Recall that by Prop 1.1 $\pi(\mathbb{R}) = \{(-\infty, c] : c \in \mathbb{R}\}$ is a π-system and $\sigma(\pi(\mathbb{R})) = \mathcal{B}(\mathbb{R})$. Moreover by Lemma 1.2 if two finite measures on a σ-algebra generated by a π-system have the same total mass and coincide on that π-system, then they coincide on the entire σ-algebra. In particular let μ_1 and μ_2 be two probability measures on $\mathcal{B}(\mathbb{R})$ such that $\mu_1((-\infty, c]) = \mu_2((-\infty, c])$ $\forall c \in \mathbb{R}$. Since both measures are probability measures $\mu_1(\mathbb{R}) = \mu_2(\mathbb{R}) = 1 < \infty$. Therefore all the assumption of Lemma 1.2 are satisfied and we conclude that $\mu_1(A) = \mu_2(A)$ $\forall A \in \mathcal{B}(\mathbb{R})$. Hence a probability measure on $\mathcal{B}(\mathbb{R})$ is uniquely determined by its values on the intervals $(-\infty, c]$ for every $c \in \mathbb{R}$. In particular the law of a random variable can be uniquely determined without specifying its value on every Borel set. It is sufficient to know $\mathcal{L}_X((-\infty, c])$ $\forall c \in \mathbb{R}$. This motivates the definition of the cumulative distribution function.

Definition CDF

Let $X : \Omega \to \mathbb{R}$ be a random variable. The CDF of X is the function $F_X : \mathbb{R} \to [0, 1]$ defined by $F_X(c) := \mathcal{L}_X((-\infty, c]) = \mathbb{P}(X \leq c)$ $\forall c \in \mathbb{R}$. Indeed by definition of the law of X

$$F_X(c) = \mathcal{L}_X((-\infty, c]) = \mathbb{P}(X^{-1}((-\infty, c])) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq c\}),$$

where the second equality follows from the definition of $\mathcal{L}_X$ and the third from the definition of pre-image.

Since the intervals $(-\infty, c]$ form a π-system generating $\mathcal{B}(\mathbb{R})$ by Lemma 1.2 the CDF uniquely determines the law $\mathcal{L}_X$. In particular $F_X(c) = F_Y(c)$ $\forall c \in \mathbb{R} \iff \mathcal{L}_X = \mathcal{L}_Y$. Thus two random variables have the same law exactly when they have the same CDF.

Properties of cdf

The cdf enjoys the following properties: **(a)** Range and monotonicity $F_X : \mathbb{R} \to [0, 1]$, $F_X(y) \leq F_X(z)$ $\forall y \leq z$.

(b) Limit at infinity. If $\mathbb{P}(|X| < \infty) = 1$ then $\lim_{z \to +\infty} F_X(z) = 1$, $\lim_{z \to -\infty} F_X(z) = 0$.

(c) Right-continuity $\forall z \in \mathbb{R}$, $\lim_{\varepsilon \downarrow 0} F_X(z + \varepsilon) = F_X(z)$.

Proof.

(a) By definition of the CDF $F_X(z) = \mathbb{P}(X \leq z)$. Since $\mathbb{P}$ is a probability measure we have $0 \leq \mathbb{P}(X \leq z) \leq 1$, $\forall z \in \mathbb{R}$. Therefore $0 \leq F_X(z) \leq 1$ $\forall z \in \mathbb{R}$ hence $F_X : \mathbb{R} \to [0, 1]$. For monotonicity, let $y, z \in \mathbb{R}$ such that $y \leq z$. Notice that $\{X \leq y\} \subseteq \{X \leq z\}$. Indeed if $\omega \in \{X \leq y\}$ then $X(\omega) \leq y \leq z$ therefore $\omega \in \{X \leq z\}$. Since X is $\mathcal{F}$-measurable both events belong to $\mathcal{F}$ and by monotonicity of probability HW 12 we obtain $\mathbb{P}(X \leq y) \leq \mathbb{P}(X \leq z)$. Hence by definition of cdf $F_X(y) \leq F_X(z)$.

(b) For the positive tail, $\forall n \in \mathbb{N}$ define $A_n := \{\omega \in \Omega : X(\omega) \leq n\}$. Since X is $\mathcal{F}$-measurable by definition 3.1 $A_n \in \mathcal{F}$ $\forall n \in \mathbb{N}$. Notice that $A_n \subseteq A_{n+1}$ $\forall n \in \mathbb{N}$, because $X(\omega) \leq n$ implies $X(\omega) \leq n + 1$. Moreover $\bigcup_{n=1}^{\infty} A_n = \{\omega \in \Omega : X(\omega) < +\infty\}$. Since $\mathbb{P}(|X| < \infty) = 1$, we have $\mathbb{P}(\bigcup_{n=1}^{\infty} A_n) = 1$. Therefore by Lemma 1.5 (continuity of probability from below)

$$\lim_{n \to +\infty} F_X(n) = \lim_{n \to +\infty} \mathbb{P}(A_n) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = 1,$$

where the first equality follows from the definition of the CDF, the second from Lemma 1.5 and third from assumption. Since F_X is non-decreasing by prop (a) the convergence $F_X(n) \to 1$ implies that $\lim_{z\to+\infty} F_X(z) = 1$. For the negative tail, $\forall n \in \mathbb{N}$ define $B_n := \{\omega \in \Omega : X(\omega) \leq -n\}$.
Again since X is measurable, $B_n \in \mathcal{F}$ $\forall n \in \mathbb{N}$. Notice that $B_n \supseteq B_{n+1}$ $\forall n \in \mathbb{N}$ because $X(\omega) \leq -(n+1)$ implies $X(\omega) \leq -n$. Moreover $\bigcap_{n=1}^{\infty} B_n = \{\omega \in \Omega : X(\omega) = -\infty\}$. By assumption $\mathbb{P}(|X| < \infty) = 1$, $\mathbb{P}(\bigcap_{n=1}^{\infty} B_n) = 0$. Since the events B_n form a non-increasing sequence and $\mathbb{P}(B_1) \leq 1 < \infty$, we may apply Lemma 1.6 (continuity from above). Therefore

$$\lim_{n\to+\infty} F_X(-n) = \lim_{n\to+\infty} \mathbb{P}(B_n) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} B_n\right) = 0,$$

where the first equality follows from the definition of CDF, the second from Lemma 1.6 and third from ass. Since F_X is non-decreasing by prop (a) the convergence $F_X(-n) \to 0$ implies that $\lim_{z\to-\infty} F_X(z) = 0$.
(c) Fix an arbitrary z. $\forall n \in \mathbb{N}$ define $A_n := \{\omega \in \Omega : X(\omega) \leq z + \frac{1}{n}\}$. Since X is $\mathcal{F}$-measurable, $A_n \in \mathcal{F}$ $\forall n \in \mathbb{N}$. Notice that $A_n \supseteq A_{n+1}$ $\forall n \in \mathbb{N}$ because $z + \frac{1}{n+1} \leq z + \frac{1}{n}$. *We now show that* $\bigcap_{n=1}^{\infty} A_n = \{X \leq z\}$ *by showing both inclusions. First we show that* $\bigcap_{n=1}^{\infty} A_n \subseteq \{X \leq z\}$. Let $\omega_0 \in \bigcap_{n=1}^{\infty} A_n$, by definition of intersection $\omega_0 \in A_n$ $\forall n \in \mathbb{N}$. Therefore by definition of A_n, $X(\omega_0) \leq z + \frac{1}{n}$ $\forall n \in \mathbb{N}$. Since $\frac{1}{n} \to 0$ we obtain $X(\omega_0) \leq z$. Hence $\omega_0 \in \{X \leq z\}$. Since ω_0 was arbitrary we conclude that $\bigcap_{n=1}^{\infty} A_n \subseteq \{X \leq z\}$. *For the reverse inclusion, we show that* $\{X \leq z\} \subseteq \bigcap_{n=1}^{\infty} A_n$. Let $\omega_0 \in \{X \leq z\}$. By definition of this set $X(\omega_0) \leq z$. Therefore $X(\omega_0) \leq z < z + \frac{1}{n}$ $\forall n \in \mathbb{N}$. Hence $\omega_0 \in A_n$ $\forall n \in \mathbb{N}$. By definition of intersection $\omega_0 \in \bigcap_{n=1}^{\infty} A_n$. Since ω_0 was arbitrary we obtain $\{X \leq z\} \subseteq \bigcap_{n=1}^{\infty} A_n$. Combining the two inclusionss we conclude that $\bigcap_{n=1}^{\infty} A_n = \{X \leq z\}$. Now since $(A_n)_{n\in\mathbb{N}}$ is a decreasing sequence of events and $\mathbb{P}(A_1) \leq 1 < \infty$, by Lemma 1.6 we obtain

$$\lim_{n\to+\infty} F_X(z + \tfrac{1}{n}) = \lim_{n\to+\infty} \mathbb{P}(A_n) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right) = \mathbb{P}(X \leq z) = F_X(z),$$

where the first equality follows from the definition of the CDF, the second from Lemma 1.6, the third from the equality of sets proved above and the last the definition of F_X. Finally, to obtain the limit for arbitrary $\varepsilon \downarrow 0$, let $\delta > 0$. From the preceding convergence, there exists $N \in \mathbb{N}$ such that $F_X(z + \frac{1}{N}) < F_X(z) + \delta$. For every $0 < \varepsilon \leq \frac{1}{N}$, by monotonicity of F_X, $F_X(z) \leq F_X(z+\varepsilon) \leq F_X(z + \frac{1}{N}) < F_X(z) + \delta$. Since $\delta > 0$ was arbitrary we conclude that

$$\lim_{\varepsilon\downarrow 0} F_X(z + \varepsilon) = F_X(z).$$

Since $z \in \mathbb{R}$ was arbitrary, F_X is right-continuous.

HW 33 — CDF tails with mass at infinity

Let X be an extended-real random variable. Prove if $\mathbb{P}(X = +\infty) = p$ and $\mathbb{P}(X = -\infty) = q$ then $\lim_{z\to+\infty} F_X(z) = 1 - p$ and $\lim_{z\to-\infty} F_X(z) = q$.

Proof.

Recall that by definition of CDF $F_X(z) = \mathbb{P}(X \leq z)$, $z \in \mathbb{R}$. For the positive tail, $\forall n \in \mathbb{N}$ define $A_n := \{X \leq n\}$. Notice that $A_n \subseteq A_{n+1}$ $\forall n \in \mathbb{N}$ since $X(\omega) \leq n \Rightarrow X(\omega) \leq n + 1$. Therefore $(A_n)_{n\in\mathbb{N}}$ is an increasing sequence of events. Moreover $\bigcup_{n=1}^{\infty} A_n = \{X < \infty\}$. Indeed, $X(\omega) \leq n$ for at least one $n \in \mathbb{N}$ if and only if $X(\omega) < +\infty$. Hence by Lemma 1.5 continuity of probability from below we obtain

$$\lim_{n\to+\infty} F_X(n) = \lim_{n\to+\infty} \mathbb{P}(A_n) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \mathbb{P}(X < +\infty) = 1 - \mathbb{P}(X = +\infty) = 1 - p,$$

where the first equality follows from the definition of the CDF, the second from Lemma 1.5, the third from the equality of sets above and the fourth from the complement rule for probability.

Since F_X is non-decreasing by prop (a) of the CDF, and $F_X(n) \leq F_X(z) \leq 1-p \ \forall z \geq n$ we conclude that $\lim_{z\to+\infty} F_X(z) = 1-p$. For the negative tail, $\forall n \in \mathbb{N}$ define $B_n := \{X \leq -n\}$. Notice that $B_n \supseteq B_{n+1} \ \forall n \in \mathbb{N}$ since $X(\omega) \leq -(n+1) \Rightarrow X(\omega) \leq -n$. Therefore $(B_n)_{n\in\mathbb{N}}$ is a decreasing sequence of events. Moreover $\bigcap_{n=1}^{\infty} B_n = \{X = -\infty\}$. Indeed $X(\omega) \leq -n \ \forall n \in \mathbb{N} \iff X(\omega) = -\infty$. Hence, since $\mathbb{P}(B_1) \leq 1 < \infty$ by Lemma 1.6 (continuity of probability from above) we obtain

$$\lim_{n\to+\infty} F_X(-n) = \lim_{n\to+\infty} \mathbb{P}(B_n) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} B_n\right) = \mathbb{P}(X = -\infty) = q,$$

where the first equality follows from the definition of the CDF, the second from Lemma 1.6, the third from the equality of sets above and the last from the assumption. Since F_X is non-decreasing by prop (a) and $q \leq F_X(z) \leq F_X(-n) \ \forall z \leq -n$ we conclude that $\lim_{z\to-\infty} F_X(z) = q$.

HW 34 — Left limits and atoms

Prove that $\forall z \in \mathbb{R}$, $\{X \leq z - \varepsilon\} \uparrow \{X < z\}$ as $\varepsilon \downarrow 0$. Then conclude that $\lim_{\varepsilon\downarrow 0} F_X(z-\varepsilon) = F_X(z) - \mathcal{L}_X(\{z\})$.

Proof.

Fix an arbitrary $z \in \mathbb{R}$. For every $n \in \mathbb{N}$ define $A_n := \{\omega \in \Omega : X(\omega) \leq z - \frac{1}{n}\}$. Since X is $\mathcal{F}$-measurable we have $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$. Notice that $A_n \subseteq A_{n+1} \ \forall n \in \mathbb{N}$. Indeed let $\omega_0 \in A_n$. By definition of A_n $X(\omega_0) \leq z - \frac{1}{n}$, since $z - \frac{1}{n} < z - \frac{1}{n+1}$ we obtain $X(\omega_0) \leq z - \frac{1}{n} < z - \frac{1}{n+1}$. Therefore $\omega_0 \in A_{n+1}$. Since ω_0 was arbitrary the inclusion follows. Hence $(A_n)_{n\in\mathbb{N}}$ is an increasing sequence of events, *we now prove that* $\bigcup_{n=1}^{\infty} A_n = \{X < z\}$. We establish the equality by proving both inclusionss. First inclusion. Let $\omega_0 \in \bigcup_{n=1}^{\infty} A_n$. By definition of union, there exists $n_0 \in \mathbb{N}$ such that $\omega_0 \in A_{n_0}$. By definition of A_{n_0}, $X(\omega_0) \leq z - \frac{1}{n_0} < z$. Therefore $X(\omega_0) < z$ which implies that $\omega_0 \in \{X < z\}$. Since ω_0 was arbitrary we conclude that $\bigcup_{n=1}^{\infty} A_n \subseteq \{X < z\}$. Reverse inclusion. Let $\omega_0 \in \{X < z\}$. By definition of this set $X(\omega_0) < z$. Therefore $z - X(\omega_0) > 0$. Since $\frac{1}{n} \to 0$ as $n \to +\infty$ there exists $n_0 \in \mathbb{N}$ such that $\frac{1}{n_0} < z - X(\omega_0)$. Rearranging this inequality we obtain $X(\omega_0) < z - \frac{1}{n_0}$ which implies $X(\omega_0) \leq z - \frac{1}{n_0}$ hence $\omega_0 \in A_{n_0}$ and consequently $\omega_0 \in \bigcup_{n=1}^{\infty} A_n$. Since ω_0 was arbitrary we conclude that $\{X < z\} \subseteq \bigcup_{n=1}^{\infty} A_n$. Combining the two inclusionss, we obtain $\bigcup_{n=1}^{\infty} A_n = \{X < z\}$. Since $(A_n)_{n\in\mathbb{N}}$ is an increasing sequence of events we have proved that $\{X \leq z - \varepsilon\} \uparrow \{X < z\}$ as $\varepsilon \downarrow 0$. Since $(A_n)_{n\in\mathbb{N}}$ is an increasing sequence of events, by Lemma 1.5 we obtain

$$\lim_{n\to+\infty} F_X(z - \tfrac{1}{n}) = \lim_{n\to+\infty} \mathbb{P}(A_n) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \mathbb{P}(X < z)$$

where the first equality follows from the definition of the CDF and of A_n, the second from Lemma 1.5 and the third from the equality of sets proved above. Moreover since F_X is non-decreasing for every $0 < \varepsilon \leq \frac{1}{n}$, $F_X(z - \frac{1}{n}) \leq F_X(z-\varepsilon) \leq \mathbb{P}(X < z)$. Since the left-hand side converges to $\mathbb{P}(X < z)$ as $n \to +\infty$ we conclude that $\lim_{\varepsilon\downarrow 0} F_X(z-\varepsilon) = \mathbb{P}(X < z)$. Finally we notice that $\{X \leq z\} = \{X < z\} \cup \{X = z\}$

where the two events are disjoint on the right-hand side. Therefore by the additivity of the probability measure $\mathbb{P}$ $\mathbb{P}(X \leq z) = \mathbb{P}(X < z) + \mathbb{P}(X = z)$. Rearranging we obtain $\mathbb{P}(X < z) = \mathbb{P}(X \leq z) - \mathbb{P}(X = z)$. Hence

$$\lim_{\varepsilon\downarrow 0} F_X(z-\varepsilon) = \mathbb{P}(X < z) = \mathbb{P}(X \leq z) - \mathbb{P}(X = z) = F_X(z) - \mathcal{L}_X(\{z\}),$$

where the first equality follows from the continuity of probability from below established above, the second from the additivity of $\mathbb{P}$, the third from the definition of the CDF and of the law of X. Since $z \in \mathbb{R}$ was arbitrary we conclude that $\lim_{\varepsilon\downarrow 0} F_X(z-\varepsilon) = F_X(z) - \mathcal{L}_X(\{z\})$.

Definition — Probability Density Function (pdf)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X : \Omega \to \mathbb{R}$ be a r.v. Recall that cdf $F_X(z) := \mathbb{P}(X \leq z)$, $z \in \mathbb{R}$. A r.v. X admits a probability density function (pdf) if there exists a non-negative measurable function $f_X : \mathbb{R} \to [0, +\infty)$ such that

$$\mathbb{P}(X \in A) = \int_A f_X(z)\, dz \quad \forall A \in \mathcal{B}(\mathbb{R}).$$

In particular taking $A = (-\infty, z]$ we obtain $F_X(z) = \int_{-\infty}^{z} f_X(t)\, dt$. If the CDF is absolutely continuous then it is differentiable almost everywhere and its density is given by $f_X(z) = F_X'(z)$ for almost every $z \in \mathbb{R}$.

Properties of pdf

(a) Non-negativity $f_X(z) \geq 0$ for almost every $z \in \mathbb{R}$. **(b)** Normalization $\int_{-\infty}^{+\infty} f_X(z)\, dz = 1$.

Proof.

(a) Recall that by prop (a) of the CDF, F_X is non-decreasing. Therefore $\forall z \in \mathbb{R}$ and $\forall h > 0$, $F_X(z + h) \geq F_X(z)$. Subtracting $F_X(z)$ and dividing by $h > 0$ we obtain $\frac{F_X(z+h) - F_X(z)}{h} \geq 0$. At every point where F_X is differentiable letting $h \downarrow 0$ gives $F_X'(z) = \lim_{h \downarrow 0} \frac{F_X(z+h) - F_X(z)}{h} \geq 0$. The integral representation $F_X(z) = \int_{-\infty}^{z} f_X(t)\, dt$ from the definition of a density makes F_X absolutely continuous, with $F_X' = f_X$ almost everywhere by the fundamental theorem for the Lebesgue integral. Consequently, $f_X(z) \geq 0$ almost everywhere, consistently with the non-negativity already required in the definition. **(b)** Recall that by def of probability density function $\mathbb{P}(X \in A) = \int_A f_X(z)\, dz$ $\forall A \in \mathcal{B}(\mathbb{R})$. In particular taking $A = \mathbb{R}$ we obtain $\int_{-\infty}^{+\infty} f_X(z)\, dz = \mathbb{P}(X \in \mathbb{R}) = \mathbb{P}(\Omega) = 1$ where the first equality follows from the definition of the pdf, the second from the fact that $X : \Omega \to \mathbb{R}$ and the third from the normalization of the probability measure $\mathbb{P}$.

Theorem – Existence of a r.v. with a prescribed CDF

Let $F : \mathbb{R} \to [0, 1]$ be a function satisfying the following properties:

- **(a)** Range and monotonicity $F : \mathbb{R} \to [0, 1]$, $F(y) \leq F(z)$ $\forall y \leq z$
- **(b)** Limits of infinity $\lim_{z \to +\infty} F(z) = 1$ $\lim_{z \to -\infty} F(z) = 0$
- **(c)** Right continuity $\lim_{\varepsilon \downarrow 0} F(z + \varepsilon) = F(z)$ $\forall z \in \mathbb{R}$

Then there exists a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a r.v. $X : \Omega \to \mathbb{R}$ such that $F_X(c) = F(c)$ $\forall c \in \mathbb{R}$

Proof

First we construct the probability space. Let $\Omega = \mathbb{R}$ and $\mathcal{F} = \mathcal{B}(\mathbb{R})$. We now introduce a measure $\mathcal{L}$ by defining $\forall a, b \in \mathbb{R}$ with $a < b$ $\mathcal{L}((a, b]) := F(b) - F(a)$. Notice that, since F is non-decreasing by property (a) we have $F(a) \leq F(b)$ whenever $a < b$. Therefore $\mathcal{L}((a, b]) = F(b) - F(a) \geq 0$. The monotonicity and right-continuity of F allow this assignment on intervals to be extended to a measure $\mathcal{L}$ on $\mathcal{B}(\mathbb{R})$ by the Lebesgue–Stieltjes measure construction. This construction is taken as a known result here; it is the same existence step used in the handwritten proof. *We now verify that $\mathcal{L}$ is a probability measure.* Since $\mathcal{L}$ is already a measure on $\mathcal{B}(\mathbb{R})$ it remains to prove that $\mathcal{L}(\mathbb{R}) = 1$. Notice that $(-n, n] \subseteq (-(n+1), n+1]$ $\forall n \in \mathbb{N}$ and $\bigcup_{n=1}^{\infty} (-n, n] = \mathbb{R}$. Therefore since $((-n, n])_{n \in \mathbb{N}}$ is an increasing sequence of Borel sets, by continuity of measures from below we obtain

$$\mathcal{L}(\mathbb{R}) = \mathcal{L}\left(\bigcup_{n=1}^{\infty} (-n, n]\right) = \lim_{n \to \infty} \mathcal{L}((-n, n]) = \lim_{n \to \infty} (F(n) - F(-n)) = 1 - 0 = 1$$

Hence $\mathcal{L}$ is a probability measure on $\mathcal{B}(\mathbb{R})$. We can therefore consider the probability space $(\Omega, \mathcal{F}, \mathbb{P}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathcal{L})$.

Second we construct the random variable. Define $X : \Omega \to \mathbb{R}$ by $X(\omega) := \omega \ \forall \omega \in \Omega$. Notice that X is identity function on $\mathbb{R}$. *We now verify that X is a r.v. namely that is $\mathcal{F}$-measurable.* Let $A \in \mathcal{B}(\mathbb{R})$. By definition of pre-image

$$X^{-1}(A) = \{\omega \in \Omega : X(\omega) \in A\} = \{\omega \in \mathbb{R} : \omega \in A\} = A$$

where the second equality follows from $\Omega = \mathbb{R}$ and $X(\omega) = \omega$. Since $A \in \mathcal{B}(\mathbb{R}) = \mathcal{F}$ we conclude that $X^{-1}(A) \in \mathcal{F}$. Since $A \in \mathcal{B}(\mathbb{R})$ was arbitrary by Definition 3.1, X is $\mathcal{F}$-measurable and therefore a random variable.

Finally we prove that the CDF of X coincides with F. Fix an arbitrary $c \in \mathbb{R}$. First notice that $(-n, c] \uparrow (-\infty, c]$ as $n \to \infty$ where we consider sufficiently large n so that $-n < c$. Therefore by continuity of measure from below

$$\mathcal{L}((-\infty, c]) = \lim_{n\to\infty} \mathcal{L}((-n, c]) = \lim_{n\to\infty} (F(c) - F(-n)) = F(c)$$

where the first equality follows from continuity of measure from below, the second from the definition of $\mathcal{L}$ and the third from property (b) since $\lim_{n\to\infty} F(-n) = 0$. Now by definition of the CDF

$$\begin{aligned} F_X(c) &= \mathcal{L}_X((-\infty, c]) = \mathbb{P}(X \le c) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \le c\}) \\ &= \mathbb{P}(\{\omega \in \mathbb{R} : \omega \le c\}) = \mathbb{P}((-\infty, c]) = \mathcal{L}((-\infty, c]) = F(c) \end{aligned}$$

where the first and second equalities follow from the definition of CDF and the law of X, the fourth from $X(\omega) = \omega$ and $\Omega = \mathbb{R}$, the sixth from $\mathbb{P} = \mathcal{L}$ and the last from calculation above. Since $c \in \mathbb{R}$ was arbitrary we conclude that $F_X(c) = F(c) \ \forall c \in \mathbb{R}$. Thus we have constructed a r.v. X whose CDF is precisely F.

Construction — Inverse Transform Method

Another way to construct a random variable with a prescribed CDF F is to start from a uniformly distributed random variable $U \sim \text{Unif}(0, 1)$.

Now, define

$$X := F^{-1}(U),$$

where F^{-1} denotes the inverse function when it exists, or the generalized inverse for a general CDF.

The idea is that applying F^{-1} to a uniform random variable transforms its distribution into the prescribed distribution F.

Thus, the inverse transform method allows us to construct a random variable with any prescribed CDF F.

Lemma 3.7 — The σ-Algebra Generated by a Random Variable

Statement

Let $Y : \Omega \to \mathbb{R}$ be a random variable, and consider the class

$$\pi(Y) := \big\{\{\omega \in \Omega : Y(\omega) \le x\} : x \in \mathbb{R}\big\} = Y^{-1}(\pi(\mathbb{R})),$$

where

$$\pi(\mathbb{R}) := \{(-\infty, x] : x \in \mathbb{R}\}.$$

Then

$$\sigma(Y) = \sigma(\pi(Y)).$$

Expanded Proof of Lemma 3.7

As we will see in HW 35, $\pi(Y)$ is a π-system on Ω.

We prove the two inclusions.

First, we prove that

$$\sigma(Y) \subseteq \sigma(\pi(Y)).$$

By definition of the pre-image of a set, for every $x \in \mathbb{R}$,

$$Y^{-1}((-\infty, x]) = \{\omega \in \Omega : Y(\omega) \in (-\infty, x]\} = \{\omega \in \Omega : Y(\omega) \leq x\} = \{Y \leq x\},$$

where the second equality follows from the fact that $Y(\omega) \in (-\infty, x]$ if and only if $Y(\omega) \leq x$.
Since $\{Y \leq x\}$ is precisely one of the sets defining $\pi(Y)$, we obtain

$$Y^{-1}((-\infty, x]) = \{Y \leq x\} \in \pi(Y).$$

Since $\sigma(\pi(Y))$ is the smallest σ-algebra containing $\pi(Y)$, we have

$$\pi(Y) \subseteq \sigma(\pi(Y)).$$

Consequently,

$$Y^{-1}((-\infty, x]) \in \sigma(\pi(Y)) \qquad \forall x \in \mathbb{R}.$$

In other words, the pre-image under Y of every set belonging to $\pi(\mathbb{R})$ belongs to $\sigma(\pi(Y))$.
Moreover, recall that, by Proposition 1.1,

$$\sigma(\pi(\mathbb{R})) = \mathcal{B}(\mathbb{R}).$$

Thus, all the assumptions of Lemma 3.2 are satisfied.
Therefore, by Lemma 3.2, Y is measurable with respect to $\sigma(\pi(Y))$, namely

$$Y \in m\sigma(\pi(Y)).$$

Now, recall that $\sigma(Y)$ is the smallest σ-algebra on Ω with respect to which Y is measurable.
Since $\sigma(\pi(Y))$ is a σ-algebra with respect to which Y is measurable, by the minimality of $\sigma(Y)$, we conclude that

$$\sigma(Y) \subseteq \sigma(\pi(Y)).$$

We now prove the reverse inclusion, namely

$$\sigma(\pi(Y)) \subseteq \sigma(Y).$$

Recall that

$$\sigma(Y) = Y^{-1}(\mathcal{B}(\mathbb{R})).$$

We first prove that

$$\pi(Y) \subseteq \sigma(Y).$$

Let $A \in \pi(Y)$.
By definition of $\pi(Y)$, there exists $x_0 \in \mathbb{R}$ such that

$$A = \{Y \leq x_0\}.$$

As shown above, by definition of pre-image,

$$A = \{Y \leq x_0\} = Y^{-1}((-\infty, x_0]).$$

Notice that $(-\infty, x_0]$ is a Borel set, since it is closed in $\mathbb{R}$. Therefore,

$$(-\infty, x_0] \in \mathcal{B}(\mathbb{R}).$$

Hence, by definition of $\sigma(Y)$,

$$A = Y^{-1}((-\infty, x_0]) \in \sigma(Y).$$

Since $A \in \pi(Y)$ was arbitrary, we conclude that

$$\pi(Y) \subseteq \sigma(Y).$$

Now, since $\sigma(Y)$ is a σ-algebra containing $\pi(Y)$, and $\sigma(\pi(Y))$ is the smallest σ-algebra containing $\pi(Y)$, we obtain

$$\sigma(\pi(Y)) \subseteq \sigma(Y).$$

Combining the two inclusions, we conclude that

$$\sigma(Y) = \sigma(\pi(Y)).$$

HW 35 — Prove That $\pi(Y)$ Is a π-System on Ω

Statement

Let $Y : \Omega \to \mathbb{R}$ be a random variable, and consider

$$\pi(Y) := \{\{Y \leq x\} : x \in \mathbb{R}\}.$$

Prove that $\pi(Y)$ is a π-system on Ω.

Proof

Recall that a π-system is a non-empty collection of sets closed under finite intersections.
It is trivial that $\pi(Y)$ is non-empty.
We now prove that $\pi(Y)$ is closed under finite intersections.
Let $A, B \in \pi(Y)$. By definition, there exist $x, y \in \mathbb{R}$ such that

$$A = \{Y \leq x\}, \qquad B = \{Y \leq y\}.$$

Recall that, as shown in HW 6,

$$(-\infty, x] \cap (-\infty, y] = (-\infty, \min\{x, y\}].$$

Therefore,

$$\begin{aligned} A \cap B &= \{Y \leq x\} \cap \{Y \leq y\} = Y^{-1}((-\infty, x]) \cap Y^{-1}((-\infty, y]) \\ &= Y^{-1}((-\infty, x] \cap (-\infty, y]) = Y^{-1}((-\infty, \min\{x, y\}]) = \{Y \leq \min\{x, y\}\}, \end{aligned}$$

where the third equality follows from HW 28-**(iii)**, and the fourth from HW 6.
Since $\min\{x, y\} \in \mathbb{R}$, by definition of $\pi(Y)$,

$$\{Y \leq \min\{x, y\}\} \in \pi(Y).$$

Hence,

$$A \cap B \in \pi(Y).$$

Since $A, B \in \pi(Y)$ were arbitrary, $\pi(Y)$ is closed under finite intersections.
Therefore,

$$\pi(Y) \text{ is a } \pi\text{-system on } \Omega.$$

The statement is proved.

Properties — Measurable Functions of Random Variables

Here are some useful facts that we state without proof.
(a) Let $Y : \Omega \to \mathbb{R}$ and $Z : \Omega \to \mathbb{R}$. We have

$$Z \in m\sigma(Y) \iff \exists f \in m\mathcal{B}(\mathbb{R}) \text{ such that } Z = f(Y),$$

where $f : \mathbb{R} \to \mathbb{R}$.
(b) Let $Y : \Omega \to \mathbb{R}^n$, i.e.,

$$Y(\omega) = (Y_1(\omega), Y_2(\omega), \ldots, Y_n(\omega)),$$

and let $Z : \Omega \to \mathbb{R}$. We have

$$Z \in m\sigma(Y) \iff \exists f \in m\mathcal{B}(\mathbb{R}^n) \text{ such that } Z = f(Y),$$

where $f : \mathbb{R}^n \to \mathbb{R}$.

(c) Let $(Y_\gamma)_{\gamma \in I}$ be a collection of functions $Y_\gamma : \Omega \to \mathbb{R}$, and let $Z : \Omega \to \mathbb{R}$. We have

$$\begin{aligned} Z \in m\sigma(Y_\gamma, \gamma \in I) \iff & \text{ there exist a countable collection} \\ & (\gamma_n)_{n\in\mathbb{N}} \text{ and a function} \\ & f \in m\mathcal{B}(\mathbb{R}^{\mathbb{N}}) \\ & \text{such that } Z = f((Y_{\gamma_n})_{n\in\mathbb{N}}), \end{aligned}$$

where $f : \mathbb{R}^{\mathbb{N}} \to \mathbb{R}$.

Financial Interpretation

Item (b) can be interpreted as follows.

Let $(Y_1, \ldots, Y_n)$ be n assets traded on the market, and let Z be the payoff of a derivative contract written on the vector $(Y_1, \ldots, Y_n)$.

Then there must exist a function f such that

$$Z = f(Y_1, \ldots, Y_n),$$

which describes the random payoff Z of the derivative contract.

4. Independence

Definition 4.1 — Independent Sigma-Algebras

Sub-sigma-algebras $\mathcal{G}$ and $\mathcal{H}$ of $\mathcal{F}$ are independent when

$$\mathbb{P}(G \cap H) = \mathbb{P}(G)\mathbb{P}(H) \qquad \forall G \in \mathcal{G},\ H \in \mathcal{H}.$$

For a family of sub-sigma-algebras, the corresponding factorisation is required for every finite selection of distinct members and one event from each selected member.

Definition 4.2 — Independent Random Variables

Random variables are independent when their generated sigma-algebras are independent. In particular, the threshold criteria below test this property using the classes $\pi(X)$ introduced in Lemma 3.7.

Definition 4.3 — Independent Events

Events are independent when the sigma-algebras generated by their indicator functions are independent. HW 36 describes each of these sigma-algebras explicitly.

HW 36 — σ-Algebra Generated by an Indicator Function

Statement

Let $E \in \mathcal{F}$. Prove that

$$\sigma(\mathbf{1}_E) = \{\emptyset, \Omega, E, E^c\}.$$

Proof

Recall that, by Lemma 3.7,

$$\sigma(\mathbf{1}_E) = \sigma(\pi(\mathbf{1}_E)),$$

where

$$\pi(\mathbf{1}_E) = \{\{\mathbf{1}_E \leq x\} : x \in \mathbb{R}\}.$$

Since the indicator function takes only the values 0 and 1, we have

$$\{\mathbf{1}_E \leq x\} = \begin{cases} \emptyset, & x < 0, \\ E^c, & 0 \leq x < 1, \\ \Omega, & x \geq 1. \end{cases}$$

Therefore,

$$\pi(\mathbf{1}_E) = \{\emptyset, E^c, \Omega\}.$$

Since every σ-algebra is closed under complements, we must also include

$$E = (E^c)^c.$$

Notice that the collection

$$\mathcal{G} := \{\emptyset, \Omega, E, E^c\}$$

is already a σ-algebra on Ω, since it contains Ω, is closed under complements and countable unions, and consequently also under intersections.
Therefore, no additional sets are needed, and we obtain

$$\sigma(\pi(\mathbf{1}_E)) = \{\emptyset, \Omega, E, E^c\}.$$

Finally, by Lemma 3.7,

$$\sigma(\mathbf{1}_E) = \sigma(\pi(\mathbf{1}_E)) = \{\emptyset, \Omega, E, E^c\}.$$

Therefore, the statement is proved.

Connection with Independence of Events

This tells us how to calculate all elements in $\sigma(\mathbf{1}_E)$. In particular, given two independent events $E, F \in \mathcal{F}$, according to the definition above we have

$$\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F),$$
$$\mathbb{P}(E^c \cap F) = \mathbb{P}(E^c)\mathbb{P}(F),$$
$$\mathbb{P}(E^c \cap F^c) = \mathbb{P}(E^c)\mathbb{P}(F^c),$$
$$\mathbb{P}(E \cap F^c) = \mathbb{P}(E)\mathbb{P}(F^c).$$

Plus analogous relations involving $\emptyset$ and Ω. So here we recognise the "old" notion of independence!
Next comes a lemma that will further allow us to deepen our understanding of independence for random variables.

Lemma 4.1 — Independence of σ-Algebras Generated by π-Systems

Statement

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $\mathcal{G}$ and $\mathcal{H}$ be sub-σ-algebras of $\mathcal{F}$, and let $\mathcal{I}$ and $\mathcal{J}$ be π-systems such that

$$\sigma(\mathcal{I}) = \mathcal{G}, \qquad \sigma(\mathcal{J}) = \mathcal{H}.$$

Then the σ-algebras $\mathcal{G}$ and $\mathcal{H}$ are independent if and only if

$$\mathbb{P}(I \cap J) = \mathbb{P}(I)\mathbb{P}(J) \qquad \forall I \in \mathcal{I},\ J \in \mathcal{J}.$$

Expanded Proof of Lemma 4.1

($\Rightarrow$) Assume that $\mathcal{G}$ and $\mathcal{H}$ are independent.
By Definition 4.1,

$$\mathbb{P}(G \cap H) = \mathbb{P}(G)\mathbb{P}(H) \qquad \forall G \in \mathcal{G},\ H \in \mathcal{H}.$$

Since

$$\mathcal{I} \subseteq \sigma(\mathcal{I}) = \mathcal{G}, \qquad \mathcal{J} \subseteq \sigma(\mathcal{J}) = \mathcal{H},$$

the equality holds, in particular, for every $I \in \mathcal{I}$ and $J \in \mathcal{J}$. Therefore,

$$\mathbb{P}(I \cap J) = \mathbb{P}(I)\mathbb{P}(J) \qquad \forall I \in \mathcal{I},\ J \in \mathcal{J}.$$

($\Leftarrow$) Assume the factorisation in the statement.
First, fix an arbitrary $I \in \mathcal{I}$, and consider the two set functions defined on $(\Omega, \mathcal{H})$ by

$$\mu_1(H) := \mathbb{P}(I \cap H), \qquad \mu_2(H) := \mathbb{P}(I)\mathbb{P}(H), \quad H \in \mathcal{H}.$$

As we will see in HW 37, these two set functions are measures on $(\Omega, \mathcal{H})$.
Notice that they have the same finite total mass. Indeed,

$$\mu_1(\Omega) = \mathbb{P}(I \cap \Omega) = \mathbb{P}(I) = \mathbb{P}(I)\mathbb{P}(\Omega) = \mu_2(\Omega) < \infty,$$

where the second equality follows from $I \cap \Omega = I$, and the third from $\mathbb{P}(\Omega) = 1$.
Moreover, we prove that the two measures coincide on the π-system $\mathcal{J}$.
Let $J \in \mathcal{J}$. By our initial assumption,

$$\mu_1(J) = \mathbb{P}(I \cap J) = \mathbb{P}(I)\mathbb{P}(J) = \mu_2(J),$$

where the second equality follows precisely from the assumption of this implication.
Since $J \in \mathcal{J}$ was arbitrary, we obtain

$$\mu_1(J) = \mu_2(J) \qquad \forall J \in \mathcal{J}.$$

Thus, the two measures have the same finite total mass and coincide on the π-system $\mathcal{J}$.
Since $\sigma(\mathcal{J}) = \mathcal{H}$, all the assumptions of Lemma 1.2 are satisfied. Therefore, by Lemma 1.2, the two measures coincide on the entire σ-algebra $\mathcal{H}$:

$$\mu_1(H) = \mu_2(H) \qquad \forall H \in \mathcal{H}.$$

Substituting their definitions, and recalling that $I \in \mathcal{I}$ was arbitrary, we conclude that

$$\mathbb{P}(I \cap H) = \mathbb{P}(I)\mathbb{P}(H) \qquad \forall I \in \mathcal{I},\ H \in \mathcal{H}.$$

We now fix an arbitrary $H \in \mathcal{H}$ and consider the two measures on $(\Omega, \mathcal{G})$ defined by

$$\nu_1(G) := \mathbb{P}(G \cap H), \qquad \nu_2(G) := \mathbb{P}(G)\mathbb{P}(H), \quad G \in \mathcal{G}.$$

By the same argument as above, these two measures have the same finite total mass:

$$\nu_1(\Omega) = \mathbb{P}(H) = \nu_2(\Omega) < \infty.$$

Moreover, by the equality established in the first part of the proof, they coincide on the π-system $\mathcal{I}$:

$$\nu_1(I) = \nu_2(I) \qquad \forall I \in \mathcal{I}.$$

Therefore, since $\sigma(\mathcal{I}) = \mathcal{G}$, by Lemma 1.2 the two measures also coincide on the entire σ-algebra $\mathcal{G}$:

$$\nu_1(G) = \nu_2(G) \qquad \forall G \in \mathcal{G}.$$

Substituting their definitions, and recalling that $H \in \mathcal{H}$ was arbitrary, we conclude that

$$\mathbb{P}(G \cap H) = \mathbb{P}(G)\mathbb{P}(H) \qquad \forall G \in \mathcal{G},\ H \in \mathcal{H}.$$

Thus, by Definition 4.1, the σ-algebras $\mathcal{G}$ and $\mathcal{H}$ are independent.

HW 37 — Prove That Two Set Functions Define Measures

Statement

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $\mathcal{H} \subseteq \mathcal{F}$ be a σ-algebra, and let $I \in \mathcal{F}$.
Prove that the two set functions

$$H \longmapsto \mathbb{P}(I \cap H), \qquad H \longmapsto \mathbb{P}(I)\mathbb{P}(H)$$

define two measures on $(\Omega, \mathcal{H})$.

Expanded Proof of HW 37

Write μ_1 and μ_2 for the first and second set functions in the statement, respectively.
First, we prove that μ_1 is a measure.
Let $H \in \mathcal{H}$. Since $\mathcal{H} \subseteq \mathcal{F}$, we have $H \in \mathcal{F}$. Moreover, since $I \in \mathcal{F}$ and $\mathcal{F}$ is closed under intersections,

$$I \cap H \in \mathcal{F}.$$

Therefore, $\mu_1(H) = \mathbb{P}(I \cap H)$ is well-defined and non-negative:

$$\mu_1(H) \geq 0 \qquad \forall H \in \mathcal{H}.$$

Furthermore,

$$\mu_1(\emptyset) = \mathbb{P}(I \cap \emptyset) = \mathbb{P}(\emptyset) = 0.$$

We now prove countable additivity.
Let $(H_n)_{n \in \mathbb{N}} \subseteq \mathcal{H}$ be a sequence of pairwise disjoint sets, namely

$$H_i \cap H_j = \emptyset \qquad \forall i \neq j.$$

Notice that the sets $I \cap H_n$ are also pairwise disjoint. Indeed, for every $i \neq j$,

$$(I \cap H_i) \cap (I \cap H_j) = I \cap (H_i \cap H_j) = I \cap \emptyset = \emptyset.$$

Moreover, since $I \in \mathcal{F}$ and $H_n \in \mathcal{H} \subseteq \mathcal{F}$, we have $I \cap H_n \in \mathcal{F}$ for every $n \in \mathbb{N}$.
Therefore,

$$\mu_1\left(\bigcup_{n=1}^{\infty} H_n\right) = \mathbb{P}\left(I \cap \bigcup_{n=1}^{\infty} H_n\right) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} (I \cap H_n)\right) = \sum_{n=1}^{\infty} \mathbb{P}(I \cap H_n) = \sum_{n=1}^{\infty} \mu_1(H_n),$$

where the second equality follows from the distributive property of intersections over unions, and the third from the countable additivity of $\mathbb{P}$, since the sets $I \cap H_n$ are measurable and pairwise disjoint.
Thus, μ_1 is a measure on $(\Omega, \mathcal{H})$.
We now prove that μ_2 is also a measure.
Since $\mathbb{P}(I) \geq 0$ and $\mathbb{P}$ is a probability measure, we immediately have

$$\mu_2(H) \geq 0 \qquad \forall H \in \mathcal{H},$$

and

$$\mu_2(\emptyset) = \mathbb{P}(I)\mathbb{P}(\emptyset) = 0.$$

Moreover, let $(H_n)_{n\in\mathbb{N}} \subseteq \mathcal{H}$ be a sequence of pairwise disjoint sets. By the countable additivity of $\mathbb{P}$,

$$\mu_2\left(\bigcup_{n=1}^{\infty} H_n\right) = \mathbb{P}(I)\mathbb{P}\left(\bigcup_{n=1}^{\infty} H_n\right) = \mathbb{P}(I)\sum_{n=1}^{\infty} \mathbb{P}(H_n) = \sum_{n=1}^{\infty} \mathbb{P}(I)\mathbb{P}(H_n) = \sum_{n=1}^{\infty} \mu_2(H_n).$$

Therefore, μ_2 is also a measure on $(\Omega, \mathcal{H})$.
We conclude that both set functions

$$\begin{aligned} \mu_1(H) &= \mathbb{P}(I \cap H), \\ \mu_2(H) &= \mathbb{P}(I)\mathbb{P}(H), \end{aligned} \qquad H \in \mathcal{H}$$

define measures on $(\Omega, \mathcal{H})$.

Corollary — Characterisation of Independence via Threshold Events

Let $X, Y : \Omega \to \mathbb{R}$ be two random variables.
Then X and Y are independent if and only if

$$\mathbb{P}(X \leq x,\ Y \leq y) = \mathbb{P}(X \leq x)\,\mathbb{P}(Y \leq y) \qquad \forall x, y \in \mathbb{R}.$$

Indeed, this is a direct application of Lemma 4.1 with

$$\mathcal{I} := \pi(X) = \big\{\{\omega \in \Omega : X(\omega) \leq x\} : x \in \mathbb{R}\big\},$$

and

$$\mathcal{J} := \pi(Y) = \big\{\{\omega \in \Omega : Y(\omega) \leq y\} : y \in \mathbb{R}\big\}.$$

Recall that, by Lemma 3.7,

$$\sigma(\pi(X)) = \sigma(X), \qquad \sigma(\pi(Y)) = \sigma(Y).$$

Therefore, the independence of the random variables X and Y is equivalent to the factorisation of the probabilities of the events $\{X \leq x\}$ and $\{Y \leq y\}$ for every $x, y \in \mathbb{R}$.

Generalisation — Independence of Finitely Many Random Variables

More generally, random variables $X_1, \dots, X_n$ are independent if and only if, for every $x_1, \dots, x_n \in \mathbb{R}$,

$$\mathbb{P}(X_1 \le x_1, \dots, X_n \le x_n) = \prod_{i=1}^{n} \mathbb{P}(X_i \le x_i).$$

This is the analogous finite-dimensional characterisation of independence, obtained by applying the same idea to the π-systems generated by the threshold events of the variables $X_1, \dots, X_n$.

HW 38 — Independence of Three σ-Algebras Generated by π-Systems

(Williams, E4.1, p. 225.)

Statement

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $\mathcal{I}_1, \mathcal{I}_2, \mathcal{I}_3$ be three π-systems such that

$$\mathcal{I}_k \subseteq \mathcal{F}, \qquad \Omega \in \mathcal{I}_k, \qquad k = 1, 2, 3.$$

Prove that $\sigma(\mathcal{I}_1)$, $\sigma(\mathcal{I}_2)$ and $\sigma(\mathcal{I}_3)$ are independent if and only if

$$\mathbb{P}(I_1 \cap I_2 \cap I_3) = \prod_{k=1}^{3} \mathbb{P}(I_k)$$

for every $I_k \in \mathcal{I}_k$, $k = 1, 2, 3$.
Why is it necessary to assume that $\Omega \in \mathcal{I}_k$, $k = 1, 2, 3$?

Part I — Expanded Proof of the Equivalence

Let

$$\mathcal{G}_k := \sigma(\mathcal{I}_k), \qquad k = 1, 2, 3.$$

($\Rightarrow$) Assume that $\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3$ are independent.
By Definition 4.1, for every $G_k \in \mathcal{G}_k$, $k = 1, 2, 3$,

$$\mathbb{P}(G_1 \cap G_2 \cap G_3) = \prod_{k=1}^{3} \mathbb{P}(G_k).$$

Since

$$\mathcal{I}_k \subseteq \sigma(\mathcal{I}_k) = \mathcal{G}_k, \qquad k = 1, 2, 3,$$

the equality holds, in particular, for every $I_k \in \mathcal{I}_k$. Therefore,

$$\mathbb{P}(I_1 \cap I_2 \cap I_3) = \prod_{k=1}^{3} \mathbb{P}(I_k).$$

($\Leftarrow$) Assume the factorisation in the statement.
First, without loss of generality, we consider $\mathcal{G}_1$ and $\mathcal{G}_2$, since the three indices play symmetric roles. In particular, since $\Omega \in \mathcal{I}_3$, we can choose $I_3 = \Omega$ in our assumption.
Therefore, for every $I_1 \in \mathcal{I}_1$ and $I_2 \in \mathcal{I}_2$,

$$\mathbb{P}(I_1 \cap I_2) = \mathbb{P}(I_1 \cap I_2 \cap \Omega) = \mathbb{P}(I_1)\mathbb{P}(I_2)\mathbb{P}(\Omega) = \mathbb{P}(I_1)\mathbb{P}(I_2),$$

where the second equality follows from our assumption, and the third from $\mathbb{P}(\Omega) = 1$.
Since $\mathcal{I}_1$ and $\mathcal{I}_2$ are π-systems generating $\mathcal{G}_1$ and $\mathcal{G}_2$, respectively, by Lemma 4.1 we conclude that

$$\mathcal{G}_1 \text{ and } \mathcal{G}_2 \text{ are independent.}$$

We now introduce the collection

$$\mathcal{I}_{1,2} := \{I_1 \cap I_2 : I_1 \in \mathcal{I}_1,\ I_2 \in \mathcal{I}_2\}.$$

We first prove that $\mathcal{I}_{1,2}$ is a π-system.
Let $A, B \in \mathcal{I}_{1,2}$. By definition, there exist $I_1, J_1 \in \mathcal{I}_1$ and $I_2, J_2 \in \mathcal{I}_2$ such that

$$A = I_1 \cap I_2, \qquad B = J_1 \cap J_2.$$

Therefore,

$$A \cap B = (I_1 \cap I_2) \cap (J_1 \cap J_2) = (I_1 \cap J_1) \cap (I_2 \cap J_2),$$

where the second equality follows from the associative and commutative properties of intersections.
Since $\mathcal{I}_1$ and $\mathcal{I}_2$ are π-systems,

$$I_1 \cap J_1 \in \mathcal{I}_1, \qquad I_2 \cap J_2 \in \mathcal{I}_2.$$

Hence,

$$A \cap B \in \mathcal{I}_{1,2}.$$

Moreover, $\Omega = \Omega \cap \Omega \in \mathcal{I}_{1,2}$, since $\Omega \in \mathcal{I}_1$ and $\Omega \in \mathcal{I}_2$.
Thus, $\mathcal{I}_{1,2}$ is a non-empty collection closed under intersections, and therefore a π-system.
Notice also that, since $\Omega \in \mathcal{I}_1$ and $\Omega \in \mathcal{I}_2$,

$$I_1 = I_1 \cap \Omega \in \mathcal{I}_{1,2} \qquad \forall I_1 \in \mathcal{I}_1,$$

and

$$I_2 = \Omega \cap I_2 \in \mathcal{I}_{1,2} \qquad \forall I_2 \in \mathcal{I}_2.$$

Therefore,

$$\mathcal{I}_1 \subseteq \mathcal{I}_{1,2}, \qquad \mathcal{I}_2 \subseteq \mathcal{I}_{1,2}.$$

Since $\sigma(\mathcal{I}_{1,2})$ is a σ-algebra containing both $\mathcal{I}_1$ and $\mathcal{I}_2$, by the minimality of generated σ-algebras we obtain

$$\mathcal{G}_1 \subseteq \sigma(\mathcal{I}_{1,2}), \qquad \mathcal{G}_2 \subseteq \sigma(\mathcal{I}_{1,2}).$$

We now prove that $\sigma(\mathcal{I}_{1,2})$ and $\mathcal{G}_3$ are independent.
Let $A \in \mathcal{I}_{1,2}$ and $I_3 \in \mathcal{I}_3$. By definition, there exist $I_1 \in \mathcal{I}_1$ and $I_2 \in \mathcal{I}_2$ such that

$$A = I_1 \cap I_2.$$

Therefore,

$$\mathbb{P}(A \cap I_3) = \mathbb{P}(I_1 \cap I_2 \cap I_3) = \mathbb{P}(I_1)\mathbb{P}(I_2)\mathbb{P}(I_3) = \mathbb{P}(I_1 \cap I_2)\mathbb{P}(I_3) = \mathbb{P}(A)\mathbb{P}(I_3),$$

where the second equality follows from our initial assumption, and the third from the independence of $\mathcal{G}_1$ and $\mathcal{G}_2$, established above.
Since $A \in \mathcal{I}_{1,2}$ and $I_3 \in \mathcal{I}_3$ were arbitrary,

$$\mathbb{P}(A \cap I_3) = \mathbb{P}(A)\mathbb{P}(I_3),$$
$$\forall A \in \mathcal{I}_{1,2},\ I_3 \in \mathcal{I}_3.$$

Since $\mathcal{I}_{1,2}$ and $\mathcal{I}_3$ are π-systems, by Lemma 4.1 we conclude that

$$\sigma(\mathcal{I}_{1,2}) \text{ and } \mathcal{G}_3 \text{ are independent.}$$

Finally, let

$$G_k \in \mathcal{G}_k, \qquad k = 1, 2, 3.$$

Since $\mathcal{G}_1, \mathcal{G}_2 \subseteq \sigma(\mathcal{I}_{1,2})$, and $\sigma(\mathcal{I}_{1,2})$ is closed under intersections, we have

$$G_1 \cap G_2 \in \sigma(\mathcal{I}_{1,2}).$$

Therefore,

$$\mathbb{P}(G_1 \cap G_2 \cap G_3) = \mathbb{P}((G_1 \cap G_2) \cap G_3) = \mathbb{P}(G_1 \cap G_2)\mathbb{P}(G_3) = \mathbb{P}(G_1)\mathbb{P}(G_2)\mathbb{P}(G_3),$$

where the second equality follows from the independence of $\sigma(\mathcal{I}_{1,2})$ and $\mathcal{G}_3$, and the third from the independence of $\mathcal{G}_1$ and $\mathcal{G}_2$.
Since $G_k \in \mathcal{G}_k$, $k = 1, 2, 3$, were arbitrary, we conclude that

$$\mathbb{P}(G_1 \cap G_2 \cap G_3) = \prod_{k=1}^{3} \mathbb{P}(G_k),$$
$$\forall G_k \in \mathcal{G}_k.$$

Thus, by Definition 4.1, the σ-algebras $\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3$ are independent.
Therefore, the equivalence is proved.

HW 38 — Part II: Why Is the Assumption $\Omega \in \mathcal{I}_k$ Necessary?

In Part I, the assumption $\Omega \in \mathcal{I}_3$ allowed us to choose $I_3 = \Omega$ and establish the independence of $\mathcal{G}_1$ and $\mathcal{G}_2$.
However, pairwise independence alone does not imply joint independence. We first illustrate this fact and then construct a counterexample showing why the assumption $\Omega \in \mathcal{I}_k$ is needed.

Counterexample 1 — Pairwise Independence Does Not Imply Joint Independence

Let X_1 and X_2 be two independent and identically distributed random variables such that

$$\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = \frac{1}{2}, \qquad i = 1, 2.$$

Define

$$X_3 := X_1 X_2.$$

We first prove that X_1, X_2, X_3 are pairwise independent.
By assumption, X_1 and X_2 are independent. Therefore, it remains to consider the pairs (X_1, X_3) and (X_2, X_3).
Notice that $X_3 = 1$ if and only if X_1 and X_2 have the same sign. Thus,

$$\{X_3 = 1\} = \big(\{X_1 = 1\} \cap \{X_2 = 1\}\big) \cup \big(\{X_1 = -1\} \cap \{X_2 = -1\}\big).$$

Since these two events are disjoint, by additivity and the independence of X_1 and X_2,

$$\begin{aligned}\mathbb{P}(X_3 = 1) &= \mathbb{P}(X_1 = 1, X_2 = 1) \quad + \mathbb{P}(X_1 = -1, X_2 = -1) \\ &= \mathbb{P}(X_1 = 1)\mathbb{P}(X_2 = 1) \quad + \mathbb{P}(X_1 = -1)\mathbb{P}(X_2 = -1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.\end{aligned}$$

We now prove that X_1 and X_3 are independent.
Since each X_i takes only the values -1 and 1, we have

$$\sigma(X_i) = \{\emptyset, \Omega, \{X_i = 1\}, \{X_i = -1\}\}, \qquad i = 1, 2, 3.$$

Therefore, the collection

$$\mathcal{J}_i := \big\{\{X_i = 1\}\big\}$$

is a π-system generating $\sigma(X_i)$. Indeed, it is closed under intersections, and its generated σ-algebra also contains

$$\{X_i = 1\}^c = \{X_i = -1\}.$$

Thus, by Lemma 4.1, it is sufficient to prove that

$$\mathbb{P}(X_1 = 1, X_3 = 1) = \mathbb{P}(X_1 = 1)\mathbb{P}(X_3 = 1).$$

Notice that, since $X_3 = X_1 X_2$,

$$\{X_1 = 1, X_3 = 1\} = \{X_1 = 1, X_2 = 1\}.$$

Indeed, when $X_1 = 1$, the equality $X_3 = 1$ holds if and only if $X_2 = 1$.
Therefore,

$$\mathbb{P}(X_1 = 1, X_3 = 1) = \mathbb{P}(X_1 = 1, X_2 = 1) = \mathbb{P}(X_1 = 1)\mathbb{P}(X_2 = 1) = \frac{1}{4} = \mathbb{P}(X_1 = 1)\mathbb{P}(X_3 = 1),$$

where the second equality follows from independence, and the last from $\mathbb{P}(X_3 = 1) = 1/2$, established above.
Hence, by Lemma 4.1, $\sigma(X_1)$ and $\sigma(X_3)$ are independent.
Similarly, exchanging X_1 and X_2, we obtain

$$\mathbb{P}(X_2 = 1, X_3 = 1) = \mathbb{P}(X_1 = 1, X_2 = 1) = \frac{1}{4} = \mathbb{P}(X_2 = 1)\mathbb{P}(X_3 = 1).$$

Therefore, again by Lemma 4.1, $\sigma(X_2)$ and $\sigma(X_3)$ are independent.
We conclude that

$$X_1, X_2, X_3 \text{ are pairwise independent.}$$

We now prove that they are not jointly independent.
Since $X_3 = X_1X_2$, we have

$$\{X_1 = 1, X_2 = 1, X_3 = 1\} = \{X_1 = 1, X_2 = 1\}.$$

Therefore,

$$\mathbb{P}(X_1 = 1, X_2 = 1, X_3 = 1) = \mathbb{P}(X_1 = 1, X_2 = 1) = \frac{1}{4}.$$

On the other hand,

$$\prod_{i=1}^{3} \mathbb{P}(X_i = 1) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

Since $\frac{1}{4} \neq \frac{1}{8}$, we obtain

$$\mathbb{P}(X_1 = 1, X_2 = 1, X_3 = 1) \neq \prod_{i=1}^{3} \mathbb{P}(X_i = 1).$$

Thus, by Definition 4.1, the three σ-algebras $\sigma(X_1)$, $\sigma(X_2)$ and $\sigma(X_3)$ are not jointly independent.
Therefore, this counterexample shows that pairwise independence does not imply joint independence.
In particular, it explains why, in Part I of HW 38, establishing the independence of $\mathcal{G}_1$ and $\mathcal{G}_2$ was not sufficient.

Counterexample 2 — Necessity of $\Omega \in \mathcal{I}_k$

We now prove that, without the assumption $\Omega \in \mathcal{I}_k$, the factorisation on the three π-systems does not necessarily imply the independence of the generated σ-algebras.
Consider the probability space

$$\Omega = \{0, 1\}, \qquad \mathcal{F} = \mathcal{P}(\Omega),$$

with

$$\mathbb{P}(\{0\}) = \mathbb{P}(\{1\}) = \frac{1}{2}.$$

Let $A := \{1\}$, so that

$$\mathbb{P}(A) = \frac{1}{2}.$$

Define

$$\mathcal{I}_1 = \mathcal{I}_2 = \{\Omega, A\}, \qquad \mathcal{I}_3 = \{\emptyset\}.$$

First, notice that these collections are π-systems.
Indeed, $\{\Omega, A\}$ is closed under intersections, since

$$\Omega \cap A = A, \qquad A \cap A = A, \qquad \Omega \cap \Omega = \Omega.$$

Moreover, $\{\emptyset\}$ is a non-empty collection closed under intersections, because

$$\emptyset \cap \emptyset = \emptyset.$$

However, although $\Omega \in \mathcal{I}_1$ and $\Omega \in \mathcal{I}_2$,

$$\Omega \notin \mathcal{I}_3.$$

We first prove that the factorisation in HW 38 holds on these three π-systems.
Let $I_k \in \mathcal{I}_k$, $k = 1, 2, 3$.
Since $\mathcal{I}_3 = \{\emptyset\}$, necessarily $I_3 = \emptyset$.
Therefore,

$$\mathbb{P}(I_1 \cap I_2 \cap I_3) = \mathbb{P}(I_1 \cap I_2 \cap \emptyset) = \mathbb{P}(\emptyset) = 0,$$

where the second equality follows from the fact that the intersection of any set with the empty set is empty.
On the other hand,

$$\mathbb{P}(I_1)\mathbb{P}(I_2)\mathbb{P}(I_3) = \mathbb{P}(I_1)\mathbb{P}(I_2) \cdot 0 = 0.$$

Consequently,

$$\mathbb{P}(I_1 \cap I_2 \cap I_3) = \prod_{k=1}^{3} \mathbb{P}(I_k).$$

for every $I_k \in \mathcal{I}_k$, $k = 1, 2, 3$.
We now prove that the generated σ-algebras are not independent.
Let

$$\mathcal{G}_k := \sigma(\mathcal{I}_k), \qquad k = 1, 2, 3.$$

Since $A \in \mathcal{I}_1$ and $A \in \mathcal{I}_2$, we have

$$A \in \mathcal{G}_1, \qquad A \in \mathcal{G}_2.$$

Moreover, although $\Omega \notin \mathcal{I}_3$, we necessarily have

$$\Omega \in \mathcal{G}_3,$$

because every σ-algebra on Ω contains Ω.
Therefore, we can choose

$$G_1 = A, \qquad G_2 = A, \qquad G_3 = \Omega.$$

If the three σ-algebras were independent, by Definition 4.1 we would have

$$\mathbb{P}(A \cap A \cap \Omega) = \mathbb{P}(A)\mathbb{P}(A)\mathbb{P}(\Omega).$$

However,

$$\mathbb{P}(A \cap A \cap \Omega) = \mathbb{P}(A) = \frac{1}{2},$$

where the first equality follows from $A \cap A \cap \Omega = A$.
On the other hand,

$$\mathbb{P}(A)\mathbb{P}(A)\mathbb{P}(\Omega) = \frac{1}{2} \cdot \frac{1}{2} \cdot 1 = \frac{1}{4}.$$

Since $\frac{1}{2} \neq \frac{1}{4}$, we obtain

$$\mathbb{P}(A \cap A \cap \Omega) \neq \mathbb{P}(A)\mathbb{P}(A)\mathbb{P}(\Omega).$$

Thus, by Definition 4.1, the σ-algebras $\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3$ are not independent.

Notice that this does not contradict the factorisation established above. Indeed, that factorisation holds only for events belonging to the original π-systems. Since $\Omega \notin \mathcal{I}_3$, we could not choose $I_3 = \Omega$ there, whereas Ω necessarily belongs to the generated σ-algebra $\mathcal{G}_3$.

Therefore, the factorisation holds on the three π-systems but fails on their generated σ-algebras.

Hence, the assumption $\Omega \in \mathcal{I}_k$, $k = 1, 2, 3$, is necessary to guarantee the equivalence in HW 38.

Therefore, the statement is proved.

Lemma 4.2 — Second Borel–Cantelli's Lemma

Statement

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$ be a sequence of independent events such that

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n) = +\infty.$$

Then

$$\mathbb{P}\left(\limsup_{n\to\infty} E_n\right) = 1.$$

In other words, with probability one, infinitely many of the events E_n occur.

Expanded Proof of Lemma 4.2

Set

$$E := \limsup_{n\to\infty} E_n.$$

By definition of the limit superior of a sequence of sets and De Morgan's laws, we have

$$E^c = \left(\bigcap_{n=1}^{\infty} \bigcup_{m\geq n} E_m\right)^c = \bigcup_{n=1}^{\infty} \left(\bigcup_{m\geq n} E_m\right)^c = \bigcup_{n=1}^{\infty} \bigcap_{m\geq n} E_m^c = \liminf_{n\to\infty} E_n^c,$$

where the second and third equalities follow from De Morgan's laws, and the last from the definition of the limit inferior.

We want to prove that $\mathbb{P}(E^c) = 0$, which will imply $\mathbb{P}(E) = 1$.

Set

$$p_n := \mathbb{P}(E_n), \qquad n \in \mathbb{N}.$$

By assumption,

$$\sum_{n=1}^{\infty} p_n = +\infty.$$

Fix an arbitrary $n \in \mathbb{N}$. We first calculate

$$\mathbb{P}\left(\bigcap_{m\geq n} E_m^c\right).$$

For every $k \geq 0$, define

$$A_k := \bigcap_{m=n}^{n+k} E_m^c.$$

Notice that $(A_k)_{k\geq 0}$ is a decreasing sequence of measurable sets. Indeed,

$$A_{k+1} = A_k \cap E_{n+k+1}^c \subseteq A_k.$$

Moreover,

$$\bigcap_{k=0}^{\infty} A_k = \bigcap_{m\geq n} E_m^c.$$

Since $\mathbb{P}$ is a probability measure, we also have

$$\mathbb{P}(A_0) \leq \mathbb{P}(\Omega) = 1 < \infty.$$

Thus, all the assumptions of Lemma 1.6 are satisfied. Therefore, by continuity of measures from above,

$$\mathbb{P}\left(\bigcap_{m\geq n} E_m^c\right) = \lim_{k\to\infty} \mathbb{P}\left(\bigcap_{m=n}^{n+k} E_m^c\right).$$

Now, since the events $(E_n)_{n\in\mathbb{N}}$ are independent, their complements are also independent. Indeed, by Definition 4.3 and HW 36,

$$E_m^c \in \sigma(\mathbf{1}_{E_m}) \qquad \forall m \in \mathbb{N},$$

and the σ-algebras generated by the indicator functions of the independent events are themselves independent.

Consequently, for every finite $k \geq 0$,

$$\mathbb{P}\left(\bigcap_{m=n}^{n+k} E_m^c\right) = \prod_{m=n}^{n+k} \mathbb{P}(E_m^c) = \prod_{m=n}^{n+k} (1 - p_m),$$

where the first equality follows from the independence of the complementary events, and the second from

$$\mathbb{P}(E_m^c) = 1 - \mathbb{P}(E_m) = 1 - p_m.$$

Combining this equality with continuity from above, we obtain

$$\mathbb{P}\left(\bigcap_{m\geq n} E_m^c\right) = \lim_{k\to\infty} \prod_{m=n}^{n+k} (1 - p_m).$$

We now use the inequality

$$1 - x \leq e^{-x}, \qquad x \geq 0.$$

Since $p_m = \mathbb{P}(E_m) \in [0, 1]$, applying this inequality with $x = p_m$ gives

$$0 \leq 1 - p_m \leq e^{-p_m}.$$

Therefore, multiplying these inequalities for $m = n, \ldots, n + k$, we obtain

$$\prod_{m=n}^{n+k} (1 - p_m) \leq \prod_{m=n}^{n+k} e^{-p_m} = \exp\left(-\sum_{m=n}^{n+k} p_m\right),$$

where the last equality follows from the properties of the exponential function.

Consequently,

$$0 \leq \mathbb{P}\left(\bigcap_{m\geq n} E_m^c\right) \leq \lim_{k\to\infty} \exp\left(-\sum_{m=n}^{n+k} p_m\right).$$

Notice that, since

$$\sum_{m=1}^{\infty} p_m = +\infty$$

and each $p_m \in [0,1]$, removing finitely many terms does not change the divergence of the series. Hence, for every fixed $n \in \mathbb{N}$,

$$\sum_{m=n}^{\infty} p_m = +\infty.$$

Therefore,

$$\lim_{k\to\infty} \exp\left(-\sum_{m=n}^{n+k} p_m\right) = 0,$$

since $e^{-t} \to 0$ as $t \to +\infty$.

Thus, by the inequality established above,

$$0 \le \mathbb{P}\left(\bigcap_{m\ge n} E_m^c\right) \le 0.$$

We conclude that

$$\mathbb{P}\left(\bigcap_{m\ge n} E_m^c\right) = 0 \qquad \forall n \in \mathbb{N}.$$

Finally, recall that we established at the beginning of the proof that

$$E^c = \bigcup_{n=1}^{\infty} \bigcap_{m\ge n} E_m^c.$$

Since each event in this countable union has probability zero, by Lemma 1.7 (a countable union of null sets is a null set), we obtain

$$\mathbb{P}(E^c) = 0.$$

Therefore,

$$\mathbb{P}(E) = 1 - \mathbb{P}(E^c) = 1.$$

Recalling that $E = \limsup_{n\to\infty} E_n$, we conclude that

$$\mathbb{P}\left(\limsup_{n\to\infty} E_n\right) = 1.$$

Therefore, the statement is proved.

Example 2 (Williams, Example 4.4) — Application of the Borel–Cantelli Lemmas Statement

Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent and identically distributed random variables such that

$$X_n \sim \mathrm{Exp}(1), \qquad n \in \mathbb{N}.$$

For $\alpha > 0$, determine the probability of the event

$$E_\alpha := \{\omega \in \Omega : X_n(\omega) > \alpha \log n, \text{ i.o.}\},$$

where i.o. means infinitely often. Then study the almost sure behaviour of

$$L(\omega) := \limsup_{n\to\infty} \frac{X_n(\omega)}{\log n}.$$

Expanded Solution of Example 2

Fix $\alpha > 0$, and define

$$E_n^\alpha := \{\omega \in \Omega : X_n(\omega) > \alpha \log n\}, \qquad n \in \mathbb{N}.$$

The survival function of X_n satisfies

$$\mathbb{P}(X_n > t) = e^{-t}, \qquad t \geq 0.$$

This follows from $X_n \sim \mathrm{Exp}(1)$. Consequently,

$$\mathbb{P}(E_n^\alpha) = \mathbb{P}(X_n > \alpha \log n) = e^{-\alpha \log n} = n^{-\alpha}.$$

The first equality uses the definition of E_n^α, the second the exponential survival function (since $\alpha \log n \geq 0$), and the last the identity $e^{a \log n} = n^a$.

Moreover,

$$E_n^\alpha = X_n^{-1}((\alpha \log n, \infty)) \in \sigma(X_n) \subseteq \mathcal{F}.$$

The equality is the definition of a pre-image; the membership follows from the definition of $\sigma(X_n)$, and the final inclusion from the $\mathcal{F}$-measurability of X_n. Since the X_n are independent, the events $(E_n^\alpha)_{n\in\mathbb{N}}$ are independent by Definition 4.2.

We now express E_α as a limit superior:

$$\begin{aligned} E_\alpha &= \{\omega \in \Omega : X_n(\omega) > \alpha \log n \text{ for infinitely many } n\} \\ &= \{\omega \in \Omega : \forall N \in \mathbb{N},\ \exists m \geq N \text{ such that } X_m(\omega) > \alpha \log m\} \\ &= \bigcap_{N=1}^{\infty} \bigcup_{m \geq N} \{\omega \in \Omega : X_m(\omega) > \alpha \log m\} = \bigcap_{N=1}^{\infty} \bigcup_{m \geq N} E_m^\alpha = \limsup_{n\to\infty} E_n^\alpha. \end{aligned}$$

The first equality states the definition of E_α. The second expresses infinitely often: after every N there must be some $m \geq N$ for which the inequality holds. The third translates "for every N" into an intersection and "there exists $m \geq N$" into a union. The fourth substitutes the definition of E_m^α, and the last is the definition of the limit superior of sets.

We will use formula (4.3):

$$\mathbb{P}(E_\alpha) = \begin{cases} 0, & \alpha > 1, \\ 1, & \alpha \in (0, 1]. \end{cases}$$

As we will prove in HW 39, this follows from the two Borel–Cantelli lemmas and the convergence criterion for the p-series $\sum_{n=1}^{\infty} n^{-\alpha}$.

We now consider

$$L(\omega) := \limsup_{n\to\infty} \frac{X_n(\omega)}{\log n},$$

where the ratios are defined for $n \geq 2$, because $\log 1 = 0$. For each $n \geq 2$, the function $X_n/\log n$ is measurable, since X_n is measurable and $\log n$ is a positive constant. Moreover,

$$L = \inf_{N\geq 2} \sup_{n \geq N} \frac{X_n}{\log n}.$$

This is the definition of the numerical limit superior; since countable suprema and infima of measurable functions are measurable, L is an extended-real-valued measurable random variable.

We first prove that $\mathbb{P}(L \geq 1) = 1$.

Let $\omega_0 \in E_1$. We have

$$\frac{X_n(\omega_0)}{\log n} > 1 \qquad \text{for infinitely many } n \geq 2.$$

Indeed, the definition of E_1 gives $X_n(\omega_0) > \log n$ infinitely often, and division by $\log n > 0$ preserves the inequality. Consequently,

$$L(\omega_0) \geq 1.$$

This follows from the definition of the limit superior: a sequence exceeding 1 infinitely often cannot have limit superior below 1. Since $\omega_0 \in E_1$ was arbitrary,

$$E_1 \subseteq \{L \geq 1\}.$$

We therefore obtain

$$1 \geq \mathbb{P}(L \geq 1) \geq \mathbb{P}(E_1) = 1.$$

The first bound holds for every probability, the second inequality follows from the inclusion just proved and monotonicity of probability, and the last equality uses formula (4.3) with $\alpha = 1$. Hence $\mathbb{P}(L \geq 1) = 1$.

We now prove that $\mathbb{P}(L > 1) = 0$.

First, notice that

$$\{L > 1\} = \bigcup_{k=1}^{\infty} \left\{ L \geq 1 + \frac{2}{k} \right\} = \bigcup_{k=1}^{\infty} A_k,$$

where, for each fixed $k \in \mathbb{N}$, we define

$$A_k := \left\{ L \geq 1 + \frac{2}{k} \right\}.$$

The first equality follows from $2/k \to 0$ and the Archimedean property of the real numbers (including the case $L = +\infty$); the second follows from the definition of A_k.

Now fix $k \in \mathbb{N}$, and define

$$B_k := \left\{ X_n > \left(1 + \frac{1}{k}\right) \log n, \text{ i.o.} \right\}.$$

We want to show that

$$A_k \subseteq B_k \quad \Longleftrightarrow \quad A_k \cap B_k^c = \emptyset.$$

The equivalence holds because an inclusion $A_k \subseteq B_k$ means that no outcome can belong to A_k without belonging to B_k.

Suppose, for a contradiction, that there exists

$$\omega_0 \in A_k \cap B_k^c.$$

From the membership $\omega_0 \in A_k$, we obtain

$$L(\omega_0) \geq 1 + \frac{2}{k}.$$

This is the definition of A_k. On the other hand,

$$X_n(\omega_0) > \left(1 + \frac{1}{k}\right) \log n \qquad \text{only finitely often.}$$

This follows from $\omega_0 \in B_k^c$: the outcome does not belong to the event where the inequality holds infinitely often. Thus, for some $N \geq 2$,

$$X_n(\omega_0) \leq \left(1 + \frac{1}{k}\right) \log n \qquad \forall n \geq N.$$

Indeed, after the last index (if any) at which the strict inequality holds, its negation must hold for every subsequent index. Dividing yields

$$\frac{X_n(\omega_0)}{\log n} \leq 1 + \frac{1}{k} \qquad \forall n \geq N.$$

The inequality is preserved because $\log n > 0$ for $n \geq 2$. Therefore,

$$L(\omega_0) \leq 1 + \frac{1}{k}.$$

This follows from the definition of limit superior, since every sufficiently late term is bounded by $1 + 1/k$, and contradicts the membership $\omega_0 \in A_k$ established above. Thus,

$$A_k \cap B_k^c = \emptyset, \qquad A_k \subseteq B_k.$$

The contradiction rules out any outcome in $A_k \cap B_k^c$, which is equivalent to the desired inclusion.

Consequently,

$$\mathbb{P}(A_k) \le \mathbb{P}(B_k) = \mathbb{P}(E_{1+1/k}) = 0.$$

The inequality follows from $A_k \subseteq B_k$ and monotonicity of probability, the first equality from the definition of E_α, and the last from formula (4.3), since $1 + 1/k > 1$.

Since k was arbitrary, we obtain

$$\mathbb{P}(L > 1) = \mathbb{P}\left(\bigcup_{k=1}^{\infty} A_k\right) = 0.$$

The first equality uses the identity for $\{L > 1\}$, and the second follows from Lemma 1.7 because each A_k has probability zero.

Finally,

$$\mathbb{P}(L = 1) = \mathbb{P}(L \ge 1) - \mathbb{P}(L > 1) = 1 - 0 = 1.$$

The first equality follows from the disjoint decomposition $\{L \ge 1\} = \{L = 1\} \cup \{L > 1\}$ and additivity of probability; the other equalities use the two probabilities already established. Recalling the definition of L, we conclude that

$$\mathbb{P}\left(\limsup_{n\to\infty} \frac{X_n}{\log n} = 1\right) = 1.$$

HW 39 — Proof of Formula (4.3)

Statement

Let $(X_n)_{n\in\mathbb{N}}$ be independent and identically distributed random variables with

$$X_n \sim \text{Exp}(1).$$

For $\alpha > 0$, let

$$E_\alpha := \{X_n > \alpha \log n, \text{ i.o.}\}.$$

Prove formula (4.3):

$$\mathbb{P}(E_\alpha) = \begin{cases} 0, & \alpha > 1, \\ 1, & \alpha \in (0, 1]. \end{cases}$$

Expanded Proof of HW 39

Fix $\alpha > 0$, and consider

$$E_n^\alpha := \{X_n > \alpha \log n\}, \qquad n \in \mathbb{N}.$$

As established in Example 2, these events are measurable and independent, and

$$\mathbb{P}(E_n^\alpha) = n^{-\alpha}, \qquad E_\alpha = \limsup_{n\to\infty} E_n^\alpha.$$

The first identity was obtained from the exponential survival function; the second from the meaning of infinitely often and the definition of the limit superior. Hence,

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n^\alpha) = \sum_{n=1}^{\infty} \frac{1}{n^\alpha}.$$

The equality follows by substituting the probability of each E_n^α.

If $\alpha > 1$, the p-series criterion gives

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n^\alpha) < \infty.$$

Consequently,

$$\mathbb{P}(E_\alpha) = \mathbb{P}\left(\limsup_{n\to\infty} E_n^\alpha\right) = 0.$$

The first equality uses the representation of E_α, and the second follows from Lemma 2.3 (First Borel–Cantelli), whose assumptions hold because the events are measurable and their probabilities have finite sum. Independence is not required here.

If $0 < \alpha \leq 1$, the same p-series diverges, so

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n^\alpha) = +\infty.$$

At $\alpha = 1$ this is the harmonic series. Therefore,

$$\mathbb{P}(E_\alpha) = \mathbb{P}\left(\limsup_{n\to\infty} E_n^\alpha\right) = 1.$$

The first equality uses the same representation of E_α, and the second follows from Lemma 4.2 (Second Borel–Cantelli), because the events are independent and the sum of their probabilities diverges.

Both cases of (4.3) follow.

Fundamental Question for Modelling

So far, we have considered sequences of random variables with arbitrary distributions. A natural question is whether we can actually construct a sequence of independent random variables $(X_n)_{n\in\mathbb{N}}$, each with a prescribed CDF F_n.

The answer is yes. Indeed, let $(U_n)_{n\in\mathbb{N}}$ be a sequence of i.i.d. random variables such that

$$U_n \sim \text{Unif}(0,1).$$

Define

$$X_n := F_n^{-1}(U_n), \qquad n \in \mathbb{N},$$

where F_n^{-1} denotes the generalised inverse of the CDF F_n.

Then

$$X_n \sim F_n,$$

by the inverse transform method. Moreover, the random variables $(X_n)_{n\in\mathbb{N}}$ are independent, since each X_n is a measurable function of U_n, and measurable transformations of independent random variables preserve independence.

Stochastic Processes

A stochastic process on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a family of random variables

$$(Y_t)_{t\in\mathcal{I}},$$

indexed by a parameter t, usually interpreted as time.

For example, the index set $\mathcal{I}$ may be $\mathbb{N}$, corresponding to discrete time, or $[0,\infty)$, corresponding to continuous time.

Existence of Stochastic Processes

A natural question is whether a stochastic process can be constructed with a prescribed distributional structure.

Under suitable consistency conditions, the answer is yes, as established by the Kolmogorov Existence Theorem.

The technical details of this theorem are beyond the scope of the present discussion.

Example 3 — Markov Chains

Statement

Let E be a finite or countable set, and let $P = (p_{ij})_{i,j\in E}$ be a square matrix with positive entries satisfying

$$\sum_{j\in E} p_{ij} = 1, \qquad \forall i \in E.$$

Such a matrix is called a stochastic matrix.

Let μ be a probability measure on E, and set

$$\mu_i := \mu(\{i\}), \qquad i \in E.$$

A time-homogeneous Markov chain $Z = (Z_n)_{n\in\mathbb{N}\cup\{0\}}$ on E, with initial distribution μ and transition matrix P, is a stochastic process satisfying

$$\mathbb{P}(Z_0 = i_0, Z_1 = i_1, \dots, Z_n = i_n) = \mu_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n}$$

for every $n \in \mathbb{N}$ and every choice of states $i_0, \dots, i_n \in E$.

We want to construct such a process using independent random variables.

Expanded Construction and Proof of Example 3

First, let Z_0 be an E-valued random variable with distribution μ, so that

$$\mathbb{P}(Z_0 = i) = \mu_i, \qquad \forall i \in E.$$

Next, consider a family of E-valued random variables

$$(Y_{i,n})_{i\in E,\ n\in\mathbb{N}}$$

such that

$$\mathbb{P}(Y_{i,n} = j) = p_{ij}, \qquad \forall i, j \in E,\ n \in \mathbb{N}.$$

We assume that all the random variables $Y_{i,n}$ are mutually independent and also independent of Z_0.

Notice that this independence is an assumption of our initial construction: **the random variables $Y_{i,n}$ are chosen independently before defining the process Z.**

For each fixed state i and time n, the random variable $Y_{i,n}$ takes values in E, according to the probabilities p_{ij}. Thus, the index i identifies which random variable we are considering, whereas j represents one of its possible values.

We now define the process recursively by setting

$$Z_n(\omega) := Y_{Z_{n-1}(\omega),n}(\omega), \qquad n \geq 1.$$

This means that the value of $Z_{n-1}(\omega)$ determines which random variable $Y_{i,n}$ is selected to obtain $Z_n(\omega)$.

In particular, if $Z_0(\omega) = i_0$, then

$$Z_1(\omega) = Y_{i_0,1}(\omega).$$

The equality follows from the recursive definition with $n = 1$, after substituting $Z_0(\omega) = i_0$.

Notice that knowing $Z_0(\omega) = i_0$ identifies the random variable $Y_{i_0,1}$, **but does not determine its realised value**. We only know that

$$\mathbb{P}(Y_{i_0,1} = j) = p_{i_0 j}, \qquad j \in E.$$

If we additionally consider an outcome ω for which $Z_1(\omega) = i_1$, then

$$Z_2(\omega) = Y_{i_1,2}(\omega).$$

This follows from the same recursive definition with $n = 2$. Again, the condition $Z_1(\omega) = i_1$ identifies the variable used at the second step, without specifying its realised value.

Notice also that, although the variables $Y_{i,n}$ are independent, the resulting variables Z_n are generally not independent, since the selection of the variable used at time n depends on the previous state Z_{n-1}.

Nevertheless, the construction defines measurable random variables. Indeed, for every $j \in E$ and $n \geq 1$,

$$\{Z_n = j\} = \bigcup_{i \in E} \left(\{Z_{n-1} = i\} \cap \{Y_{i,n} = j\}\right).$$

The equality follows from the recursive definition: $Z_n = j$ precisely when the previous state equals some $i \in E$ and the corresponding variable $Y_{i,n}$ takes the value j. Since E is finite or countable, the right-hand side is a countable union of measurable events. Starting from Z_0, this proves the measurability of each Z_n by induction.

We now verify that the constructed process has the required joint distribution.

We first consider the first two transitions.

Fix arbitrary states $i_0, i_1, i_2 \in E$. We obtain

$$\begin{aligned}
&\{Z_0 = i_0, Z_1 = i_1, Z_2 = i_2\} \\
&\quad = \{Z_0 = i_0, Y_{i_0,1} = i_1, Z_2 = i_2\} \\
&\quad = \{Z_0 = i_0, Y_{i_0,1} = i_1, Y_{i_1,2} = i_2\}.
\end{aligned}$$

The first equality follows from the recursive definition: on the event $\{Z_0 = i_0\}$, we have $Z_1 = Y_{i_0,1}$. Therefore, requiring $Z_1 = i_1$ is equivalent to requiring $Y_{i_0,1} = i_1$.

The second equality follows by applying the same reasoning to the next step. Once the first two conditions hold, we have $Z_1 = i_1$, and therefore $Z_2 = Y_{i_1,2}$. Hence, requiring $Z_2 = i_2$ is equivalent to requiring $Y_{i_1,2} = i_2$.

The important point is that i_0, i_1, i_2 are fixed states. We are considering the event in which the process follows the particular path

$$i_0 \longrightarrow i_1 \longrightarrow i_2.$$

Consequently, the event on the right-hand side involves the three specific random variables

$$Z_0, \qquad Y_{i_0,1}, \qquad Y_{i_1,2},$$

which are independent by our initial assumption. We are not using independence of Z_0, Z_1, Z_2.

Therefore,

$$\begin{aligned}
&\mathbb{P}(Z_0 = i_0, Z_1 = i_1, Z_2 = i_2) \\
&\quad = \mathbb{P}(Z_0 = i_0, Y_{i_0,1} = i_1, Y_{i_1,2} = i_2) \\
&\quad = \mathbb{P}(Z_0 = i_0)\mathbb{P}(Y_{i_0,1} = i_1)\mathbb{P}(Y_{i_1,2} = i_2) \\
&\quad = \mu_{i_0} p_{i_0 i_1} p_{i_1 i_2}.
\end{aligned}$$

The first equality follows from the equality of events established above. The second follows from the independence of $Z_0, Y_{i_0,1}, Y_{i_1,2}$, and the last from their prescribed distributions.

We now extend the argument to an arbitrary number of transitions.

Fix $n \in \mathbb{N}$ and arbitrary states $i_0, \ldots, i_n \in E$.

Repeatedly applying the recursive definition gives

$$\begin{aligned}
&\{Z_0 = i_0, Z_1 = i_1, \ldots, Z_n = i_n\} \\
&\quad = \{Z_0 = i_0\} \cap \bigcap_{r=1}^{n} \{Y_{i_{r-1},r} = i_r\}.
\end{aligned}$$

This follows from the same argument used for the first two transitions. For a fixed path, the condition $Z_{r-1} = i_{r-1}$ implies that the variable selected at time r is $Y_{i_{r-1},r}$. Therefore, requiring $Z_r = i_r$ is equivalent to requiring $Y_{i_{r-1},r} = i_r$.

Since the states $i_0, \dots, i_n$ are fixed, the event on the right-hand side involves the random variables

$$Z_0,\, Y_{i_0,1},\, Y_{i_1,2},\, \dots,\, Y_{i_{n-1},n}.$$

These are independent by construction. In particular, the selected Y-variables are distinct because their time indices are different, even if some of the states coincide.

Consequently,

$$\begin{aligned}
&\mathbb{P}(Z_0 = i_0, Z_1 = i_1, \dots, Z_n = i_n) \\
&\quad = \mathbb{P}\left(\{Z_0 = i_0\} \cap \bigcap_{r=1}^{n} \{Y_{i_{r-1},r} = i_r\}\right) \\
&\quad = \mathbb{P}(Z_0 = i_0) \prod_{r=1}^{n} \mathbb{P}(Y_{i_{r-1},r} = i_r) \\
&\quad = \mu_{i_0} \prod_{r=1}^{n} p_{i_{r-1} i_r} \\
&\quad = \mu_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n}.
\end{aligned}$$

The first equality follows from the representation of the event just established. The second follows from the independence of the selected random variables, the third from their prescribed distributions, and the last from expanding the finite product.

Since $n \in \mathbb{N}$ and the states $i_0, \dots, i_n \in E$ were arbitrary, we have proved that

$$\mathbb{P}(Z_0 = i_0, Z_1 = i_1, \dots, Z_n = i_n) = \mu_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n}.$$

Thus, the constructed process satisfies the required joint distribution and is a time-homogeneous Markov chain with initial distribution μ and transition matrix P.

Therefore, the statement is proved.

A Monkey Types Shakespeare

Statement and construction

Consider a fixed text, such as the complete works of William Shakespeare (WS), consisting of $N \in \mathbb{N}$ symbols in a specified order. Write the target sequence as

$$(w_1, w_2, \dots, w_N).$$

Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of independent random variables, where X_n is the symbol typed by the monkey at the n-th keystroke. We assume that each symbol needed to type WS has a probability uniformly bounded below by a constant $\varepsilon > 0$ at every keystroke:

$$\mathbb{P}(X_n = x) \geq \varepsilon > 0 \qquad \text{for every } n \in \mathbb{N} \text{ and every symbol } x \text{ appearing in WS.}$$

This makes explicit the uniform-in-time meaning of the positive lower bound used in the lecture notes. No assumption that the X_n have identical distributions is needed.

For $s \in \mathbb{N} \cup \{0\}$, let

$$C_s := \bigcap_{r=1}^{N} \{X_{s+r} = w_r\}.$$

The event C_s means that a complete copy of WS starts at keystroke $s+1$ and ends at keystroke $s+N$. Different occurrences may overlap; each occurrence is identified by its starting position.

Following the lecture notes, introduce the events

$$\begin{aligned} H &:= \{\text{the monkey types infinitely many copies of WS}\},\\ H_k &:= \{\text{the monkey types at least } k \text{ copies of WS}\},\\ H_{m,k} &:= \{\text{the monkey types at least } k \text{ copies of WS by time } m\},\\ H^{(m)} &:= \{\text{the monkey types infinitely many copies of WS after time } m\}. \end{aligned}$$

Here $H_{m,k}$ concerns copies completed no later than the m-th keystroke, whereas $H^{(m)}$ concerns copies whose starting positions are after that keystroke. In particular, $H_{m,k}$ depends only on $X_1, \dots, X_m$, while $H^{(m)}$ depends only on $X_{m+1}, X_{m+2}, \dots$. These are measurable events: each occurrence C_s is a finite intersection of events determined by random variables, and the conditions involving finitely or infinitely many occurrences can be expressed through countable unions and intersections.

We want to prove

$$\mathbb{P}(H) = 1.$$

Expanded proof

We first show that $\mathbb{P}(H)$ *is either* 0 *or* 1.

Fix $m, k \in \mathbb{N}$. We obtain

$$\mathbb{P}\big(H_{m,k} \cap H^{(m)}\big) = \mathbb{P}(H_{m,k})\mathbb{P}\big(H^{(m)}\big).$$

The equality follows from independence: $H_{m,k}$ is determined by the first m symbols, whereas $H^{(m)}$ is determined by symbols typed strictly after time m; the corresponding families of random variables are independent.

As we will see in HW 40, deleting finitely many possible starting positions does not change whether infinitely many copies occur, so $H^{(m)} = H$. Consequently,

$$\mathbb{P}(H_{m,k} \cap H) = \mathbb{P}(H_{m,k})\mathbb{P}(H).$$

This equality follows by substituting $H^{(m)} = H$ into the preceding identity.

As we will also see in HW 40,

$$H_{m,k} \uparrow H_k \quad \text{as } m \to \infty, \qquad H_k \cap H = H.$$

Therefore,

$$\mathbb{P}(H) = \mathbb{P}(H_k \cap H) = \lim_{m\to\infty} \mathbb{P}(H_{m,k} \cap H) = \lim_{m\to\infty} \big[\mathbb{P}(H_{m,k})\mathbb{P}(H)\big] = \mathbb{P}(H_k)\mathbb{P}(H).$$

The first equality uses $H_k \cap H = H$. The second follows from Lemma 1.5 (continuity from below), since $H_{m,k} \cap H \uparrow H_k \cap H$. The third uses the independence identity already established, and the last follows from continuity from below applied to $H_{m,k} \uparrow H_k$.

As we will prove in HW 40,

$$H_k \downarrow H \quad \text{as } k \to \infty.$$

Consequently,

$$\mathbb{P}(H) = \mathbb{P}(H)^2.$$

The equality follows by taking $k \to \infty$ in $\mathbb{P}(H) = \mathbb{P}(H_k)\mathbb{P}(H)$ and applying Lemma 1.6 (continuity from above). Since $\mathbb{P}(H) \in [0, 1]$, we conclude that

$$\mathbb{P}(H) \in \{0, 1\}.$$

Indeed, the equation $p = p^2$ is equivalent to $p(1 - p) = 0$.

We now prove that $\mathbb{P}(H) = 1$.

Divide the successive keystrokes into disjoint blocks of length N. For every $k \in \mathbb{N}$, define

$$E_k := \bigcap_{r=1}^{N} \{X_{(k-1)N+r} = w_r\}.$$

Thus E_k is the event that the monkey types an entire copy of WS in the k-th block, consisting precisely of keystrokes $(k-1)N+1, \ldots, kN$.

We obtain

$$\mathbb{P}(E_k) = \prod_{r=1}^{N} \mathbb{P}(X_{(k-1)N+r} = w_r) \geq \varepsilon^N > 0.$$

The equality follows from independence of the N random variables used in the block, while the inequality follows from the uniform lower bound on the probability of each required symbol. In particular, the same positive bound holds for every k.

Consequently,

$$\sum_{k=1}^{\infty} \mathbb{P}(E_k) \geq \sum_{k=1}^{\infty} \varepsilon^N = +\infty.$$

The inequality follows from the preceding bound, and the equality holds because the sum contains infinitely many copies of the fixed positive constant ε^N.

Moreover, the events $(E_k)_{k\in\mathbb{N}}$ are independent, because distinct blocks involve disjoint sets of the independent random variables $(X_n)_{n\in\mathbb{N}}$. Thus all the assumptions of Lemma 4.2, the Second Borel–Cantelli Lemma, are satisfied, and we obtain

$$\mathbb{P}\left(\limsup_{k\to\infty} E_k\right) = 1.$$

As we will see in HW 40,

$$\limsup_{k\to\infty} E_k \subseteq H.$$

Therefore,

$$1 = \mathbb{P}\left(\limsup_{k\to\infty} E_k\right) \leq \mathbb{P}(H) \leq 1.$$

The first equality is the conclusion of the Second Borel–Cantelli Lemma, the first inequality follows from the inclusion and monotonicity of probability, and the last inequality follows from $H \in \mathcal{F}$.

Hence,

$$\mathbb{P}(H) = 1.$$

Therefore, the statement is proved.

HW 40 — Set-Theoretic Verification of the Events in the Monkey Example

Statement

With the notation introduced in A Monkey Types Shakespeare, prove that

- **(i)** $H = H^{(m)}$;
- **(ii)** $H_k \cap H = H$;
- **(iii)** $\limsup_{k\to\infty} E_k \subseteq H$;
- **(iv)** $H_k = \bigcup_{m=1}^{\infty} H_{m,k}$;
- **(v)** $H = \bigcap_{k=1}^{\infty} H_k$.

Expanded proof of HW 40

Recall that $C_s := \bigcap_{r=1}^{N} \{X_{s+r} = w_r\}$, $s \in \mathbb{N}\cup\{0\}$, is the event that the monkey types one complete copy of WS starting at keystroke $s+1$.

(i) Recall that H is the event of infinitely many occurrences of WS in the entire sequence, while $H^{(m)}$ concerns infinitely many occurrences whose first keystrokes are strictly after time m. We also allow $m = 0$.

For every fixed threshold $q \geq 0$, consider the event $\bigcup_{s \geq q} C_s$. This is the event that at least one complete copy of WS starts at a keystroke $s+1$ with $s \geq q$. Here s labels a possible starting position and q is a variable threshold; all indices are nonnegative integers.

For fixed $m \geq 0$, we have

$$H^{(m)} = \bigcap_{q=m}^{\infty} \bigcup_{s \geq q} C_s,$$

$$\omega \in H^{(m)} \iff \forall q \geq m,\ \exists s \geq q :\ \omega \in C_s.$$

The outer intersection requires an occurrence beyond every threshold $q \geq m$, which is precisely the meaning of infinitely many occurrences after time m.

In particular,

$$H = H^{(0)} = \limsup_{s \to \infty} C_s.$$

The first equality uses the case $m = 0$, when the entire sequence is considered; the second is the definition of the limit superior of events.

Fix $m \in \mathbb{N}$. *First, we prove that $H^{(m)} \subseteq H$.*

For every $0 \leq q < m$, we have

$$H^{(m)} \subseteq \bigcup_{s \geq m} C_s \subseteq \bigcup_{s \geq q} C_s.$$

The first inclusion follows by taking the threshold m in the intersection defining $H^{(m)}$; the second follows because $s \geq m$ implies $s \geq q$. Hence,

$$H^{(m)} \subseteq H^{(0)} = H.$$

The inclusion holds because the conditions with $q < m$ follow from the preceding inclusions, while those with $q \geq m$ already enter the definition of $H^{(m)}$; the equality was established above.

Conversely, we prove that $H \subseteq H^{(m)}$ by contradiction.

Recall that

$$H \subseteq H^{(m)} \quad \iff \quad H \cap \left(H^{(m)}\right)^c = \emptyset.$$

The equivalence follows because the inclusion means that no outcome can belong to H without belonging to $H^{(m)}$.

Suppose, for a contradiction, that there exists

$$\omega_0 \in H \cap \left(H^{(m)}\right)^c.$$

Since $\omega_0 \in H$, the monkey types infinitely many copies of WS in the entire sequence. On the other hand, since $\omega_0 \in \left(H^{(m)}\right)^c$, only finitely many copies start strictly after time m.

However, at most m distinct copies can start at or before time m, corresponding to the starting indices $s = 0, \ldots, m-1$. Therefore, infinitely many copies in total and only finitely many copies starting after time m are incompatible.

We now express this contradiction formally.

We obtain

$$\left(H^{(m)}\right)^c = \left(\bigcap_{q=m}^{\infty} \bigcup_{s \geq q} C_s \right)^c = \bigcup_{q=m}^{\infty} \bigcap_{s \geq q} C_s^c.$$

The first equality uses the representation of $H^{(m)}$, while the second follows from De Morgan's laws.

Consequently, since $\omega_0 \in \left(H^{(m)}\right)^c$, there exists $q_0 \geq m$ such that

$$\omega_0 \notin C_s \qquad \forall s \geq q_0.$$

By the definitions of union and intersection, membership in the preceding event gives such a threshold q_0. This is the meaning of finitely often: no further copy starts beyond that threshold.

On the other hand, since $\omega_0 \in H$, we obtain

$$\omega_0 \in C_{s_0} \qquad \text{for some } s_0 \geq q_0.$$

This follows by taking $q = q_0$ in the membership characterisation of H, since $q_0 \geq m \geq 0$. It contradicts the fact that $\omega_0 \notin C_s$ for every $s \geq q_0$.
Hence,

$$H \cap \left(H^{(m)}\right)^c = \emptyset,$$

and therefore

$$H \subseteq H^{(m)}.$$

Combining the two inclusions, we conclude that

$$H = H^{(m)}.$$

□

(iv) We first express the two events in terms of the C_s:

$$H_k = \bigcup_{0 \leq s_1 < \cdots < s_k} \bigcap_{j=1}^{k} C_{s_j}$$

and

$$H_{m,k} = \bigcup_{\substack{0 \leq s_1 < \cdots < s_k \\ s_k + N \leq m}} \bigcap_{j=1}^{k} C_{s_j}.$$

In the first expression, the intersection requires k distinct occurrences of WS, while the union considers every possible selection of their starting positions. In the second, the additional restriction $s_k + N \leq m$ requires all selected copies to be completed by time m.
Consequently,

$$\bigcup_{m=1}^{\infty} H_{m,k} = \bigcup_{m=1}^{\infty} \bigcup_{\substack{0 \leq s_1 < \cdots < s_k \\ s_k + N \leq m}} \bigcap_{j=1}^{k} C_{s_j} = \bigcup_{0 \leq s_1 < \cdots < s_k} \bigcap_{j=1}^{k} C_{s_j} = H_k.$$

The first equality substitutes the definition of $H_{m,k}$. The second follows because, for every finite selection of starting positions, we can choose $m \geq s_k + N$, so that selection eventually appears in the union. The last equality uses the representation of H_k.
Moreover, $H_{m,k} \subseteq H_{m+1,k}$, because increasing the deadline from m to $m+1$ does not exclude any previously completed copies.
Therefore,

$$H_{m,k} \uparrow H_k \text{ as } m \to \infty.$$

(v) Using complements, we obtain

$$\left(\bigcap_{k=1}^{\infty} H_k\right)^c = \bigcup_{k=1}^{\infty} H_k^c = \bigcup_{q=0}^{\infty} \bigcap_{s \geq q} C_s^c = \left(\bigcap_{q=0}^{\infty} \bigcup_{s \geq q} C_s\right)^c = H^c.$$

The first equality follows from De Morgan's law. The second expresses the equivalence between having fewer than k occurrences for some k and having only finitely many occurrences, which means that from some position q onward no further C_s occurs. The third equality follows again from De Morgan's law, while the last uses the representation of H established above.
Taking complements, we conclude that

$$H = \bigcap_{k=1}^{\infty} H_k.$$

Moreover, $H_{k+1} \subseteq H_k$, because having at least $k+1$ copies implies having at least k copies. Hence,

$$H_k \downarrow H \text{ as } k \to \infty.$$

(ii) In particular,

$$H \subseteq H_k,\ H_k \cap H = H.$$

The inclusion follows because an intersection is contained in each of its constituent sets, while the equality follows from the identity $A \cap B = B$ whenever $B \subseteq A$.

(iii) Recall that $E_k = C_{(k-1)N}$. The equality follows because E_k requires one complete copy of WS in the k-th block of N keystrokes.

For every $r \in \mathbb{N}$,

$$\bigcup_{k \geq r} E_k = \bigcup_{k \geq r} C_{(k-1)N} \subseteq \bigcup_{s \geq (r-1)N} C_s.$$

The equality substitutes the definition of E_k, while the inclusion follows because the positions $(k-1)N$, for $k \geq r$, form a subset of all starting positions $s \geq (r-1)N$.

Therefore,

$$\limsup_{k \to \infty} E_k = \bigcap_{r=1}^{\infty} \bigcup_{k \geq r} E_k \subseteq \bigcap_{r=1}^{\infty} \bigcup_{s \geq (r-1)N} C_s = \bigcap_{q=0}^{\infty} \bigcup_{s \geq q} C_s = H.$$

The first equality is the definition of the limit superior, and the inclusion follows from the preceding inclusion of unions. The next equality holds because every omitted threshold q is below some retained threshold $(r-1)N$, whose union is smaller and hence imposes a stronger condition. The last equality follows from the representation of H.

HW 41 — Independence of Products of Random Variables

Statement

Let $(Y_n)_{n \in \mathbb{N} \cup \{0\}}$ be a sequence of i.i.d. random variables such that

$$\mathbb{P}(Y_n = 1) = \mathbb{P}(Y_n = -1) = \frac{1}{2}.$$

Define

$$X_n := \prod_{i=0}^{n} Y_i, \qquad n \in \mathbb{N} \cup \{0\}.$$

Prove that $(X_n)_{n \in \mathbb{N} \cup \{0\}}$ is a sequence of i.i.d. random variables.

Expanded Proof of HW 41

We first notice that

$$X_0 = Y_0, \qquad X_n = X_{n-1} Y_n, \quad n \geq 1.$$

The first equality follows from the definition of X_0, while the second follows by separating the last factor in the product defining X_n.

Moreover, $X_n \in \{-1, 1\}$ for every n, since each Y_i takes values in $\{-1, 1\}$.

We now use the same idea as in Example 3 on Markov chains: we fix a possible path of the process and calculate its probability.

Fix $n \in \mathbb{N} \cup \{0\}$ and arbitrary states $x_0, \ldots, x_n \in \{-1, 1\}$. We obtain

$$\begin{aligned} &\{X_0 = x_0, \ldots, X_n = x_n\} \\ &\quad = \{Y_0 = x_0, Y_1 = x_0 x_1, \ldots, Y_n = x_{n-1} x_n\}. \end{aligned}$$

The equality follows from the recursive relation $X_k = X_{k-1} Y_k$. Indeed, since $X_{k-1}^2 = 1$, the conditions $X_{k-1} = x_{k-1}$ and $X_k = x_k$ are equivalent to requiring $Y_k = x_{k-1} x_k$, once the previous states have been fixed.

Therefore,

$$\begin{aligned}
&\mathbb{P}(X_0 = x_0, \dots, X_n = x_n) \\
&\quad = \mathbb{P}(Y_0 = x_0, Y_1 = x_0x_1, \dots, Y_n = x_{n-1}x_n) \\
&\quad = \mathbb{P}(Y_0 = x_0) \prod_{k=1}^{n} \mathbb{P}(Y_k = x_{k-1}x_k) \\
&\quad = \left(\frac{1}{2}\right)^{n+1}.
\end{aligned}$$

The first equality follows from the identity of events just established, the second from independence of the Y_k, and the last from their common distribution, since every $x_{k-1}x_k$ belongs to $\{-1, 1\}$.

We now relate this result to Example 3.

Consider the state space $E = \{-1, 1\}$, with initial distribution and transition matrix given by

$$\mu_{-1} = \mu_1 = \frac{1}{2}, \qquad P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

The joint distribution obtained above can be written as

$$\mathbb{P}(X_0 = x_0, \dots, X_n = x_n) = \mu_{x_0} \prod_{k=1}^{n} p_{x_{k-1}x_k}.$$

This is precisely the joint-distribution formula for a Markov chain introduced in Example 3, with $\mu_i = 1/2$ and $p_{ij} = 1/2$ for every $i, j \in E$.

In this particular case, all rows of the transition matrix are identical: the probability of the next state does not depend on the current state.

We now determine the marginal distributions. For every $m \geq 0$ and $a \in \{-1, 1\}$,

$$\mathbb{P}(X_m = a) = \sum_{x_0, \dots, x_{m-1} \in E} \mathbb{P}(X_0 = x_0, \dots, X_{m-1} = x_{m-1}, X_m = a) = 2^m \left(\frac{1}{2}\right)^{m+1} = \frac{1}{2}.$$

The first equality follows by summing over all possible values of the preceding variables, while the second uses the joint distribution already established and the fact that there are 2^m possible choices of $x_0, \dots, x_{m-1}$.

Consequently,

$$\mathbb{P}(X_0 = x_0, \dots, X_n = x_n) = \left(\frac{1}{2}\right)^{n+1} = \prod_{k=0}^{n} \mathbb{P}(X_k = x_k).$$

The first equality was proved above, and the second follows from the marginal distributions. The factorisation also holds for any finite selection of the X_n, by summing over the remaining coordinates. Since the variables take values only in $\{-1, 1\}$, this proves their independence according to Definition 4.2.

Moreover, all the X_n have the same distribution. Therefore,

$$(X_n)_{n \in \mathbb{N} \cup \{0\}} \text{ is an i.i.d. sequence.}$$

Therefore, the statement is proved.

5. Integration

Motivation — Expectation as an Integral

A key quantity in probability theory is the expected value $\mathbb{E}[X]$ of a random variable X. We already know how to calculate it in some familiar cases, but we now want to understand its general mathematical meaning.

If X takes countably many possible values $x_1, x_2, \ldots$, then

$$\mathbb{E}[X] = \sum_{i=1}^{\infty} x_i \, \mathbb{P}(X = x_i).$$

This is the usual formula for the expectation of a discrete random variable, whenever the sum is well-defined (for instance, when X is integrable).

If X is real-valued and admits a probability density function f_X, then

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} z f_X(z) \, dz.$$

This is the familiar formula for a random variable with a density, again provided the expectation is well-defined.

These two expressions do not cover every random variable: its distribution need not be discrete or admit a density. We therefore seek a general notion of integration that allows us to write

$$\mathbb{E}[X] = \int_{\Omega} X(\omega) \, d\mathbb{P}(\omega).$$

Here $\mathbb{P}$ is the probability measure on $(\Omega, \mathcal{F})$, and the integral is understood whenever it is well-defined. The Lebesgue integral will provide the meaning of this expression.

Recall — Riemann Integral

Let $g \in C([a,b])$, and for each $n \in \mathbb{N}$ consider the dyadic partition

$$z_i^{(n)} = a + \frac{i}{2^n}(b-a), \qquad i = 0, 1, \ldots, 2^n.$$

The Riemann integral can be expressed as

$$\int_a^b g(z) \, dz = \lim_{n \to \infty} \sum_{i=0}^{2^n - 1} g\big(z_i^{(n)}\big)\big(z_{i+1}^{(n)} - z_i^{(n)}\big).$$

The sum is formed by multiplying the value of g at the left endpoint of each subinterval by its length. There are 2^n subintervals, so the last summation index is $2^n - 1$.

However, a random variable $X : \Omega \to \mathbb{R}$ is only assumed to be $\mathcal{F}$-measurable. Measurability alone does not guarantee Riemann integrability, and the sample space Ω need not be an interval of real numbers. We therefore need an integral defined for measurable functions on general measure spaces. For continuous functions on a real interval, the new integral should agree with the familiar Riemann integral; recall that continuous functions are Borel-measurable.

Notation — Lebesgue Integral with Respect to a Measure

Let (S, Σ, μ) be a measure space. For a Σ-measurable function f for which the integral is well-defined (in particular, for any non-negative measurable f), we use the notation

$$\mu(f) := \int_S f(s) \, d\mu(s) = \int_S f \, d\mu.$$

The measure μ determines how the integral assigns weight to measurable subsets of S. The precise construction of this integral will be developed next.

For any $A \in \Sigma$, we also write

$$\mu(f; A) := \mu(f\mathbf{1}_A) = \int_S f\mathbf{1}_A \, d\mu = \int_A f \, d\mu.$$

Here $\mathbf{1}_A$ is the indicator function of A: multiplying by $\mathbf{1}_A$ keeps the values of f on A and makes the integrand zero outside A. Thus the last expression denotes integration restricted to A.

Example — Counting Measure and Infinite Series

Consider the measure space

$$(S, \Sigma, \mu) = \big(\mathbb{N}, \mathcal{P}(\mathbb{N}), \#\big),$$

where $\#$ is the counting measure, so that $\mu(\{k\}) = 1$ for every $k \in \mathbb{N}$.
Let $f : \mathbb{N} \to [0, \infty)$ be given by $f(k) = a_k$. Then

$$\int_S f \, d\mu = \sum_{k=1}^{\infty} f(k)\, \mu(\{k\}) = \sum_{k=1}^{\infty} a_k.$$

The first equality is the integral formula for counting measure: each point k contributes its value $f(k)$ multiplied by the measure of the singleton $\{k\}$. The second follows from $\mu(\{k\}) = 1$ and $f(k) = a_k$. The identity also holds for signed a_k when the series is absolutely convergent.
Thus, the Lebesgue integral must recover the usual infinite series in this special case, just as it must agree with the Riemann integral when both notions apply.

Integrals of Positive Simple Functions

Definition — Integral of an Indicator Function

Let (S, Σ, μ) be a measure space. For any $A \in \Sigma$, recall that the indicator function of A is

$$\mathbf{1}_A(s) = \begin{cases} 1, & s \in A, \\ 0, & s \notin A. \end{cases}$$

We first define its integral by

$$\mu_0(\mathbf{1}_A) := \int_S \mathbf{1}_A \, d\mu := \mu(A).$$

Thus, integrating an indicator function gives the measure of the set on which it equals 1. The notation μ_0 emphasizes that, at this stage, the integral has been defined only for indicator functions. The value $\mu(A)$ may be infinite.

Definition — Positive Simple Function

A non-negative function $f : S \to [0, \infty)$ is called a simple function if it can be written as

$$f(s) = \sum_{k=1}^{m} a_k \mathbf{1}_{A_k}(s), \qquad s \in S,$$

for some $m \in \mathbb{N}$, coefficients $a_k \in [0, \infty)$ and sets $A_k \in \Sigma$.
For each $s \in S$, only the terms with $s \in A_k$ contribute to the sum. The sets A_k need not be disjoint, so their contributions may add up. Since there are only finitely many indicators, f can take only finitely many distinct values; this explains the word simple.
We now verify that every function with this representation belongs to $(m\Sigma)^+$, the class of non-negative Σ-measurable functions.

HW 42 — Measurability of a Simple Function

Statement. Let

$$f(s) = \sum_{k=1}^{m} a_k \mathbf{1}_{A_k}(s), \qquad a_k \in [0, \infty), \quad A_k \in \Sigma.$$

Check that $f \in m\Sigma$.

Expanded Proof.

Fix $k \in \{1, \dots, m\}$. For any Borel set $B \in \mathcal{B}(\mathbb{R})$, we have

$$(\mathbf{1}_{A_k})^{-1}(B) \in \{\emptyset, A_k, A_k^c, S\} \subseteq \Sigma.$$

The membership follows because $\mathbf{1}_{A_k}$ takes only the values 0 and 1: its pre-image of B depends only on which of these two values belong to B. The inclusion follows from $A_k \in \Sigma$ and the properties of a σ-algebra. Hence $\mathbf{1}_{A_k} \in m\Sigma$ by Definition 3.1.

We now obtain

$$a_k \mathbf{1}_{A_k} \in m\Sigma, \qquad k = 1, \dots, m.$$

This follows from Lemma 3.4**(iii)**, since each a_k is a real constant and $\mathbf{1}_{A_k}$ is measurable.

Consequently,

$$f = \sum_{k=1}^{m} a_k \mathbf{1}_{A_k} \in m\Sigma.$$

The membership follows by applying Lemma 3.4**(i)** repeatedly to the finite sum of measurable functions. Moreover, $f(s) \geq 0$ for all $s \in S$, because every coefficient and indicator is non-negative. Thus $f \in (m\Sigma)^+$.

Therefore, the statement is proved.

Different Representations and Canonical Form of Simple Functions

A simple function may admit different representations, corresponding to different choices of coefficients a_k and measurable sets A_k.

For example, let $S = [0, 1]$ and $\Sigma = \mathcal{B}([0, 1])$. Then

$$f(s) = \mathbf{1}_{[1/2,1]}(s) = \mathbf{1}_{[1/2,2/3)}(s) + \mathbf{1}_{[2/3,1]}(s).$$

The equality follows because the two intervals on the right are disjoint and their union is $[1/2, 1]$. Thus, the same function can be represented using different measurable sets.

A simple function is expressed in canonical form if

$$f(s) = \sum_{k=1}^{m} a_k \mathbf{1}_{A_k}(s),$$

where

$$a_1 < a_2 < \cdots < a_m, \qquad A_k = f^{-1}(\{a_k\}), \quad k = 1, \dots, m.$$

In this representation, the coefficients are precisely the distinct values attained by f, and the sets A_k are their corresponding level sets. Consequently, these sets are pairwise disjoint and form a partition of S.

Every nonnegative simple function can be expressed in canonical form by collecting all points where it takes the same value. The corresponding level sets are measurable, pairwise disjoint, and form a finite partition of S. All values actually attained by f must be included, including zero whenever it is attained.

Definition — Integral of a Positive Simple Function

We denote the collection of non-negative simple functions by SF^+. For $f \in SF^+$, represented as

$$f = \sum_{k=1}^{m} a_k \mathbf{1}_{A_k},$$

we define

$$\mu_0(f) := \int_S f \, d\mu := \sum_{k=1}^{m} a_k \mu(A_k).$$

This extends the definition of the integral of an indicator function, since for $f = \mathbf{1}_A$ we recover $\mu_0(f) = \mu(A)$.

Observation — The Convention $0 \cdot \infty = 0$

Let $S = \mathbb{R}$, $\Sigma = \mathcal{B}(\mathbb{R})$, $\mu = \text{Leb}$, and consider the zero function $f = 0$.
We obtain

$$\int_S f \, d\mu = 0 \cdot \mu(\mathbb{R}) = 0 \cdot \infty.$$

Since the integral of the zero function must equal zero, we adopt the convention

$$0 \cdot \infty := 0.$$

This ensures that zero coefficients contribute nothing to the integral, even when the corresponding sets have infinite measure.
Finally, since a simple function may admit different representations, we must verify that the value of $\mu_0(f)$ does not depend on the chosen representation.
This is the well-definedness of the integral of simple functions, which will be established in the next homework.

HW 43 — Well-Definedness of the Integral of Simple Functions

Statement. Fix (S, Σ, μ) and let $f \in SF^+$ have the following two representations:

$$f = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i} \qquad \text{and} \qquad f = \sum_{j=1}^{m} b_j \mathbf{1}_{B_j},$$

for numbers $(a_i)_{i=1}^n, (b_j)_{j=1}^m \subseteq [0, \infty)$ and sets $(A_i)_{i=1}^n, (B_j)_{j=1}^m \subseteq \Sigma$. Show that

$$\mu_0(f) = \sum_{i=1}^{n} a_i \mu(A_i) = \sum_{j=1}^{m} b_j \mu(B_j).$$

Proof.

Reduction to disjoint partitions.

A finite measurable partition of S is a family of nonempty measurable sets that are pairwise disjoint and whose union is S. For instance, two sets $A, B \in \Sigma$ give the pieces

$$A \cap B, \qquad A \cap B^c, \qquad A^c \cap B, \qquad A^c \cap B^c,$$

with empty pieces omitted. In particular,

$$A = (A \cap B) \cup (A \cap B^c), \qquad B = (A \cap B) \cup (A^c \cap B).$$

Thus each original set can be recovered as a union of disjoint pieces, while the remaining piece completes the partition of S.

For the first representation, start with S and consider $A_1, \ldots, A_n$ one at a time. At the ith step, split every current piece E into the nonempty sets among

$$E \cap A_i, \qquad E \cap A_i^c.$$

Indeed,

$$E = (E \cap A_i) \cup (E \cap A_i^c), \qquad (E \cap A_i) \cap (E \cap A_i^c) = \emptyset.$$

Each step preserves disjointness and the union S; measurability follows from closure of Σ under complements and finite intersections. After finitely many steps we obtain a partition $C_1, \ldots, C_r$ of S. Each C_k lies entirely inside or outside every A_i, so each A_i is a union of some C_k and f is constant on each C_k.

Applying the same construction to $B_1, \ldots, B_m$ gives another partition $D_1, \ldots, D_t$ of S. Each B_j is a union of some D_ℓ, and f is constant on each D_ℓ.

Choose $s_k \in C_k$ and $t_\ell \in D_\ell$, and set

$$c_k = f(s_k), \qquad d_\ell = f(t_\ell), \qquad k = 1, \dots, r, \quad \ell = 1, \dots, t.$$

From the two representations of f,

$$c_k = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i}(s_k), \qquad d_\ell = \sum_{j=1}^{m} b_j \mathbf{1}_{B_j}(t_\ell),$$

for every $k = 1, \dots, r$ and $\ell = 1, \dots, t$.

Since the C_k partition S, we obtain

$$\begin{aligned} \sum_{i=1}^{n} a_i \mu(A_i) &= \sum_{i=1}^{n} a_i \mu\left(A_i \cap \bigcup_{k=1}^{r} C_k\right) = \sum_{k=1}^{r} \sum_{i=1}^{n} a_i \mu(A_i \cap C_k) \\ &= \sum_{k=1}^{r} \left(\sum_{i=1}^{n} a_i \mathbf{1}_{A_i}(s_k)\right) \mu(C_k) = \sum_{k=1}^{r} c_k \mu(C_k). \end{aligned}$$

The second equality uses finite additivity on the disjoint pieces $A_i \cap C_k$ and rearranges finite nonnegative sums. The third follows because each C_k lies entirely inside or outside A_i, and the last uses the expression for c_k. Likewise,

$$\sum_{j=1}^{m} b_j \mu(B_j) = \sum_{\ell=1}^{t} d_\ell \mu(D_\ell).$$

These identities show that we may assume, without loss of generality, that the representing families are finite measurable partitions of S, since the replacement preserves both f and the weighted sums. Moreover, by merging partition elements with equal coefficients, any such representation can be reduced to canonical form, again without changing the weighted sum.

Proof for partition representations.

Assume now that $(A_i)_{i=1}^n$ and $(B_j)_{j=1}^m$ are finite measurable partitions of S, with the corresponding coefficients a_i and b_j. As proved above, this entails no loss of generality.

The nonempty intersections $A_i \cap B_j$ form a common finite measurable partition of S. Indeed, they cover S because every $s \in S$ belongs to some A_i and some B_j. Moreover, for $(i, j) \neq (k, \ell)$,

$$(A_i \cap B_j) \cap (A_k \cap B_\ell) = (A_i \cap A_k) \cap (B_j \cap B_\ell) = \emptyset.$$

The last equality follows because either $i \neq k$ or $j \neq \ell$, so at least one intersection is empty. Finally, $A_i \cap B_j \in \Sigma$ by closure under finite intersections.

For every $i = 1, \dots, n$ and $j = 1, \dots, m$, we distinguish two cases.

Case 1 — Empty intersection.

If $A_i \cap B_j = \emptyset$, then $\mu(A_i \cap B_j) = 0$.

Case 2 — Nonempty intersection.

If $A_i \cap B_j \neq \emptyset$, choose $s \in A_i \cap B_j$. Since the two families are partitions, s belongs to exactly one set in each family. Hence,

$$a_i = \sum_{k=1}^{n} a_k \mathbf{1}_{A_k}(s) = f(s) = \sum_{\ell=1}^{m} b_\ell \mathbf{1}_{B_\ell}(s) = b_j.$$

The first and last equalities follow because only the indicators of A_i and B_j equal 1 in their respective sums; the middle equalities use the two representations of f.

Thus $a_i = b_j$ holds for pairs whose intersection is nonempty. For the remaining pairs the coefficients need not agree, but the intersection has measure zero. In both cases, therefore,

$$a_i \mu(A_i \cap B_j) = b_j \mu(A_i \cap B_j), \qquad \forall\, i = 1, \dots, n, \quad j = 1, \dots, m.$$

Using $S = \bigcup_{j=1}^{m} B_j = \bigcup_{i=1}^{n} A_i$, we obtain

$$\sum_{i=1}^{n} a_i \mu(A_i) = \sum_{i=1}^{n} a_i \mu(A_i \cap S) = \sum_{i=1}^{n} a_i \mu\left(A_i \cap \bigcup_{j=1}^{m} B_j\right) = \sum_{i=1}^{n}\sum_{j=1}^{m} a_i \mu(A_i \cap B_j)$$
$$= \sum_{j=1}^{m}\sum_{i=1}^{n} b_j \mu(A_i \cap B_j) = \sum_{j=1}^{m} b_j \mu\left(B_j \cap \bigcup_{i=1}^{n} A_i\right) = \sum_{j=1}^{m} b_j \mu(B_j).$$

The third and fifth equalities use finite additivity on disjoint intersections. The fourth follows from the equality of the contributions in both cases and rearrangement of finite sums. All expressions remain valid when some measures are infinite, under the convention $0 \cdot \infty = 0$.

Thus the weighted sum is independent of the chosen representation of f, and $\mu_0(f)$ is well-defined. Therefore, the statement is proved.

HW 44 — Properties of Integrals of Simple Functions

Statement. Fix a measure space (S, Σ, μ). Prove the following properties.

(i) If $f, g \in SF^+$ and $\mu(\{s \in S : f(s) \neq g(s)\}) = 0$, then $\mu_0(f) = \mu_0(g)$.

(ii) If $f, g \in SF^+$ and $c \geq 0$, then $f + g, cf \in SF^+$; moreover,

$$\mu_0(f + g) = \mu_0(f) + \mu_0(g), \qquad \mu_0(cf) = c\mu_0(f).$$

(iii) If $f, g \in SF^+$ and $f \leq g$, then $\mu_0(f) \leq \mu_0(g)$.

(iv) If $f, g \in SF^+$, then $f \vee g := \max\{f, g\}$ and $f \wedge g := \min\{f, g\}$ also belong to SF^+.

Recall — μ-almost everywhere.

A property $P(s)$ holds μ-almost everywhere (μ-a.e.) if it fails only on a set of μ-measure zero. In particular, for measurable f, g,

$$f = g \quad \mu\text{-a.e.} \quad \Longleftrightarrow \quad \mu\big(\{s \in S : f(s) \neq g(s)\}\big) = 0.$$

Thus f and g need not agree at every point.

When constructing a common partition as in HW 43, we discard only empty intersections, since partition elements must be nonempty. We do not discard nonempty sets of measure zero: for example, on $S = [0, 1]$ with $\mu = \text{Leb}$,

$$D = \{\tfrac{1}{2}\} \neq \emptyset, \qquad \mu(D) = 0.$$

Such a set remains a partition element, although its contribution to the integral is zero.

Proof.

By HW 43, both f and g admit canonical representations, whose level sets form finite measurable partitions of S. Intersecting these partitions and omitting empty intersections gives a common finite measurable partition $D_1, \ldots, D_r$, on which both functions are constant.

$$f = \sum_{k=1}^{r} a_k \mathbf{1}_{D_k}, \qquad g = \sum_{k=1}^{r} b_k \mathbf{1}_{D_k},$$

where $a_k, b_k \geq 0$. By HW 43, the integral is independent of the representation, so

$$\mu_0(f) = \sum_{k=1}^{r} a_k \mu(D_k), \qquad \mu_0(g) = \sum_{k=1}^{r} b_k \mu(D_k).$$

(i) Equality almost everywhere. Suppose $f = g$ μ-a.e., and let

$$N := \{s \in S : f(s) \neq g(s)\}, \qquad \mu(N) = 0.$$

Fix k such that $a_k \neq b_k$. For any $s \in D_k$, the partition representations give

$$f(s) = a_k \neq b_k = g(s) \quad \Longrightarrow \quad s \in N.$$

Since $s \in D_k$ was arbitrary, $D_k \subseteq N$. By monotonicity of μ,

$$0 \le \mu(D_k) \le \mu(N) = 0 \quad \Longrightarrow \quad \mu(D_k) = 0.$$

Consequently, for every $k = 1, \dots, r$,

$$a_k \mu(D_k) = b_k \mu(D_k).$$

Indeed, if $a_k = b_k$ the equality is immediate; otherwise $\mu(D_k) = 0$ and both sides vanish. Summing gives

$$\mu_0(f) = \sum_{k=1}^{r} a_k \mu(D_k) = \sum_{k=1}^{r} b_k \mu(D_k) = \mu_0(g).$$

(ii) Additivity and positive homogeneity. Using the common partition,

$$f + g = \sum_{k=1}^{r} (a_k + b_k) \mathbf{1}_{D_k}, \qquad cf = \sum_{k=1}^{r} (ca_k) \mathbf{1}_{D_k}.$$

Thus $f + g, cf \in SF^+$, since their coefficients are nonnegative and the D_k are measurable. Moreover,

$$\mu_0(f+g) = \sum_{k=1}^{r} (a_k + b_k) \mu(D_k) = \sum_{k=1}^{r} a_k \mu(D_k) + \sum_{k=1}^{r} b_k \mu(D_k) = \mu_0(f) + \mu_0(g),$$

and

$$\mu_0(cf) = \sum_{k=1}^{r} (ca_k) \mu(D_k) = c \sum_{k=1}^{r} a_k \mu(D_k) = c\mu_0(f).$$

The equalities follow from the definition of μ_0 and distributivity of finite nonnegative sums, with $0 \cdot \infty = 0$ when necessary.

(iii) Monotonicity. Suppose $f \le g$. Since $D_k \neq \emptyset$ and both functions are constant on it, $a_k \le b_k$ for every $k = 1, \dots, r$. Hence

$$\mu_0(f) = \sum_{k=1}^{r} a_k \mu(D_k) \le \sum_{k=1}^{r} b_k \mu(D_k) = \mu_0(g).$$

The inequality follows from $a_k \le b_k$ and $\mu(D_k) \ge 0$ for every k.

(iv) Maximum and minimum. Since $f = a_k$ and $g = b_k$ on D_k,

$$f \vee g = \sum_{k=1}^{r} \max\{a_k, b_k\} \mathbf{1}_{D_k},$$

$$f \wedge g = \sum_{k=1}^{r} \min\{a_k, b_k\} \mathbf{1}_{D_k}.$$

Both functions belong to SF^+, because their coefficients are nonnegative and their partition elements are measurable.

Therefore, the statement is proved.

Example — Equality Lebesgue-Almost Everywhere

Let $S = [0,1]$, $\Sigma = \mathcal{B}([0,1])$, $\mu = \text{Leb}$, and consider

$$f = \mathbf{1}_{[\frac{1}{2},1]}, \qquad g = \mathbf{1}_{(\frac{1}{2},1]}.$$

The functions differ only at $s = \frac{1}{2}$, so

$$\{s \in S : f(s) \neq g(s)\} = \{\tfrac{1}{2}\}, \qquad \text{Leb}\big(\{\tfrac{1}{2}\}\big) = 0.$$

Hence $f = g$ Leb-a.e. and, by HW 44(i),

$$\mu_0(f) = \text{Leb}\big([\tfrac{1}{2}, 1]\big) = \tfrac{1}{2} = \text{Leb}\big((\tfrac{1}{2}, 1]\big) = \mu_0(g).$$

The equalities follow from the definition of the integral of an indicator function and the length of the two intervals.

Definition — Integral of a Positive Measurable Function

In this chapter, $(m\Sigma)^+$ includes all measurable maps $f : S \to [0, +\infty]$, where measurability is understood through the real threshold tests. Simple functions in SF^+ still have only finitely many finite values.

For signed functions, $m\Sigma$ retains the real-valued meaning of Definition 3.1 unless stated otherwise.

Let (S, Σ, μ) be a measure space. For any $f \in (m\Sigma)^+$, define

$$\mu(f) := \int_S f \, d\mu := \sup\{\mu_0(h) : h \in SF^+,\ h \le f\}.$$

Thus the integral of f is the supremum of the integrals of all nonnegative simple functions lying below f.

Observation — Agreement with the Simple Integral

If $f \in SF^+$, then

$$\mu(f) = \mu_0(f).$$

Indeed, f itself belongs to the family over which the supremum is taken, so $\mu(f) \ge \mu_0(f)$. Conversely, for every $h \in SF^+$ with $h \le f$, monotonicity of μ_0 (HW 44(iii)) gives $\mu_0(h) \le \mu_0(f)$. Taking the supremum yields $\mu(f) \le \mu_0(f)$, proving the equality.

HW 45 — Monotonicity of the Positive Lebesgue Integral

Statement. Prove that if $f, g \in (m\Sigma)^+$ satisfy $f \le g$, then

$$\mu(f) \le \mu(g).$$

Proof.

By definition, for every $f \in (m\Sigma)^+$,

$$\mu(f) := \sup\{\mu_0(h) : h \in SF^+,\ h \le f\}.$$

Let $f, g \in (m\Sigma)^+$ with $f \le g$. For any $h \in SF^+$ such that $h \le f$, we have

$$h \le f \le g \quad \Longrightarrow \quad h \le g.$$

Hence,

$$\{h \in SF^+ : h \le f\} \subseteq \{h \in SF^+ : h \le g\}.$$

The inclusion follows because every nonnegative simple function lying below f also lies below g.

Taking the supremum over the two families, we obtain

$$\mu(f) = \sup\{\mu_0(h) : h \in SF^+,\ h \le f\} \le \sup\{\mu_0(h) : h \in SF^+,\ h \le g\} = \mu(g).$$

The inequality follows from the inclusion above, since the supremum over a smaller family cannot exceed the supremum over a larger one. Both families are nonempty because the zero function is an admissible simple function.

Therefore, the statement is proved.

Lemma 5.1 — Zero Integral Implies Zero Almost Everywhere

Statement. Let (S, Σ, μ) be a measure space and let $f \in (m\Sigma)^+$ satisfy $\mu(f) = 0$. Then

$$\mu(\{f > 0\}) = 0.$$

Expanded Proof.

We argue by contradiction. Suppose $\mu(\{f > 0\}) > 0$ and define

$$A_n := \left\{ s \in S : f(s) > \frac{1}{n} \right\} \in \Sigma, \qquad n \in \mathbb{N}.$$

These sets are measurable since f is measurable. Moreover, $A_n \subseteq A_{n+1}$ because $1/(n+1) < 1/n$. As we will see in HW 46,

$$\{f > 0\} = \bigcup_{n=1}^{\infty} A_n = \lim_{n\to\infty} A_n.$$

Hence, by continuity from below (Lemma 1.5),

$$0 < \mu(\{f > 0\}) = \lim_{n\to\infty} \mu(A_n).$$

Since $\mu(A_n)$ is nondecreasing and its limit is strictly positive, there exists $\bar{n} \in \mathbb{N}$ such that $\mu(A_n) > 0$ for every $n \geq \bar{n}$.

For any $n \geq \bar{n}$, the sets A_n and A_n^c partition S, so $f = f\mathbf{1}_{A_n} + f\mathbf{1}_{A_n^c}$ pointwise. Moreover, $f\mathbf{1}_{A_n^c} \geq 0$ and

$$f\mathbf{1}_{A_n} \geq \frac{1}{n}\mathbf{1}_{A_n}.$$

Indeed, if $s \in A_n$, then $f(s) > 1/n$ by definition; if $s \notin A_n$, both sides of the latter inequality vanish. Hence, using monotonicity of the positive integral (HW 45),

$$\begin{aligned} 0 = \mu(f) &= \int_S \left(f\mathbf{1}_{A_n} + f\mathbf{1}_{A_n^c} \right) d\mu \\ &\geq \int_S f\mathbf{1}_{A_n}\, d\mu \geq \int_S \frac{1}{n}\mathbf{1}_{A_n}\, d\mu = \frac{1}{n}\mu(A_n) > 0. \end{aligned}$$

The first equality is the hypothesis $\mu(f) = 0$; the second uses the pointwise decomposition of f, without linearity of the integral. The next two inequalities follow from the pointwise bounds above and HW 45. The final equality is the integral of the nonnegative simple function $(1/n)\mathbf{1}_{A_n}$, and the strict inequality holds since $\mu(A_n) > 0$ and $1/n > 0$, even if $\mu(A_n) = \infty$.

This contradicts $\mu(f) = 0$. Therefore $\mu(\{f > 0\}) = 0$.

Therefore, the statement is proved.

HW 46 — Level-Set Representation of $\{f > 0\}$

Statement. Let $f \in (m\Sigma)^+$. Prove that

$$\{f > 0\} = \bigcup_{n=1}^{\infty} \left\{ f > \frac{1}{n} \right\} = \lim_{n\to\infty} \left\{ f > \frac{1}{n} \right\}.$$

Proof.

We first prove that

$$\{f > 0\} = \bigcup_{n=1}^{\infty} \left\{ f > \frac{1}{n} \right\}$$

by showing both inclusions.

($\subseteq$) Let $s_0 \in \{f > 0\}$. Then $f(s_0) > 0$, so there exists $n_0 \in \mathbb{N}$ such that

$$\frac{1}{n_0} < f(s_0).$$

This follows from the Archimedean property of $\mathbb{R}$, and is immediate if $f(s_0) = \infty$. Therefore,

$$s_0 \in \left\{ f > \frac{1}{n_0} \right\} \subseteq \bigcup_{n=1}^{\infty} \left\{ f > \frac{1}{n} \right\}.$$

Since s_0 was arbitrary, the first inclusion follows.

(⊇) Conversely, let

$$s_0 \in \bigcup_{n=1}^{\infty} \left\{ f > \frac{1}{n} \right\}.$$

By definition of union, there exists $n_0 \in \mathbb{N}$ such that

$$f(s_0) > \frac{1}{n_0} > 0.$$

Hence $s_0 \in \{f > 0\}$, proving the reverse inclusion.

Finally, define

$$A_n := \left\{ f > \frac{1}{n} \right\}, \qquad n \in \mathbb{N}.$$

Since

$$\frac{1}{n+1} < \frac{1}{n} \quad \Longrightarrow \quad A_n \subseteq A_{n+1},$$

the sequence $(A_n)_{n\in\mathbb{N}}$ is increasing. Therefore, by the definition of the limit of an increasing sequence of sets,

$$\lim_{n\to\infty} A_n = \bigcup_{n=1}^{\infty} A_n = \{f > 0\}.$$

Combining the two identities, we obtain

$$\{f > 0\} = \bigcup_{n=1}^{\infty} \left\{ f > \frac{1}{n} \right\} = \lim_{n\to\infty} \left\{ f > \frac{1}{n} \right\}.$$

Therefore, the statement is proved.

Example — Application of Lemma 5.1

Let $(S, \Sigma, \mu) = ([0,1], \mathcal{B}([0,1]), \mathrm{Leb})$ and

$$f(s) = \mathbf{1}_{\{\frac{1}{2}\}}(s).$$

Since $f \in SF^+$,

$$\mu(f) = \mu_0(f) = \mathrm{Leb}\left(\left\{ \frac{1}{2} \right\} \right) = 0.$$

The equalities follow from the agreement of μ with μ_0 on simple functions, the integral of an indicator, and the fact that a singleton has zero Lebesgue measure. By Lemma 5.1,

$$\mu(\{f > 0\}) = \mathrm{Leb}\left(\left\{ \frac{1}{2} \right\} \right) = 0.$$

Thus $f = 0$, Leb-a.e., although $f(\frac{1}{2}) = 1$.

The next theorem is probably the most useful single tool in the whole course.

Observation — Reduction from Almost-Everywhere to Pointwise Convergence

Statement. Let $N \in \Sigma$ satisfy $\mu(N) = 0$. For any $u \in (m\Sigma)^+$, define

$$u^N(s) := \begin{cases} u(s), & s \in N^c, \\ 0, & s \in N. \end{cases}$$

Then $u^N \in (m\Sigma)^+$ and

$$\mu(u^N) = \mu(u).$$

Consequently, if $f_n \uparrow f$ μ-almost everywhere, we may modify all f_n and f on a common null set so that the convergence holds everywhere, without changing their integrals.

Expanded Proof.

Fix $u \in (m\Sigma)^+$. The function u^N is measurable and nonnegative because u and $\mathbf{1}_{N^c}$ are measurable.

Let $h \in SF^+$ satisfy $h \le u$. Consider $h\mathbf{1}_{N^c}$. We have $h\mathbf{1}_{N^c} \in SF^+$, $h\mathbf{1}_{N^c} \le u^N$. Indeed, outside N, the inequality follows from $h \le u$, while on N both functions are zero.

Moreover,

$$\mu_0(h) = \mu_0(h\mathbf{1}_{N^c}) \le \mu(u^N).$$

The equality follows from HW 44(i), since the two simple functions agree outside the null set N. The inequality follows from the definition of the positive integral, because $h\mathbf{1}_{N^c}$ is an admissible simple function below u^N.

Taking the supremum over all $h \in SF^+$ with $h \le u$, we obtain

$$\mu(u) = \sup\{\mu_0(h) : h \in SF^+,\ h \le u\} \le \mu(u^N).$$

Conversely,

$$u^N \le u \quad \Longrightarrow \quad \mu(u^N) \le \mu(u).$$

The implication follows from HW 45. Combining the inequalities gives

$$\mu(u^N) = \mu(u).$$

Now suppose $f_n \uparrow f$ μ-almost everywhere. There exists a single measurable set N, with $\mu(N) = 0$, outside which both monotonicity and convergence hold. Define

$$\widehat{f}_n := f_n\mathbf{1}_{N^c}, \qquad \widehat{f} := f\mathbf{1}_{N^c}.$$

Then

$$\widehat{f}_n(s) \uparrow \widehat{f}(s) \qquad \forall s \in S.$$

On N^c, this is the original hypothesis; on N, all the modified functions are identically zero. By the result just proved,

$$\mu(\widehat{f}_n) = \mu(f_n) \quad \forall n \in \mathbb{N}, \qquad \mu(\widehat{f}) = \mu(f).$$

Thus, when proving a statement involving these integrals, we may work with the modified functions and assume pointwise monotone convergence.

This reduction is important in MON. Without it, the sets $E_n = \{ch \le f_n\}$ need not increase to S. For example, on $S = [0,1]$ with Lebesgue measure, take $f = h = 1$, $c = 1/2$, and $f_n = \mathbf{1}_{(0,1]}$ for every n. Then $f_n \uparrow f$ almost everywhere, but $E_n = (0,1]$ for every n. Monotonicity of the original E_n can also fail on the exceptional set. The reduction above eliminates both difficulties before E_n is defined.

Therefore, the statement is proved.

Theorem 5.2 — Monotone Convergence Theorem (MON)

Statement.

Let $(f_n)_{n\in\mathbb{N}} \subseteq (m\Sigma)^+$ and let $f \in (m\Sigma)^+$ satisfy $f_n(s) \uparrow f(s)$ for μ-almost every $s \in S$. Then

$$\int_S f_n \, d\mu \uparrow \int_S f \, d\mu.$$

Expanded Proof.

By the preceding Observation, we may assume that $f_n \uparrow f$ everywhere, without changing the integrals.

Recall that $\mu(f) = \sup\{\mu_0(h) : h \in SF^+,\ h \le f\}$. We first obtain

$$\mu(f_n) \le \mu(f_{n+1}) \le \mu(f).$$

Both inequalities follow from HW 45, since $f_n \le f_{n+1} \le f$ pointwise. Thus $(\mu(f_n))_{n\in\mathbb{N}}$ is nondecreasing, and

$$\alpha := \lim_{n\to\infty} \mu(f_n) \le \mu(f).$$

It remains to prove the reverse inequality.

Fix $c \in (0,1)$ and an arbitrary $h \in SF^+$ such that $h \le f$. Define

$$E_n := \{s \in S : ch(s) \le f_n(s)\} \in \Sigma.$$

The sets are measurable because h and f_n are measurable. As we will see in HW 47, $E_n \uparrow S$. Writing

$$h = \sum_{k=1}^{m} a_k \mathbf{1}_{A_k}, \qquad a_k \ge 0, \quad A_k \in \Sigma,$$

we obtain

$$\mu(f_n) \ge \int_S f_n \mathbf{1}_{E_n}\, d\mu \ge c \int_S h\mathbf{1}_{E_n}\, d\mu = c\sum_{k=1}^{m} a_k \mu(E_n \cap A_k).$$

The first inequality follows from $f_n \ge f_n\mathbf{1}_{E_n}$ and HW 45. The second follows from $f_n \ge ch$ on E_n, again using HW 45. The equality follows from the definition of the integral of a nonnegative simple function.

Since $E_n \uparrow S$, we have $E_n \cap A_k \uparrow A_k$. By continuity from below (Lemma 1.5),

$$\mu(E_n \cap A_k) \uparrow \mu(A_k).$$

Letting $n \to \infty$ in the previous inequality yields

$$\alpha \ge \lim_{n\to\infty} c\sum_{k=1}^{m} a_k \mu(E_n \cap A_k) = c\sum_{k=1}^{m} a_k \mu(A_k) = c\mu_0(h).$$

The first equality follows because the sum has finitely many nonnegative terms; the final equality follows from the definition of $\mu_0(h)$. The calculation also allows $\mu(A_k) = \infty$, with the convention $0 \cdot \infty = 0$.

Since $h \in SF^+$ with $h \le f$ was arbitrary, taking the supremum gives

$$\alpha \ge c \sup\{\mu_0(h) : h \in SF^+,\ h \le f\} = c\mu(f).$$

The equality follows from the definition of $\mu(f)$.

Finally, letting $c \uparrow 1$, we obtain $\alpha \ge \mu(f)$. If $\mu(f) = \infty$, the same conclusion already follows from $\alpha \ge c\mu(f) = \infty$ for any $c \in (0,1)$.

Combining this with $\alpha \le \mu(f)$, we conclude that

$$\mu(f_n) \uparrow \mu(f).$$

Therefore, the statement is proved.

HW 47 — Increasing Sets in MON

Statement. In the proof of Theorem 5.2, after the reduction to pointwise convergence, let $f_n \uparrow f$ everywhere. Fix $c \in (0,1)$, $h \in SF^+$ with $h \le f$, and define

$$E_n := \{s \in S : ch(s) \le f_n(s)\} \in \Sigma.$$

Prove that

$$E_n \subseteq E_{n+1}, \qquad E_n \uparrow S.$$

Expanded Proof.

We first show that $E_n \subseteq E_{n+1}$. Let $s_0 \in E_n$. Then

$$ch(s_0) \le f_n(s_0) \le f_{n+1}(s_0).$$

The first inequality follows from the definition of E_n, while the second follows from the monotonicity of (f_n). Hence $s_0 \in E_{n+1}$, so $E_n \subseteq E_{n+1}$ $\forall n \in \mathbb{N}$.

We now prove that $\bigcup_{n=1}^{\infty} E_n = S$. Fix $s_0 \in S$.

If $h(s_0) = 0$, then

$$ch(s_0) = 0 \leq f_n(s_0) \qquad \forall n \in \mathbb{N},$$

since $f_n \geq 0$. Thus $s_0 \in E_n$ for every n.

If $h(s_0) > 0$, then

$$f(s_0) \geq h(s_0) > ch(s_0).$$

The first inequality follows from $h \leq f$, and the second from $c \in (0,1)$. Since $f_n(s_0) \uparrow f(s_0)$, there exists $n_0 \in \mathbb{N}$ such that

$$f_n(s_0) \geq ch(s_0) \qquad \forall n \geq n_0.$$

This also holds when $f(s_0) = \infty$, because $ch(s_0)$ is finite and $f_n(s_0) \to \infty$.

In either case, $s_0 \in E_n$ for some n. Since $s_0 \in S$ was arbitrary,

$$\bigcup_{n=1}^{\infty} E_n = S.$$

Together with $E_n \subseteq E_{n+1}$, this gives

$$E_n \uparrow S.$$

Therefore, the statement is proved.

Approximation by Positive Simple Functions

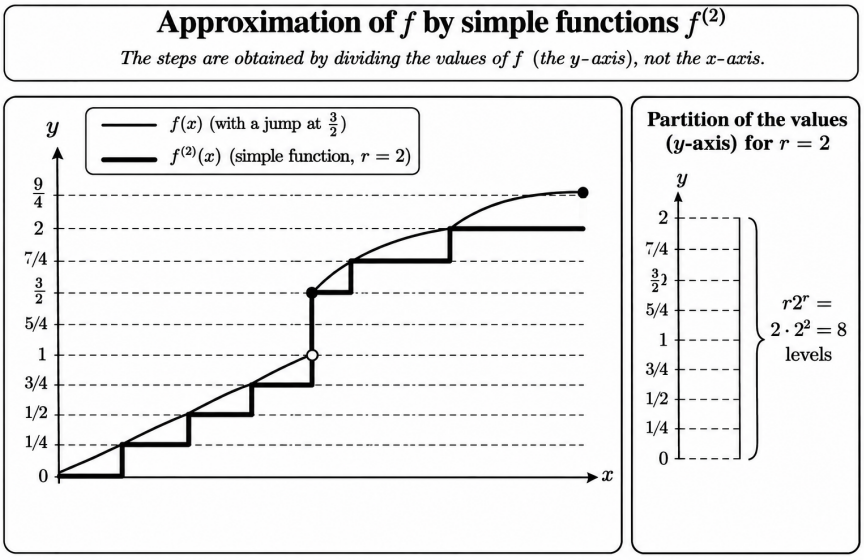
From the monotone convergence theorem (Theorem 5.2), the next question arises: given $f \in (m\Sigma)^+$, how can we construct an increasing sequence $(f^{(r)})_{r\in\mathbb{N}} \subseteq SF^+$ that approximates f?

For $k, r \in \mathbb{N}$, define

$$A_k^r := \left\{ s \in S : \frac{k-1}{2^r} \leq f(s) < \frac{k}{2^r} \right\}.$$

Set

$$f^{(r)}(s) = \sum_{k=1}^{r2^r} \frac{k-1}{2^r} \mathbf{1}_{A_k^r}(s) + r\mathbf{1}_{\{f \geq r\}}(s). \tag{5.1}$$



Verification.

First, $A_k^r, \{f \geq r\} \in \Sigma$ by measurability of f. The sets $A_1^r, \ldots, A_{r2^r}^r, \{f \geq r\}$ are pairwise disjoint and cover S: if $0 \leq f(s) < r$, its value belongs to exactly one half-open interval $[(k-1)/2^r, k/2^r)$; if $f(s) \geq r$, it belongs to the last set. Hence $f^{(r)} \in SF^+$, since (5.1) is a finite sum of measurable indicators with nonnegative coefficients.

We now prove that $f^{(r)} \leq f^{(r+1)} \leq f$. For any $s \in S$ with $x := f(s) < \infty$, formula (5.1) gives

$$f^{(r)}(s) = \min\left\{r, \frac{\lfloor 2^r x \rfloor}{2^r}\right\}.$$

Indeed, below r the coefficient is the left endpoint of the interval containing x, while at and above r the value is truncated at r. For every $x \geq 0$,

$$\frac{\lfloor 2^r x \rfloor}{2^r} \leq \frac{\lfloor 2^{r+1} x \rfloor}{2^{r+1}} \leq x.$$

The first inequality follows from $\lfloor 2^{r+1} x \rfloor \geq 2\lfloor 2^r x \rfloor$; the second is the defining property of the integer part. Since the truncation level also increases from r to $r+1$, we obtain $f^{(r)}(s) \leq f^{(r+1)}(s) \leq f(s)$. If $f(s) = \infty$, then $f^{(r)}(s) = r$, so the same inequalities hold.

Finally, for $f(s) < \infty$, choose $r > f(s)$. The truncation no longer applies and

$$0 \leq f(s) - f^{(r)}(s) < \frac{1}{2^r},$$

by the defining bounds of the interval A_k^r containing s. Thus $f^{(r)}(s) \to f(s)$. If $f(s) = \infty$, then $f^{(r)}(s) = r \to \infty$. Consequently,

$$f^{(r)}(s) \uparrow f(s), \qquad \forall s \in S.$$

In particular, the monotone convergence theorem applies to this sequence.

Lemma 5.3 — Equality Almost Everywhere

Statement. If $f, g \in (m\Sigma)^+$ and $\mu(\{f \neq g\}) = 0$, then $\mu(f) = \mu(g)$.

Proof.

Let $(f^{(r)})_{r\in\mathbb{N}}$ and $(g^{(r)})_{r\in\mathbb{N}}$ be the simple-function approximations given by (5.1). By the construction above,

$$f^{(r)} \uparrow f, \qquad g^{(r)} \uparrow g.$$

Set $N := \{s \in S : f(s) \neq g(s)\} \in \Sigma$, so that $\mu(N) = 0$. For every $r \in \mathbb{N}$ and $s \notin N$,

$$f(s) = g(s) \quad \Longrightarrow \quad f^{(r)}(s) = g^{(r)}(s),$$

since (5.1) applies the same dyadic approximation to both values. Therefore,

$$\{f^{(r)} \neq g^{(r)}\} \subseteq N \quad \Longrightarrow \quad 0 \leq \mu(\{f^{(r)} \neq g^{(r)}\}) \leq \mu(N) = 0.$$

The second implication follows from monotonicity of μ. Thus $f^{(r)} = g^{(r)}$ μ-a.e., and HW 44(i) gives

$$\mu(f^{(r)}) = \mu_0(f^{(r)}) = \mu_0(g^{(r)}) = \mu(g^{(r)}), \qquad \forall r \in \mathbb{N}.$$

The first and last equalities use the agreement of the positive Lebesgue integral with the simple integral.

Applying the monotone convergence theorem (Theorem 5.2), we conclude that

$$\mu(f) = \lim_{r\to\infty} \mu(f^{(r)}) = \lim_{r\to\infty} \mu(g^{(r)}) = \mu(g).$$

The first and last equalities follow from monotone convergence, and the middle one from equality of the integrals for every r.

Therefore, the statement is proved.

Lemma 5.4 — Fatou's Lemma for Functions

Statement. Let (S, Σ, μ) be a measure space. For any $(f_n)_{n\in\mathbb{N}} \subseteq (m\Sigma)^+$, it holds

$$\liminf_{n\to\infty} \int_S f_n \, d\mu \geq \int_S \liminf_{n\to\infty} f_n \, d\mu.$$

Proof.

Recall that

$$\liminf_{n\to\infty} f_n = \sup_{n\geq 1} \inf_{m\geq n} f_m.$$

Define

$$g_n(s) := \inf_{m\geq n} f_m(s), \qquad n \in \mathbb{N}.$$

First, $g_n \in (m\Sigma)^+$, since $g_n \geq 0$ and, for every $a \in \mathbb{R}$,

$$\{g_n < a\} = \bigcup_{m=n}^{\infty} \{f_m < a\} \in \Sigma.$$

The equality follows from the definition of infimum; the membership follows from measurability of the f_m and closure of Σ under countable unions.

Moreover,

$$g_n \leq g_{n+1}, \qquad g_n \leq f_m \quad \forall m \geq n.$$

The first inequality follows because the infimum defining g_{n+1} is taken over a smaller family; the second follows from the definition of g_n. Consequently,

$$g_n(s) \uparrow \sup_{n\geq 1} g_n(s) = \liminf_{n\to\infty} f_n(s), \qquad \forall s \in S.$$

By the monotone convergence theorem (Theorem 5.2),

$$\int_S \liminf_{n\to\infty} f_n \, d\mu = \lim_{n\to\infty} \int_S g_n \, d\mu.$$

Since $g_n \leq f_m$ for every $m \geq n$, monotonicity of the positive Lebesgue integral (HW 45) gives

$$\int_S g_n \, d\mu \leq \int_S f_m \, d\mu, \qquad \forall m \geq n.$$

Taking the infimum over $m \geq n$, we obtain

$$\int_S g_n \, d\mu \leq \inf_{m\geq n} \int_S f_m \, d\mu.$$

Therefore,

$$\int_S \liminf_{n\to\infty} f_n \, d\mu = \lim_{n\to\infty} \int_S g_n \, d\mu \leq \lim_{n\to\infty} \inf_{m\geq n} \int_S f_m \, d\mu = \liminf_{n\to\infty} \int_S f_n \, d\mu.$$

The first equality follows from MON, the inequality from the preceding bound, and the last equality from the definition of the numerical $\liminf$.

Therefore, the statement is proved.

Lemma 5.5 — Reverse Fatou's Lemma for Functions

Statement. Let $(f_n)_{n\in\mathbb{N}} \subseteq (m\Sigma)^+$. Suppose there exists $g \in (m\Sigma)^+$ such that $f_n \leq g$ for all but finitely many $n \in \mathbb{N}$ and $\mu(g) < \infty$. Then

$$\limsup_{n\to\infty} \int_S f_n \, d\mu \leq \int_S \limsup_{n\to\infty} f_n \, d\mu.$$

HW 48 — Reverse Fatou's Lemma for Functions

Statement. Let $(f_n)_{n\in\mathbb{N}} \subseteq (m\Sigma)^+$ and suppose there is $g \in (m\Sigma)^+$ such that $f_n \le g$ for all but finitely many n, with $\mu(g) < \infty$. Prove that

$$\limsup_{n\to\infty} \int_S f_n \, d\mu \le \int_S \limsup_{n\to\infty} f_n \, d\mu.$$

Observation.

The hypothesis "$f_n \le g$ for all but finitely many n" means that there is $n_0 \in \mathbb{N}$ such that

$$0 \le f_n(s) \le g(s), \qquad n \ge n_0,\ s \in S.$$

Removing finitely many initial terms changes neither $\liminf$ nor $\limsup$, pointwise or for the sequence of integrals, because their tail definitions ignore any finite initial segment.

Let $N := \{s \in S : g(s) = \infty\} \in \Sigma$. For every $M > 0$,

$$M\mu(N) = \mu_0(M\mathbf{1}_N) \le \mu(g) < \infty.$$

The set N is measurable since g is measurable. The equality is the simple-function integral, and the inequality follows from $M\mathbf{1}_N \le g$ and HW 45. Since M is arbitrary, $\mu(N) = 0$; thus $g < \infty$ μ-almost everywhere. By Lemma 5.3, setting g and every f_n to zero on N changes none of their integrals, nor the integral of $\limsup_n f_n$, since the latter also agrees outside N. We use these modified functions below, with the same notation.

Expanded Proof (HW 48).

Discard the initial terms not dominated by g and relabel the tail; we may now assume $0 \le f_n \le g < \infty$ everywhere. Define

$$h_n := g - f_n \in (m\Sigma)^+.$$

Each difference is well-defined, measurable and nonnegative. For $s \in S$,

$$\begin{aligned}\liminf_{n\to\infty}(g(s) - f_n(s)) &= \sup_{k\ge1} \inf_{m\ge k} (g(s) - f_m(s)) \\ &= g(s) - \inf_{k\ge1} \sup_{m\ge k} f_m(s) = g(s) - \limsup_{n\to\infty} f_n(s).\end{aligned}$$

The first and last equalities follow from the tail definitions of $\liminf$ and $\limsup$. The middle equality follows from HW 22, since subtracting from the finite constant $g(s)$ reverses infima and suprema.

To avoid invoking the later Lemma 5.6, we establish integral subtraction directly. For $u \in (m\Sigma)^+$ with $0 \le u \le g$, put $v = g - u$ and take increasing simple approximations $u^{(r)} \uparrow u$ and $v^{(r)} \uparrow v$. Then

$$\begin{aligned}\mu(g) &= \lim_{r\to\infty} \mu_0(u^{(r)} + v^{(r)}) \\ &= \lim_{r\to\infty} \big(\mu_0(u^{(r)}) + \mu_0(v^{(r)})\big) = \mu(u) + \mu(g-u).\end{aligned}$$

The approximations are those of formula (5.1). The first equality follows from MON (Theorem 5.2), since $u^{(r)} + v^{(r)} \uparrow g$; the second is HW 44(ii); the last uses MON for each summand and addition of limits. All integrals are finite by HW 45 and $\mu(g) < \infty$, so rearranging gives

$$\mu(g-u) = \mu(g) - \mu(u), \qquad 0 \le u \le g.$$

With $0 \le \limsup_n f_n \le g$, we obtain

$$\begin{aligned}\mu(g) - \mu\Big(\limsup_{n\to\infty} f_n\Big) &= \mu\Big(\liminf_{n\to\infty} h_n\Big) \\ &\le \liminf_{n\to\infty} \mu(h_n) \\ &= \liminf_{n\to\infty} \big(\mu(g) - \mu(f_n)\big) = \mu(g) - \limsup_{n\to\infty} \mu(f_n).\end{aligned}$$

The first equality uses the pointwise identity and integral subtraction proved above. The inequality is Fatou's lemma (Lemma 5.4), applicable because each h_n is nonnegative and measurable. The next equality uses integral subtraction with $u = f_n$; the last follows from HW 22 with the finite constant $\mu(g)$. Subtracting $\mu(g)$ and multiplying by -1 reverses the inequality:

$$\limsup_{n\to\infty} \int_S f_n \, d\mu \le \int_S \limsup_{n\to\infty} f_n \, d\mu.$$

Tail invariance from the Observation gives the claim for the original sequence. Therefore, the statement is proved.

Lemma 5.6 — Linearity of the Positive Lebesgue Integral

Statement. Let $\alpha, \beta \in [0, \infty)$ and $f, g \in (m\Sigma)^+$. Then

$$\int_S (\alpha f + \beta g) \, d\mu = \alpha \int_S f \, d\mu + \beta \int_S g \, d\mu.$$

Expanded Proof.

Let $(f^{(r)})_{r\in\mathbb{N}}$ and $(g^{(r)})_{r\in\mathbb{N}}$ be the sequences of nonnegative simple functions constructed in (5.1), so that

$$f^{(r)} \uparrow f, \qquad g^{(r)} \uparrow g.$$

Since $\alpha, \beta \ge 0$, we have

$$\alpha f^{(r)} + \beta g^{(r)} \uparrow \alpha f + \beta g.$$

Moreover, $\alpha f^{(r)} + \beta g^{(r)} \in SF^+$ by HW 44(ii).

Applying the monotone convergence theorem (Theorem 5.2) and HW 44(ii), we obtain

$$\begin{aligned}\int_S (\alpha f + \beta g) \, d\mu &= \lim_{r\to\infty} \int_S (\alpha f^{(r)} + \beta g^{(r)}) \, d\mu = \lim_{r\to\infty} \left(\alpha \int_S f^{(r)} \, d\mu + \beta \int_S g^{(r)} \, d\mu \right) \\ &= \alpha \lim_{r\to\infty} \int_S f^{(r)} \, d\mu + \beta \lim_{r\to\infty} \int_S g^{(r)} \, d\mu = \alpha \int_S f \, d\mu + \beta \int_S g \, d\mu.\end{aligned}$$

The first and last equalities follow from MON. The second uses HW 44(ii), since the approximating functions are simple, and the third follows from the limit rules for nonnegative sequences. The convention $0 \cdot \infty = 0$ applies when necessary.

Therefore, the statement is proved.

Positive and Negative Parts of a Measurable Function

We now extend the definition of the Lebesgue integral to measurable functions that can change sign.

Given $f \in m\Sigma$, define its positive and negative parts by

$$f^+ := \max\{f, 0\}, \qquad f^- := \max\{-f, 0\}.$$

Then

$$f = f^+ - f^-, \qquad |f| = f^+ + f^-.$$

Both functions are nonnegative. Their measurability and the preceding identities are verified in HW 49.

HW 49 — Positive and Negative Parts

Statement. Let $f \in m\Sigma$. Prove that

$$f^+, f^- \in (m\Sigma)^+, \qquad |f| = f^+ + f^-.$$

Expanded Proof.

By definition,

$$f^+(s) = \max\{f(s), 0\}, \qquad f^-(s) = \max\{-f(s), 0\},$$

so $f^+, f^- \geq 0$.

We first verify measurability. For every $a \in \mathbb{R}$,

$$\{f^+ > a\} = \begin{cases} S, & a < 0, \\ \{f > a\}, & a \geq 0, \end{cases} \qquad \{f^- > a\} = \begin{cases} S, & a < 0, \\ \{f < -a\}, & a \geq 0. \end{cases}$$

All these sets belong to Σ, since f is measurable. Thus

$$f^+, f^- \in (m\Sigma)^+.$$

Finally, for every $s \in S$,

$$(f^+(s), f^-(s)) = \begin{cases} (f(s), 0), & f(s) \geq 0, \\ (0, -f(s)), & f(s) < 0. \end{cases}$$

In both cases,

$$f(s) = f^+(s) - f^-(s), \qquad |f(s)| = f^+(s) + f^-(s).$$

Since $s \in S$ was arbitrary, we conclude that

$$f = f^+ - f^-, \qquad |f| = f^+ + f^-.$$

Therefore, the statement is proved.

Definition — Lebesgue Integral of a Signed Measurable Function

Let $f \in m\Sigma$. If

$$\min\{\mu(f^+), \mu(f^-)\} < \infty,$$

the Lebesgue integral of f is defined by

$$\mu(f) := \int_S f \, d\mu = \int_S f^+ \, d\mu - \int_S f^- \, d\mu.$$

The condition ensures that at least one of the two nonnegative integrals is finite, so the subtraction is well-defined in the extended real numbers.

If both integrals are infinite, the expression would be $\infty - \infty$, which is undefined. If both are finite, then $\mu(f) \in \mathbb{R}$.

Definition 5.1 — L^1-Space

Let (S, Σ, μ) be a measure space. We denote by $L^1(S, \Sigma, \mu)$ the collection of measurable functions $f \in m\Sigma$ such that

$$\mu(|f|) = \int_S f^+ \, d\mu + \int_S f^- \, d\mu < \infty.$$

Recall from HW 49 that

$$f = f^+ - f^-, \qquad |f| = f^+ + f^-,$$

where $f^+, f^- \in (m\Sigma)^+$. The equality in the definition follows from Lemma 5.6.

In particular, if $f \in L^1(S, \Sigma, \mu)$, both $\mu(f^+)$ and $\mu(f^-)$ are finite. Therefore,

$$\mu(f) = \mu(f^+) - \mu(f^-) \in \mathbb{R}.$$

Such functions are called Lebesgue integrable.

We work with measurable representatives. When $L^1(S, \Sigma, \mu)$ is viewed as a normed space, functions equal μ-almost everywhere are identified; the norm of their common equivalence class is $\|f\|_1 := \mu(|f|)$. Before this identification, the same formula defines a seminorm on the collection of integrable functions.

HW 50 — Absolute Value of the Integral

Statement. For $f \in L^1(S, \Sigma, \mu)$, prove that

$$|\mu(f)| \leq \mu(|f|).$$

Expanded Proof.

Since $f \in L^1(S, \Sigma, \mu)$, we have

$$\mu(f^+) + \mu(f^-) = \mu(|f|) < \infty.$$

Thus both integrals are finite. By the definition of the signed Lebesgue integral,

$$|\mu(f)| = |\mu(f^+) - \mu(f^-)| \leq \mu(f^+) + \mu(f^-) = \mu(|f|).$$

The inequality follows from the triangle inequality for real numbers, since $\mu(f^+), \mu(f^-) \geq 0$, and the last equality follows from Definition 5.1.

Therefore, the statement is proved.

HW 51 — Closure of L^1 under Linear Combinations

Statement. Let $f, g \in L^1(S, \Sigma, \mu)$ and $\alpha, \beta \in \mathbb{R}$. Prove that

$$\alpha f + \beta g \in L^1(S, \Sigma, \mu).$$

Expanded Proof.

Set $u := \alpha f + \beta g$. By Lemma 3.4, $u \in m\Sigma$, since f, g are measurable and $\alpha, \beta \in \mathbb{R}$.

For every $s \in S$, the triangle inequality gives

$$|u(s)| = |\alpha f(s) + \beta g(s)| \leq |\alpha|\,|f(s)| + |\beta|\,|g(s)|.$$

By monotonicity (HW 45) and nonnegative linearity (Lemma 5.6),

$$\mu(|u|) \leq \mu(|\alpha|\,|f| + |\beta|\,|g|) = |\alpha|\,\mu(|f|) + |\beta|\,\mu(|g|) < \infty.$$

The last inequality follows from $f, g \in L^1(S, \Sigma, \mu)$ and the finiteness of α, β. Therefore $u \in L^1(S, \Sigma, \mu)$.

Therefore, the statement is proved.

Linearity of the Integral on L^1

Linearity extends to signed integrable functions. If $f, g \in L^1(S, \Sigma, \mu)$ and $\alpha, \beta \in \mathbb{R}$, then

$$\int_S (\alpha f + \beta g)\, d\mu = \alpha \int_S f\, d\mu + \beta \int_S g\, d\mu.$$

Expanded Proof.

By HW 51, $u := \alpha f + \beta g \in L^1(S, \Sigma, \mu)$. Define the positive and negative parts of the scalars by

$$\alpha^+ = \max\{\alpha, 0\}, \quad \alpha^- = \max\{-\alpha, 0\}, \qquad \beta^+ = \max\{\beta, 0\}, \quad \beta^- = \max\{-\beta, 0\}.$$

Consider the nonnegative measurable functions

$$P := \alpha^+ f^+ + \alpha^- f^- + \beta^+ g^+ + \beta^- g^-,$$
$$Q := \alpha^+ f^- + \alpha^- f^+ + \beta^+ g^- + \beta^- g^+.$$

Since $f, g \in L^1$ and all coefficients are nonnegative, Lemma 5.6 gives $\mu(P), \mu(Q) < \infty$. Moreover,

$$P - Q = (\alpha^+ - \alpha^-)(f^+ - f^-) + (\beta^+ - \beta^-)(g^+ - g^-) = \alpha f + \beta g = u.$$

Together with $u = u^+ - u^-$, this yields the pointwise identity

$$u^+ + Q = u^- + P.$$

By Lemma 5.6,

$$\mu(u^+) + \mu(Q) = \mu(u^-) + \mu(P).$$

All terms are finite, since $u \in L^1$ and $\mu(P), \mu(Q) < \infty$. Thus,

$$\begin{aligned}\mu(\alpha f + \beta g) &= \mu(u^+) - \mu(u^-) = \mu(P) - \mu(Q) \\ &= (\alpha^+ - \alpha^-)(\mu(f^+) - \mu(f^-)) \quad + (\beta^+ - \beta^-)(\mu(g^+) - \mu(g^-)) = \alpha\mu(f) + \beta\mu(g).\end{aligned}$$

The second equality follows from the preceding integrated identity; the third uses Lemma 5.6 and rearrangement of finite quantities, while the last follows from the signed-integral definition.

Therefore, the statement is proved.

We will often need to interchange limits and integrals. One result of this kind is MON; the next theorem provides another important tool for doing so.

Theorem 5.7 — Dominated Convergence (DOM)

Statement. Suppose $(f_n)_{n\in\mathbb{N}} \subseteq m\Sigma$ and $f \in m\Sigma$ satisfy $f_n(s) \to f(s)$ for μ-almost every $s \in S$. If

$$|f_n| \le g, \qquad \forall n \in \mathbb{N},$$

for some $g \in L^1(S, \Sigma, \mu)$, then

$$\lim_{n\to\infty} \int_S |f_n - f|\, d\mu = 0, \qquad \lim_{n\to\infty} \int_S f_n\, d\mu = \int_S f\, d\mu.$$

Proof.

Since $|f_n| \le g$ and $g \in L^1(S, \Sigma, \mu)$, monotonicity (HW 45) gives

$$0 \le \mu(|f_n|) \le \mu(g) < \infty, \qquad \forall n \in \mathbb{N}.$$

Thus every f_n is integrable. Let $N \in \Sigma$ be a null set outside which $f_n(s) \to f(s)$. Passing to the pointwise limit in $|f_n(s)| \le g(s)$ yields

$$|f(s)| \le g(s), \qquad \forall s \in N^c.$$

As in Lemma 5.3, modifying functions on N does not affect their integrals. We may therefore replace f_n and f by $f_n\mathbf{1}_{N^c}$ and $f\mathbf{1}_{N^c}$, respectively. After this replacement, $f_n \to f$ everywhere and $|f_n|, |f| \le g$ everywhere. In particular,

$$\mu(|f|) \le \mu(g) < \infty,$$

so $f \in L^1(S, \Sigma, \mu)$ as well. The original functions have the same integrals as these modifications, by Lemma 5.3 applied to their positive and negative parts.

By the triangle inequality,

$$0 \le |f_n - f| \le |f_n| + |f| \le 2g, \qquad \forall n \in \mathbb{N}.$$

Since $g \ge 0$ and $g \in L^1$, Lemma 5.6 gives $\mu(2g) = 2\mu(g) < \infty$. We may therefore apply reverse Fatou's lemma (Lemma 5.5) to the sequence $|f_n - f|$:

$$0 \le \limsup_{n\to\infty} \int_S |f_n - f|\, d\mu \le \int_S \limsup_{n\to\infty} |f_n - f|\, d\mu = \int_S 0\, d\mu = 0.$$

The first inequality is nonnegativity of the integrals, the second follows from reverse Fatou and the bound by $2g$, and the equality uses $f_n \to f$ pointwise. Thus

$$\lim_{n\to\infty} \int_S |f_n - f|\, d\mu = 0.$$

Finally, by linearity of the integral on L^1 and HW 50,

$$\left| \int_S f_n\, d\mu - \int_S f\, d\mu \right| = \left| \int_S (f_n - f)\, d\mu \right| \le \int_S |f_n - f|\, d\mu \longrightarrow 0.$$

The inequality is valid since $f_n - f \in L^1$ by HW 51. This proves the second assertion.

Therefore, the statement is proved.

HW 52 — Why the Domination Hypothesis Is Necessary

Statement. Let $(S, \Sigma, \mu) = ([0,1], \mathcal{B}([0,1]), \text{Leb})$ and define

$$f_n(s) = n\mathbf{1}_{(0,\,1/n)}(s), \qquad n \in \mathbb{N}.$$

Show that the conclusion of DOM fails without an integrable dominating function.

Proof.

Fix $s \in (0,1]$. For every $n \ge 1/s$, we have $s \notin (0, 1/n)$, so $f_n(s) = 0$. Moreover, $f_n(0) = 0$ for every n. Hence

$$f_n(s) \longrightarrow f(s) := 0, \qquad \forall s \in [0,1].$$

Nevertheless,

$$\int_0^1 f_n\, d\text{Leb} = n\,\text{Leb}\big((0,1/n)\big) = 1, \qquad \forall n \in \mathbb{N},$$
$$\int_0^1 f\, d\text{Leb} = 0.$$

The first equality uses the integral of a nonnegative simple function; the second is the integral of the zero function. Consequently,

$$\lim_{n\to\infty} \int_0^1 f_n\, d\text{Leb} = 1 \neq 0 = \int_0^1 f\, d\text{Leb}.$$

We now verify directly which hypothesis of DOM fails. Suppose that a function

$$g \in L^1([0,1], \mathcal{B}([0,1]), \text{Leb})$$

satisfies $f_n \le g$ for every n. For each $k \in \mathbb{N}$, set

$$I_k := \left(\frac{1}{k+1}, \frac{1}{k} \right).$$

On I_k we have $f_k = k$, so $g \ge k$. Since the intervals I_k are pairwise disjoint, for every $N \in \mathbb{N}$,

$$\mu(g) \ge \mu\left(\sum_{k=1}^{N} k\mathbf{1}_{I_k} \right) = \sum_{k=1}^{N} k\,\text{Leb}(I_k) = \sum_{k=1}^{N} k\left(\frac{1}{k} - \frac{1}{k+1} \right) = \sum_{k=1}^{N} \frac{1}{k+1}.$$

The inequality follows from HW 45, and the first equality uses the definition of the simple integral. As $N \to \infty$, the last sum diverges, contradicting $\mu(g) < \infty$.

Thus no single integrable function dominates all f_n. The example does not contradict DOM: its domination hypothesis is not satisfied.

Therefore, the statement is proved.

The Standard Machinery

Many properties of Lebesgue integrals can be established through the following four steps:

1. Prove the property for indicators $\mathbf{1}_A$, where $A \in \Sigma$.
2. Use linearity to extend it to nonnegative simple functions.
3. Approximate nonnegative measurable functions by increasing simple functions and apply the monotone convergence theorem (MON, Theorem 5.2).
4. Use $f = f^+ - f^-$ to extend the result to signed measurable functions whenever the relevant integrals are well-defined.

This is the standard machinery used throughout the course.

A Measure Defined by a Nonnegative Density

Different measures on the same measurable space yield different Lebesgue integrals. Fix $f \in (m\Sigma)^+$ and define, for every $A \in \Sigma$,

$$\lambda(A) := \int_S f\mathbf{1}_A \, d\mu = \int_A f \, d\mu.$$

The following homework proves that $\lambda : \Sigma \to [0, \infty]$ is a measure. In addition, λ is absolutely continuous with respect to μ, written $\lambda \ll \mu$, meaning

$$\mu(N) = 0 \quad \Longrightarrow \quad \lambda(N) = 0, \qquad \forall N \in \Sigma.$$

HW 53 — Countable Additivity of λ

Statement. Let $f \in (m\Sigma)^+$ and define $\lambda(A) = \int_S f\mathbf{1}_A \, d\mu$ for every $A \in \Sigma$. Prove that λ is countably additive and hence is a measure on (S, Σ).

Proof.

First, $\lambda(A) \geq 0$ for every $A \in \Sigma$, since $f\mathbf{1}_A \geq 0$, and

$$\lambda(\emptyset) = \int_S f\mathbf{1}_\emptyset \, d\mu = \int_S 0 \, d\mu = 0.$$

The equality follows from $\mathbf{1}_\emptyset = 0$ and the definition of the positive integral.

Let $(A_n)_{n\in\mathbb{N}} \subseteq \Sigma$ be pairwise disjoint. Set

$$U_r := \bigcup_{n=1}^{r} A_n, \qquad U := \bigcup_{n=1}^{\infty} A_n.$$

Since the sets are pairwise disjoint and $f \geq 0$,

$$f\mathbf{1}_{U_r} = \sum_{n=1}^{r} f\mathbf{1}_{A_n} \uparrow f\mathbf{1}_U \qquad \text{pointwise on } S.$$

The equality follows from $\mathbf{1}_{U_r} = \sum_{n=1}^{r} \mathbf{1}_{A_n}$; the monotone convergence follows from $U_r \uparrow U$.

Therefore,

$$\lambda\left(\bigcup_{n=1}^{\infty} A_n\right) = \int_S f\mathbf{1}_U \, d\mu = \lim_{r\to\infty} \int_S f\mathbf{1}_{U_r} \, d\mu = \lim_{r\to\infty} \sum_{n=1}^{r} \int_S f\mathbf{1}_{A_n} \, d\mu = \sum_{n=1}^{\infty} \lambda(A_n).$$

The second equality follows from MON (Theorem 5.2), the third from nonnegative linearity (Lemma 5.6), and the last from the definition of an infinite nonnegative sum. Hence λ is countably additive and therefore is a measure on (S, Σ).

Moreover, let $N \in \Sigma$ satisfy $\mu(N) = 0$. Define

$$h_r := (f \wedge r)\mathbf{1}_N, \qquad r \in \mathbb{N}.$$

Then $h_r \in (m\Sigma)^+$, $h_r \uparrow f\mathbf{1}_N$, and $0 \leq h_r \leq r\mathbf{1}_N$. By monotonicity (HW 45),

$$0 \leq \mu(h_r) \leq \mu(r\mathbf{1}_N) = r\mu(N) = 0.$$

The equality uses the integral of a simple indicator. Applying MON once more gives

$$\lambda(N) = \int_S f\mathbf{1}_N \, d\mu = \lim_{r\to\infty} \mu(h_r) = 0.$$

Thus $\lambda \ll \mu$.

Therefore, the statement is proved.

Theorem 5.8 — Radon–Nikodym Theorem

Statement. Let μ and λ be σ-finite measures on (S, Σ) such that $\lambda \ll \mu$. Then there exists $f \in (m\Sigma)^+$ satisfying

$$\lambda(A) = \int_S f\mathbf{1}_A \, d\mu, \qquad \forall A \in \Sigma.$$

The function

$$f =: \frac{d\lambda}{d\mu}$$

is called the Radon–Nikodym derivative of λ with respect to μ.

The density is unique up to μ-almost-everywhere equality.

Recall.

A measure ν is σ-finite if there exist measurable sets $(E_n)_{n\in\mathbb{N}}$ such that

$$S = \bigcup_{n=1}^{\infty} E_n, \qquad \nu(E_n) < \infty \quad \forall n \in \mathbb{N}.$$

HW 53 constructs a measure λ from a given nonnegative density f; the Radon–Nikodym theorem gives the converse under σ-finiteness and absolute continuity. As specified in the lecture notes, the theorem is stated without proof.